Derived Category of GITs

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1 Geometric Invariant Theory

In topology, for $G \curvearrowright X$ with X a topological space and G a group, the quotient space $X \longrightarrow X/G$ parametrizing G-orbits in X. In geometry, consider the action of a Lie group G on a manifold X. We'd like to consider X/G quotient. This is usually not a well-behaved space (it is not even Hausdroff). However, if G acts on X properly and freely, then X/G is a manifold.

In algebraic geometry? Consider the action of G a group (algebraic group) on X a scheme (of finite type over the field of char 0). How to construct quotients of X by G? Still studied in algebraic geometry.

We start by the definition of algebraic groups.

Definition 1.1. An **algebraic group** G is a scheme over k with morphisms $m : G \times G \longrightarrow G$, $e : \mathrm{Spec}(k) \longrightarrow G$ and $l : G \longrightarrow G$ such that the usual composition between them satisfied:

$$\begin{array}{ccc} G \times G \times G & \xrightarrow[m\times \mathrm{id}]{} & G \times G \\ \downarrow{\scriptstyle \mathrm{id}\times m} & & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow{m} & G \end{array}$$

$$\begin{array}{ccccc} G & \xrightarrow[(l,\mathrm{id})]{} & G \times G & \xleftarrow{(\mathrm{id},l)} & G \\ \downarrow & & \downarrow{\scriptstyle m} & & \downarrow \\ \mathrm{Spec}\ k & \xrightarrow{e} & G & \xleftarrow[e]{} & \mathrm{Spec}\ k \end{array}$$

$$\begin{array}{ccccc} \mathrm{Spec\ k} \times G & \xrightarrow[e\times \mathrm{id}]{} & G \times G & \xleftarrow{\mathrm{id}\times \mathrm{e}} & G \times \mathrm{Spec\ k} \\ & \searrow & \downarrow{\scriptstyle m} & \swarrow & \\ & & G & & \end{array}$$

We restrict to affine algebraic groups.

Example 1.2. 1). Consider $G = \mathrm{Spec}\ k[t]$ with

$$\begin{aligned} m^* : k[t] &\longrightarrow k[t] \otimes k[t] \\ t &\longmapsto t \otimes 1 + 1 \otimes t \end{aligned}$$

$$\begin{aligned} i^* : k[t] &\longrightarrow k[t] \\ t &\longmapsto -t \end{aligned}$$

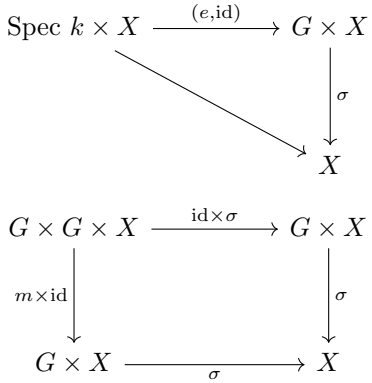
2). $\mathrm{G}_m = \mathrm{Spec}\ k[t^{\pm 1}]$

$$\begin{aligned} m^* : k[t^{\pm 1}] &\longrightarrow k[t^{\pm 1}] \otimes k[t^{\pm 1}] \\ t &\longmapsto t \otimes t \end{aligned}$$

$$\begin{aligned} i^* : k[t^{\pm 1}] &\longrightarrow k[t^{\pm 1}] \\ t &\longmapsto t^{-1} \end{aligned}$$

3). $\mathrm{GL}_n = \mathrm{Spec}\ k[x_{ij} \big| 1 \leq i, j \leq n]$ with $m^*(x_{ij}) = \sum_{k=1}^{n} x_{ik} \otimes x_{kj}$.
4). Other examples: Product of these: $\mathbb{G}_m^{\times d}$ is called torus.

Definition 1.3. The group action is a morphism $\sigma : G \times X \longrightarrow X$ with G an affine algebraic group such that



commute.

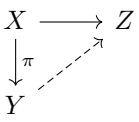
Example 1.4. 1). $X = V$, a finite dimensional representation of G. $G \longrightarrow \mathrm{GL}(V) \cong \mathrm{GL}_n$ with $n = \dim V$;

2). $X = G \curvearrowleft G$ multiplication on the left $g \cdot x = gx$.

3). $X = G \curvearrowleft G$ by conjugation $g \cdot x = gxg^{-1}$.

Now, we consider the **quotients** of these group actions. To start with, it is natural to define the quotient as follows:

Definition 1.5. For $G \curvearrowright X$, the categorical quotient is a scheme Y with morphism $\pi : X \longrightarrow Y$ which is G-equivariant and satisfies the universal property



for any G-equivariant morphism $X \longrightarrow Z$.

Remark 1.6. If (Y, π) is a categorical quotient, then ϕ is invariant so it maps the entire orbit to a single point. However, $(Y\,\pi)$ **can not** distinguish different orbit. Since π is continuous, the preimage of that point is a closed subset. Therefore, π is constant on the **closure** of any orbit.

Example 1.7. Consider the action of $\mathbb{G}_m$ on $\mathbb{A}^n$ with weight one. The point 0 is an orbit, and for any nonzero orbit $\lambda k^\times$ (with $\lambda \neq 0$), its closure contains 0. Thus, (Spec, k, π) is the categorical quotient. However, we see that (Spec, k, π) does not distinguish between the different orbits.

If we want each point of (Y, π) to correspond to a unique orbit, by the 1.7, all of the orbits should be closed. Hence, we need to refine the concept of a categorical quotient in order to obtain a variety with additional geometric properties.

Definition 1.8. For $G \curvearrowright X$, a G-equivariant morphism $\pi : X \longrightarrow Y$ is called **good quotient** if

1).π is surjective;

2). Y parametrizes G-orbits, that is, if $\pi(x) = \pi(x')$, there exists $g \in G$ such that $x' = g \cdot x$,

3). $\mathcal{O}_Y = (\pi_* \mathcal{O}_X)^G$,

4). If $W \subset X$ is a closed G-invariant subset then $\pi(W)$ is closed;

5). If W_1, W_2 are closed G-invariant subsets of X such that $W_1 \cap W_2 = \emptyset$, then $\pi(W_1) \cap \pi(W_2) = \emptyset$.

Moreover, if $\pi^{-1}(y)$ is a single orbit of the G-action, then (Y, π) is called a **geometric quotient**.

Immediately, we have:

Lemma 1.9. *$G \curvearrowright X$, with (Y, π) a good quotient. For any $x_1, x_2 \in X$, we have $\pi(x_1) = \pi(x_2)$ if and only if $\overline{Gx_1} = \overline{Gx_2}$.*

Proof. Since π is G-equivariant, then it is constant on orbit closures, and therefore $\overline{Gx_1} \cap \overline{Gx_2} \neq \emptyset$ implies $\pi(x_1) = \pi(x_2)$.

Conversely, if $\pi(x_1) = \pi(x_2)$, then by definition, $\overline{Gx_1} \cap \overline{Gx_2} \neq \emptyset$. □

Corollary 1.10. *If all the obits of an action with a good quotient are closed, then the quotient is geometric.*

David Mumford developed **Geometric Invariant Theory** (GIT) to construct good/geometric quotients of algebraic varieties by group actions in a systematic and geometrically meaningful way. The central idea is to replace naive orbit spaces—which are often ill-behaved—with well-structured quotients obtained from invariant functions.

Consider $G \curvearrowright X$ with G a reductive group and X a quasi-projective variety (with some linearization). Then there exists

$$X \supset U \supset V$$

with each inclusion open such that both of them are G-invariant and $U \longrightarrow U//G$ is a categorical quotient and $V \longrightarrow V//G$ is a geometric quotient.

Example 1.11. 1). Consider the action of $\mathbb{G}_m$ over $\mathbb{A}^n$ weigh weight one. The categorical quotient exists and this is just Spec k. However, the geometric quotient does not exist.

2). Consider the cation of $\mathbb{G}_m$ over $\mathbb{A}^n \setminus \{0\}$. The categorical quotient is just the geometric quotient, which is $\mathbb{P}^n$.

1.1 Affine GIT

Let's look at categorical quotient of affine schemes $X = \text{Spec } A \longrightarrow Y$. Any G-invariant $X \longrightarrow Z$ induces $\mathcal{O}_Z \longrightarrow \mathcal{O}_X^G = A^G$. If A were finitely generated, then $Y = \text{Spec}(A^G)$.

If A is finitely generated, is A^G finitely generated? Hilbert first proved that for $G = \text{GL}_n$, A^G is finitely generated. However, Nataga gave an example of G of dimension 13 and X of dimension $23 (\cong \mathbb{A}^{32})$ such that $k[x_1, ..., x_{32}]^G$ is not finitely generated.

There is a notation of reductive groups such that

Theorem 1.12 (Nagata)**.** *If A is finitely generated and G is a reductive group, then A^G is finitely generated.*

Examples of reductive groups are GL_n, SL_n, finite groups and products of them, etc.

$n = 1$, $\text{GL}_1 = \mathbb{G}_m$. Examples of groups which are not reductive $\mathbb{G}_a$, extensions of $\mathbb{G}_m$ such that

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Theorem 1.13 (Weyl, Nagata, Mumford)**.** char $k = 0$. *The followings are equivalent:*

1). Reductive: unipotent radical (the maximal connected unipotent normal subgroup) is trivial;

2). geometric reductive: $\forall$ finite dimensional G representation V, for all $v \in V^G \setminus \{0\}$, there exists $f \in \mathcal{O}(V)^G$ such that $f(v) \neq 0$;

3). linearly reductive: every finite dimensional representation of G splits as a direct sum of irreducible representations.

For example, no unipotent normal subgroups extension of $\mathbb{G}_a$'s.

Now, we set $X//G$ to be the affine variety Spec A^G. Then the inclusion $i : A^G \longrightarrow A$ induces an affine morphism of varieties $\pi : X \longrightarrow X//G$. We call $(X//G, \pi)$ the **affine GIT quotient**.

Here are some examples of affine GIT quotients:

Example 1.14. 1).Consider $\mathbb{G}_m \curvearrowright \mathbb{A}^n = \text{Spec } k[x_1, ..., x_n]$ with $z \cdot x_i = x_i$. The the affine GIT quotient is Spec $k[x_1, ..., x_n]^{\mathbb{G}_m}$:

$f(x_1, .., x_n)$ is $\mathbb{G}_m$-invariant: Write $f = \sum c_{a_1,a_2,..,a_n} x^{a_1} x^{a_2} \cdots x^{a_n}$. Then

$$z \cdot f = f(zx_1, ..., zx_n) = \sum c_{a_1,a_2,..,a_n} z^{a_1+a_2+\cdots+a_n} x^{a_1} x^{a_2} \cdots x^{a_n}$$

which implies $c_{a_1,...,a_n} = c_{a_1,...,a_n} z^{a_1+\cdots+a_n}$. So all the coefficients vanish unless $a_1 + \cdots + a_n = 0$. Thus, $a_1 = a_1 = \cdots = a_n = 0$. So the only $\mathbb{G}_m$-invariant functions are the constants.

2). $\mathbb{G}_m \curvearrowright \mathbb{A}^2 = \text{Spec } k[x, y]$ with wight 1 and -1. $z \cdot x = zx, z \cdot y = z^{-1}y$. Then Spec $k[x, y]^{\mathbb{G}_m} =$ Spec $k[xy]$:

For $f(x, y) = \sum_{i,j\geq 0} c_{ij} x^i y^j$, $z \cdot f(x, y) = \sum_{i,j\geq 0} c_{ij} z^{i-j} x^i y^j$. $c_{ij} = c_{ij} z^{i-j}$ implies $c_{ij} = 0$ if $i \neq j$. Thus, $f(x, y) = h(xy)$. So Spec $k[x, y]^{\mathbb{G}_m} =$ Spec $k[xy] \cong \mathbb{A}^1$.

3). Consider $\mathbb{G}_m \curvearrowright \mathbb{A}^4$ with weight $1, 1, -1, -1$. Write

$$\mathbb{A}^4 = \text{Spec } k[x_1, x_2, y_1, y_2]$$

with $z \cdot x_i = zx_i$ and $z \cdot y_i = z^{-1}y_i$. One can show that all $\mathbb{G}_m$-invariant functions on x_1, x_2, y_1, y_2 are polynomials in $x_1y_1, x_2y_1, x_1y_2, x_2y_2$.Thus,

$$\text{Spec } (k[x_1, x_2, y_1, y_2])^{\mathbb{G}_m} = \text{Spec } k[a, b, c, d]/(ab - cd)$$

with $a = x_1y_1, b = x_2y_1, c = x_1y_2, d = x_2y_2$.

4). Consider $\text{GL}_n \curvearrowright \mathfrak{gl}_n = \text{End}(V)$ by conjugation with V a k-vector space of dimension n.

Categorical quotient is $\text{Spec}(k[x_{ij}\big|1 \leq i, j \leq n]^{\text{GL}_n}) = \text{Spec}(k[\text{tr}(A^i)]) \cong \mathbb{A}^n$. Here $k[x_{ij}\big|1 \leq i, j \leq n]^{\text{GL}_n}$ is the invariants of conjugate classes of $A = (x_{ij})_{1\leq i,j\leq n}$, they are coefficients of the characteristic polynomials. In fact, we can show that $k[x_{ij}\big|1 \leq i, j \leq n]^{\text{GL}_n} = k[x_1, ..., x_n]^{S_n} = k[e_1, ..., e_n]$. Thus, the categorical quotient is just $\mathbb{A}^n$.

For an algebraic affine group G, consider its action on $V = \mathbb{A}^n$, then $V//G$ parametrizes closed G-orbits. For example, in 4), closed orbits correspond to diagonal matrices.

Theorem 1.15. *Let G be a reductive group acting on an affine variety X. Then affine GIT quotient $(X//G, \pi)$ is a good quotient.*

Proof. See Hoskins, [Hos15, Theorem 4.30]. □

Corollary 1.16. *Let G be a reductive group acting on an affine variety X such that $(X//G, \pi)$ is the affine GIT quotient. We have $\pi(x_1) = \pi(x_2)$ if and only if $\overline{Gx_1} = \overline{Gx_2} \neq \emptyset$.*

Furthermore, for any $y \in X//G$, the preimage $\pi^{-1}(y)$ contains a unique closed orbit. If all G-orbits under the action on X are closed, then $(X//G, \pi)$ is a geometric quotient.

Proof. By 1.9. □

By 1.14 1), an affine GIT quotient $X//G$ cannot always be an geometric quotient. So we would like there at least to be a large enough subset of X such that the **restriction of the quotient morphism π to this subset** defines a geometric quotient. In order to characterize the points, we have :

Definition 1.17. A point $x \in X$ is called **stable** if the orbit Gx is closed and the stabilizer $\dim G_x = 0$.

Remark 1.18. When k is a infinite field, $\dim G_x = 0$ if and only if G_x is finite.

Denote the set of stable points by X_s. Then we have

Proposition 1.19. *Let G be a reductive group acting on an affine variety X with affine GIT quotient $(X//G, \pi)$. Then $X_s \subset X$ is an open and G-invariant subset of X such that*

1). $X_s = \pi^{-1}(\pi(X_s))$;

2). $\pi(X_s)$ is open;

3). $(\pi(X_s), \pi|_{X_s})$ is a geometric quotient.

Proof. See Hoskins, [Hos15, Proposition 4.36]. □

Example 1.20. 1). In 1.14 1), there are no stable points.

2). The stable points of $\mathbb{G}_m \curvearrowright \mathbb{A}^2$ with weight ± 1 are points (x, y) belonging to the curves $xy = a$ for $a \in k^\times$: $\mathbb{G}_{m(x,y)} = \{e\}$ which is of dimension 0 and $\mathbb{G}_m(x, y) = \{(s, t)\big|st = xy = a\}$, which is closed. While if $x = 0, y \neq 0$, then $G_{(x,y)}$ is not closed for its closure contains $(0, 0)$. So is for the case $x \neq 0, y = 0$. When $x = 0, y = 0$, $\dim G_{(x,y)} = 1$. Thus,

$$\mathbb{A}^2_s = \mathbb{A}^2 \setminus \{(x, y)\big|xy = 0\}$$

and the quotient $\pi : \mathbb{A}^2 \setminus \{(x, y)\big|xy = 0\} \longrightarrow \mathbb{A}^1 \setminus \{0\}$ given by $(x, y) \longmapsto xy$ defines a geometric quotient.

3). The action by conjugation of GL_n over $\mathfrak{gl}_n$ has on stable points, since very Jordan normal form has a stabilizer of dimension greater then 1.

1.2 Projective GIT

Let X be a projective scheme with an action of reductive group G. A central idea of projective GIT is that not all points of X behave well under the group action. This leads to the introduction of stable and semistable loci, defined via the behavior of invariant sections.

Now we introduce projective GIT first for projective varieties, then extend it to projective schemes, and finally generalize the theory to quasi-projective schemes.

Projective GIT for projective varieties

Let X be a projcetive variety with an action of a reductive group G.

Definition 1.21. Let X be a projective variety with an action of affine algebraic group G. A linear G-equivariant projective embedding of X is a group homomoprhism $G \longrightarrow \mathrm{GL}_{n+1}$ and a G-equivariant projective embedding $X \hookrightarrow \mathbb{P}^n$.

If we have a linear action of G over X, then the action of G on $\mathbb{P}^n$ lifts to an action of G on the affine cone $\mathbb{A}^{n+1}$ over $\mathbb{P}^n$. More precisely, if X is the closed subscheme associated to a homogeneous ideal $I(X) \subset k[x_0, ..., x_n]$, then $\widetilde{X} = \mathrm{Spec}(R(X))$ with $R(X) = k[x_0, ..., x_n]/I(X)$.

Example 1.22. 1). We consider the action of $\mathbb{G}_m$ over $X = \mathbb{P}^1$, via

$$t \cdot [x_0 : x_1 : x_2] = [x_0 : x_1 : tx_2]$$

then this action lifts to $\mathbb{G}_m$ over $\mathbb{A}^3$ by

$$t \cdot (x_0, x_1, x_2) = (x_0, x_1, tx_2)$$

2). For $X = V_+(x_0x_2 - x_1^2) \subset \mathbb{P}^2$, $I(X) = (x_0x_2 - x_1^2)$ and $\widetilde{X} = \mathrm{Spec}(k[x_0, x_1, x_2]/(x_0x_2 - x_1^2))$. Let $\mathbb{G}_m$ act on X by

$$t \cdot [x_0 : x_1 : x_2] = [x_0 : tx_1 : t^2x_2]$$

It lifts to $\widetilde{X} = V(x_0x_2 - x_1^2) \subset \mathbb{A}^3$:

$$t \cdot (x_0, x_1, x_2) = (x_0, tx_1, t^2x_2)$$

Since G acts on $\mathbb{A}^{n+1}$ through GL_{n+1}, G-action over $\mathcal{O}(\mathbb{A}^{n+1})$ preserves the graded pieces, so

$$\mathcal{O}(\mathbb{A}^{n+1})^G = \bigoplus_{r\geq 0} k[x_0, ..., x_n]_r^G$$

is a graded algebra and similarly, $R(X)^G = \bigoplus_{r\geq 0} R(X)_r^G$ has a graded structure. By Nagata's theorem, $R(X)^G$ is finitely generated, as G is reductive. Thus, we can consider $\mathrm{Proj}(R(X)^G)$.

Since $R(X)^G \hookrightarrow R(X)$ is an inclusion preserving the grading, there exists a rational morphism of projective schemes [Har77, Exe II.3.12]:

$$X \dashrightarrow \mathrm{Proj}(R(X)^G)$$

where indeterminacy locus is the closed subscheme X defined by $R(X)_+^G = \bigoplus_{r>0} R(X)_r^G$.

Definition 1.23. The **Nullcone** N is the closed subscheme of X defined by homomorphism ideal $R(X)_+^R$ in $R(X)$. The **semistable** set $X^{ss} = X \setminus N$. The morphism $\varphi : X^{ss} \longrightarrow \mathrm{Proj}(R(X)^G)$ is called **projective GIT quotient**.

Remark. By the definition of N, $x \in X^{ss}$ if there exists $f \in R(X)_r^G$ for some $r > 0$ such that $f(x) \neq 0$.

Example 1.24. 1). For 1) in 1.22, $R(X)^G = k[x_0, x_1] \hookrightarrow k[x_0, x_1, x_2]$. Then we get

$$\begin{aligned} \mathbb{P}^2 &\dashrightarrow \mathbb{P}^1 \\ [x_0 : x_1 : x_2] &\longmapsto [x_0 : x_1] \end{aligned}$$

and the indeterminacy locus over $\mathbb{P}^2$ is $V_+(x_0, x_1)$. So $N = V_+(x_0, x_1)$ and $X^{ss} = D_+(x_0, x_1)$.

2). Let $X = \mathbb{P}^2$ with $\mathbb{G}_m$-action:

$$t \cdot [x_0 : x_1 : x_2] = [tx_0 : t^{-1}x_1 : x_2]$$

Then $R[X]^G = k[x_0x_1, x_2]$. Note that $\deg(x_0x_1) = 2$ and $\deg x_2 = 1$ over $R[X]^G$. So by [Har77, Exe. II.5.13], we have

$$\mathrm{Proj}(R[X]^G) \cong \mathrm{Proj}((R[X]^G)^{(2)})$$

where $(R[X]^G)^{(2)} = k[x_0x_1, x_2^2]$. Through $k[x_0x_1, x_2^2] \hookrightarrow k[x_0, x_1, x_2]$, we get

$$\begin{aligned} \mathbb{P}^2 &\dashrightarrow \mathbb{P}^1 \\ [x_0 : x_1 : x_2] &\longmapsto [x_0x_1, x_2^2] \end{aligned}$$

where the indeterminacy locus is $V(x_0x_1, x_2^2) = \{[0 : 1 : 0], [1 : 0 : 0]\} \subset \mathbb{P}^1$ and hence $X^{ss} = D_+(x_2) \cup (D_+(x_1) \cap D_+(x_0))$.

Theorem 1.25. *The projective GIT quotient $\varphi : X^{ss} \longrightarrow X//G$ is a good quotient.*

Proof. [Hos15, Theorem 5.3]. □

Since $\varphi : X^{ss} \longrightarrow X//G$ is a good quotient, for $x_1, x_2 \in X^{ss}$, we have:

$$\overline{Gx_1} \cap \overline{Gx_2} \cap X_{ss} \neq \emptyset \iff \varphi(x_1) = \varphi(x_2)$$

Thus the preimage of each point in $X//G$ contains a unique closed orbit. So we can ask if there is an open subset X^s of X^{ss} on which this quotient becomes a geometric quotient.

Remark. For $x_1, x_2 \in X^{ss}$, the closures of their orbits may meet at N. For example, in 2) of 1.24,

$$\overline{G \cdot [1 : 0 : 1]} \cap \overline{G \cdot [1 : 1 : 0]} = \{[1 : 0 : 0]\} \subset N$$

Definition 1.26. A point x of X is

• stable if $\dim G_x = 0$ and there exists $f \in R(X)^G_+$ homogeneous such that $x \in X_f$ and the action of G on X_f is closed;

• unstable if it is not semistable.

Theorem 1.27. *The stable set X^s is open, $Y := \varphi(X^s)$ is open and $\varphi|_{\varphi^{-1}(Y)} : \varphi^{-1}(Y) \longrightarrow Y$ is a geometric quotient.*

Example 1.28. 1). Consider the linear action of $\mathbb{G}_m$ on $\mathbb{P}^n$ by

$$t \cdot [x_0 : x_1 : x_n] = [t^{-1}x_0 : tx_1 : tx_2 : .... : tx_n]$$

Then $R[X]^G = k[x_0x_1, ..., x_0x_n] \cong k[y_1, ..., y_n]$. Hence, we have

$$\begin{aligned} \mathbb{P}^n &\dashrightarrow X//G = \mathbb{P}^{n-1} \\ [x_0 : \cdots : x_n] &\longmapsto [x_0x_1 : \cdots : x_ox_n] \end{aligned}$$

with $N = V_+(x_0) \cup \{[1 : 0 : \cdots : 0]\} \subset \mathbb{P}^n$ and so $X^{ss} = D_+(x_0) - \{[1 : 0 : \cdots : 0]\} \cong \mathbb{A}^n \setminus \{0\}$. Therefore,

$$\begin{aligned} \mathbb{A}^n \setminus \{0\} &\longrightarrow \mathbb{P}^{n-1} \\ (x_1 : \cdots : x_n) &\longmapsto [x_1 : \cdots : x_n] \end{aligned}$$

is a good quotient of action on X^{ss}. Moreover, as the preimage of each point in $X//G$ is a single orbit, this is also a geometric quotient.

2). In 2) of 1.24, let's find the stable points in $X^{ss} = D_+(x_2) \cup (D_+(x_1) \cap D_+(x_0))$. Now, let's compute the stabilizers: if $[tx_0 : t^{-1}x_1 : x_2] = [x_0 : x_1 : x_2]$, then there exists $\lambda \in k^\times$ such that

$$tx_0 = \lambda x_0,\ t^{-1}x_1 = \lambda x_1,\ x_2 = \lambda x_2$$

When $x_2 \neq 0$, $\lambda = 1$, if $x_0 = x_1 = 0$, then for any $t \in \mathbb{G}_m$, $[tx_0 : t^{-1}x_1 : x_2] = [x_0 : x_1 : x_2]$; if $x_0 \neq 0$ or $x_1 \neq 0$, we can get $t = \lambda$ or $t = \lambda^{-1}$ respectively. Hence, $\dim \mathbb{G}_{m,[0:0:1]} = 1$, which implies $[0 : 0 : 1]$ is unstable. When $x_2 = 0$, by the same argument, $t = \lambda$ or $t = \lambda^{-1}$.

Now, we fix the condition that $x_0 \neq 0$ or $x_1 \neq 0$. If $x_0, x_1 \neq 0$, then

$$\mathcal{G}_m[x_0 : x_1 : x_2] = \{[x_0 : x_1 : x_2] \big| x_0x_1 = c, x_2 = d\}$$

which is closed; if $x_0 \neq 0, x_1 = 0$, then $x_2 \neq 0$ in X^{ss}

$$\mathbb{G}_m[x_0 : x_1 : x_2] = \{[tx_0^{-1} : 0 : x_2] \big| t \in \mathbb{G}_m\}$$

which is not closed as $[1 : 0 : 0] \in \overline{\mathbb{G}_m[x_0 : x_1 : x_2]} \setminus \mathbb{G}_m[x_0 : x_1 : x_2]$. Thus, these points are not stable; if $x_0, x_1 \neq 0$, then also, these points are not stable in the same reason.

Thus, $X^s = \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 \big| x_0x_1 \neq 0\} \subset \mathbb{A}^2 \setminus \{0\}$. Therefore,

$$\begin{aligned} \mathbb{A}^2 \setminus \{0\} &\longrightarrow \mathbb{P}^1 \\ (x, y) &\longrightarrow [x : y^2] \end{aligned}$$

is a geometric quotient with $t \cdot (x, y) = (t^2x, ty)$. (Here, under the coordinates of $\mathbb{P}^2$, $x = \frac{x_1}{x_0}$, $y = \frac{x_2}{x_0}$, which gives the map above.)

Projective GIT for projective scheme

Now let X be a projective scheme. First, we embed X into a projective space, which requires choosing a very ample line bundle $\mathcal{L}$ on X. More precisely, the projective scheme X and ample line bundle L, determines a finitely generated graded k-algebra.

$$R(X,\mathcal{L}) := \bigoplus_{r\geq 0} H^0(X, L^{\otimes r})$$

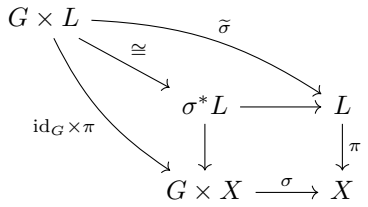
We can choose generators of this k-algebra $s_0, ..., s_n \in H^0(X, L)$. Then this section determines a closed embedding:

$$\begin{aligned} X &\hookrightarrow \mathbb{P}^n = \mathbb{P}(R(X,L)) \\ x &\longmapsto [s_0(x) : s_1(x) : \cdots : s_n(x)] \end{aligned}$$

Definition 1.29. Let X be a scheme and G be an affine algebraic group acting on X via a morphism $\sigma : G \times X \longrightarrow X$. Then a linearization of the G-action on X is a line bundle $\pi_X : L \longrightarrow X$ over X with an isomorphism of line bundles:

$$\pi_X^* L = G \times L \cong \sigma^* L$$

where $\pi_X : G\times X \longrightarrow X$ is the projection such that the induced bundle homomorphism $\widetilde{\sigma} : G\times L \longrightarrow L$ is given by:



induces an action of G on L.

Proposition 1.30. *We have:*

1). *The projective $\pi : L \longrightarrow X$ is G-equivariant;*

2). *The action of G on the fibers of L is linear: For $g \in G$ and $x \in X$, the map $L_x \longrightarrow: L_{gx}$ is linear.*

Example 1.31. 1). For G acts on Spec k trivially. The only line bundle is $\mathbb{A}^1 \longrightarrow$ Spec k, but there are many linearizations. In fact, any $G \times \mathbb{A}^1 \longrightarrow \mathbb{A}^1$ corresponds to a map $G \longrightarrow \mathrm{GL}_1 = \mathbb{G}_m$, a character of G.

2). For any scheme X, with an action of an affine algebraic group G and any character χ. We can construct a linearization on the trivial line bundle $X \times \mathbb{A}^1 \longrightarrow X$ by

$$g \cdot (x, z) \longmapsto (g \cdot x, \chi(g)^{-1} z)$$

More generally, for any linearization $\widetilde{\sigma} : L \longrightarrow X$, we can twist the linearization by a character $\chi : G \longrightarrow \mathbb{G}_m$ to get a linearization $\widetilde{\sigma}^{\chi}$.

Lemma 1.32. *Let G act on X via $\sigma : G\times X \longrightarrow X$ and let $\widetilde{\sigma} : G\times L \longrightarrow L$ be a linearization of the action on a line bundle L on X. Then there is a natural linear representation $G \longrightarrow \mathrm{GL}(H^0(X, L))$.*

Proof. We construct the comodule $H^0(X, L) \longrightarrow \mathcal{O}(G) \otimes_k H^0(X, L)$ via

$$H^0(X,L) \xrightarrow{\sigma^*} H^0(G\times L, \sigma^* L) \cong H^0(G\times X, G\times L) \longrightarrow H^0(G, \mathcal{O}_G) \otimes H^0(X, L)$$

where the last $\cong$ is Künneth formula. □

Example 1.33. Consider the action of $\mathbb{G}_m$ over $\mathbb{P}^1$. For $L = \mathcal{O}(1)$, the action of $\mathbb{G}_m$ over $H^0(\mathbb{P}^1, L) = \text{span}\{x, y\}$ is

$$t \longmapsto \begin{pmatrix} t^{-1} & 0 \\ 0 & 1 \end{pmatrix}$$

Let G be a reductive group acting on a projective scheme X and L be an ample linearization of the G-action on X. Then consider

$$R := R(X, L) = \bigoplus_{r \geq 0} H^0(X, L^{\otimes r})$$

Since each line bundle $L^{\otimes r}$ has an induced linearization, there is an action of G on $H^0(X, L^{\otimes r})$ by 1.32. Now, consider

$$R^G = \bigoplus_{r \geq 0} H^0(X, L^{\otimes r})^G$$

which is a finitely generated k-algebra by Nagata and $\text{Proj}(R^G)$ is projective over k.

Definition 1.34. We say that $x \in X$ is

1). semi-stable w.r.t. L if there exists $\sigma \in H^0(X, L^{\otimes r})^G$ for some $r > 0$ such that $\sigma(x) \neq 0$;

2). stable with respect to L if $\dim G \cdot x = \dim G$ and there is an invariant section $\sigma \in H^0(X, L^{\otimes r})^G$ for some $r > 0$ such that $\sigma(x) \neq 0$ and the action of G on X_σ is closed.

We let $X^{ss}(L)$ and $X^s(L)$ denote the open subset of semi-stable and stable points in X, respectively. Then we define the projective GIT quotient with respect to L to the morphism:

$$X^{ss} \longrightarrow X//_L G := \text{Proj}(R(X, L)^G)$$

associated to the inclusion $R(X, L)^G \hookrightarrow R(X, L)$.

Example 1.35. Let's say more about the linearization over the trivial bundle via a character $\chi : G \longrightarrow \mathbb{G}_m$. We have $\mathcal{O}(X \times \mathbb{A}^1) = \mathcal{O}(X) \otimes_k k[z] = \mathcal{O}(X)[z]$. Then

$$g \cdot f(x, z) = g \cdot (\sum_{n=0}^{l} f_n(x) z^n) = \sum_{n=0}^{l} \chi(g)^{-n} f_n(g^{-1} \cdot x) z^n$$

Therefore, if $f(x, z)$ is G-invariant, then $f_n(g^{-1} \cdot x) = \chi^n(g) f_n(x)$ for all $0 \leq n \leq l$. We call $f_n(x) \in \mathcal{O}(X)$ such that $f_n(g^{-1} \cdot x) = \chi(g)^n f_n(x)$, **$\chi^{\mathbf{n}}$-semi-invariant** and denote the set of $\chi^{\mathbf{n}}$-semi-invariant by $\mathcal{O}(X)^{\chi^n}$. Now, there is a graded structure on

$$\mathcal{O}(X \times \mathbb{A}^1)^G = \bigoplus_{n \geq 0} \mathcal{O}(X)^{\chi^n}$$

At this case, a point x is said to be χ-semi-stable if for some $n > 0$, there exists semi-invariant $f \in \mathcal{O}(X)^{\chi^n}$ such that $f(x) \neq 0$ and a point x is said to be χ-stable if there exists a semi-invariant $f \in \mathcal{O}(X)^{\chi^n}$ such that $f(x) \neq 0$, and the action of G on X_f is closed and $\dim G \cdot x = \dim G$. Moreover, we denote the projective GIT by $X^{ss}(\chi) \longrightarrow X//_\chi G$.

Theorem 1.36. *Let G be a reductive group acting on a projective scheme X and L be an ample linearization of this action. Then the GIT quotient*

$$\varphi : X^{ss}(L) \longrightarrow X//_L G = \text{Proj}(R(X, L))$$

is a good quotient and $X//_L G$ is a projective scheme with a natural ample line bundle L' such that $\varphi^* L' = L$.

Furthermore, there exists an open subset $Y^s \subset X//_L G$ such that $\varphi^{-1}(Y^s) = X^s(L)$ and $\varphi : X^s(L) \longrightarrow Y^s$ is a geometric quotient for the G-action on $X^s(L)$.

Proof. [Hos15, Theorem 5.23] □

To summarize we have

$$\begin{array}{ccccc} X^s(L) & \hookrightarrow & X^{ss}(L) & \hookrightarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ X^s(L)/G & \hookrightarrow & X//_L G & \longrightarrow & X//G \end{array}$$

Definition 1.37. We say two semi-stable k-points x and x' in X are S-equivalent if the orbit closure of x and x' meet in $X^{ss}(L)$. We say a semi-stable k-point is polystable if the orbit is closed in $X^{ss}(L)$.

Corollary 1.38. *Let x and x' be k-points in $X^{ss}(L)$. Then $\varphi(x) = \varphi(x')$ if and only if x and x' are S-equivalent.*

Example 1.39. Consider 2) in 1.24. The closure of orbits of $[1:0:1]$ and $[1:1:0]$ meets outside X^{ss}. So they are not S-equivalent. Moreover, consider the projective GIT quotient in 1.24:

$$\begin{aligned} \mathbb{P}^2 &\dashrightarrow \mathbb{P}^1 \\ [x_0 : x_1 : x_2] &\longmapsto [x_0 x_1 : x_2^2] \end{aligned}$$

$[1:0:1]$ is mapped to $[0:1]$ and $[1:1:0]$ is mapped to $[1:0]$.

Projective GIT for Quasi-projective Schemes

It is also important to consider projective GIT over affine schemes. For example, one can apply projective GIT to representation spaces of quivers, leading, for instance, to the construction of Nakajima quiver varieties.

Let X be a quasi-projective scheme with the action of a reductive group G with a linearization with respect to $\mathcal{L}$ (**Not necessarily ample!**).

Definition 1.40. Let X be a quasi-projective scheme with the action of a reductive group G with a linearization with respect to $\mathcal{L}$. Then

1). A point $x \in X$ is semi-stable with respect to $\mathcal{L}$ if there is an invariant section $\sigma \in H^0(X, L^{\otimes r})^G$ for some $r > 0$ such that $\sigma(x) \neq 0$ and X_σ is affine;

2). A point $x \in X$ is stable $\mathcal{L}$ if $\dim G_x = 0$ and there is an invariant section $\sigma \in H^0(X, L^{\otimes r})^G$ for some $r > 0$ such that $\sigma(x) \neq 0$ and X_σ is affine and the action of G on X_σ is closed.

Example 1.41. Consider $G = \mathrm{GL}(V_1) \times \mathrm{GL}(V_2)$ acts on $X = \mathrm{Hom}(V_1, V_2)$ by

$$(g_1, g_2) \cdot f = g_2 \circ f \circ g_1^{-1}$$

We linearize X by $\chi : G \longrightarrow \mathbb{G}_m, (g_1, g_2) \longmapsto \det(g_1)^{-1} \det(g_2)$.

Then by 1.35, M is semi-stable if and only if there exists $f \in \mathcal{O}(X)^{\chi^n}, n \geq 1$ satisfies $f(M) \neq 0$. Note X_f is always affine.

Similarly, we can define $R(X, \mathcal{L})^G$ to be

$$\bigoplus_{r \geq 0} H^0(X, \mathcal{L}^{\otimes r})^G$$

and define $X//_{\mathcal{L}} G$ to be $\mathrm{Proj}(R(X, \mathcal{L}))$. Moreover, we have

Theorem 1.42 (Mumford)**.** *Let G be a reductive group acting on a quasi-projective scheme X and L be a linearization of this action. Then there exists a good quotient $\varphi : X^{ss}(\mathcal{L}) \longrightarrow X//_{\mathcal{L}}G$ of the G-action of $X^{ss}(\mathcal{L})$. Furthermore, there is an open subset $Y^s \subset X//_{\mathcal{L}}G$ such that $\varphi^{-1}(X^s) = X^s(\mathcal{L})$ and $\varphi : X^s(\mathcal{L}) \longrightarrow Y^s$ is a geometric quotient.*

Sketch of proof. [Hos15, Theorem 5.31]. □

1.3 Hilbert-Mumford Criterion

We have seen that, in general like 1.28, it is not easy to tell which points are semi-stable or stable. In this section, we will give some useful criteria for semi-stability and stability.

Projective Varieties/Schemes

Definition 1.43. A 1-parameter subgroup of G is a non-trivial group homomorphism $\lambda : \mathbb{G}_m \longrightarrow G$.

Fix $x \in X(k)$ and 1-PS $\lambda : \mathbb{G}_m \longrightarrow G$. We let $\lambda_x : \mathbb{G}_m \longrightarrow X$ be the morphism given by

$$\lambda_x(t) = \lambda(t) \cdot x$$

We have a natural embedding of $\mathbb{G}_m = \mathbb{A}^1 \setminus \{0\} \hookrightarrow \mathbb{P}^1$ given by $t \longmapsto [1 : t]$. Since X is projective, it is proper over Spec k and so by the valuation criteria for properness, the morphism $\lambda_x : \mathbb{G}_m \longrightarrow X$ extends to a morphism $\hat{\lambda}_x : \mathbb{P}^1 \longrightarrow X$.

$$\begin{array}{ccc} \mathbb{G}_m & \xrightarrow{\lambda_x} & X \\ \downarrow & \overset{\exists!\hat{\lambda}_x}{\nearrow} & \downarrow \\ \mathbb{P}^1 & \longrightarrow & \text{Spec } k \end{array}$$

Here, we apply valuation criterion to $\mathcal{O}_{\mathbb{P}^1,0}$, $\mathcal{O}_{\mathbb{P}^1,\infty}$ with fraction field $k(t)$. Since $\mathbb{G}_m = \text{Spec}(k[t, t^{-1}])$, we have $\text{Spec}(k[t]) \longrightarrow \mathbb{G}_m$, which givens $\lambda|_{\text{Spec}(k(t))} : \text{Spec}(k(t)) \longrightarrow X$.

Example 1.44. 1). Consider $X = V(x_0x_1 - x_1^2) \subset \mathbb{P}^2$. We have

$$\begin{aligned} \mathbb{G}_m \times X &\longrightarrow X \\ (t, [x : y : z]) &\longmapsto [t^2x : ty : z] \end{aligned}$$

At $[x : y : z] = [1 : 1 : 1]$, we get $\lambda : t \longmapsto [t^2 : t : 1]$. Note that when $t \to 0$, $\lambda_x(t) \to [0 : 0 : 1] \in X$ and when $t \to \infty$, $\lambda_x(t) \to [1 : 0 : 0] \in X$.

2). For X non-projective, for example $\mathbb{A}^1$, we can not lift it to $\mathbb{P}^1$. For example, consider $\mathbb{G}_m \hookrightarrow \mathbb{A}^1$.

In general, we set $\lim\limits_{t\to 0}\lambda(t) \cdot x = \hat{\lambda}_x([1 : 0])$ and $\lim\limits_{t\to 0}\lambda(t) \cdot x = \hat{\lambda}_x([1 : 0])$. Fix any $x \in X$. Let's consider $\lim\limits_{t\to 0}\lambda_x(t) = y$. Then y is fixed by $\mathbb{G}_m$: $\forall s \in \mathbb{G}_m$,

$$\lambda(s) \cdot y = \lim_{t\to 0}\lambda_x(st) = y$$

Therefore, over the fiber over y of the line bundle $\mathcal{O}(1) := \mathcal{O}_{\mathbb{P}^n}(1)|_X \cong \mathbb{A}^1$, the group $\lambda(\mathbb{G}_m)$ acts by a character $t \mapsto t^r$.

Definition 1.45. We define the Hilbert-Mumford weight of the action of 1-PS λ in $x \in X(k)$ to be

$$\mu^{\mathcal{O}(1)}(x, \lambda) = r$$

where r is the weight of $\lambda(\mathbb{G}_m)$ on the fiber $\mathcal{O}(1)_y$ over $y := \lim\limits_{t\to 0}\lambda(t) \cdot x$.

Example 1.46. 1). Consider the example in 1.44. We have

$$\lim_{t\to 0}[t^2 : t : 1] = [0 : 0 : 1]$$

For $\mathbb{G}_m$, it acts on $\mathcal{O}(1)_y$ trivially, with $y = [0 : 0 : 1]$.

2). Consider the action of $\mathbb{G}_m$ over $\mathbb{P}^n$ via

$$t \cdot [x_0 : x_1 : \cdots : x_n] = [t^{a_0}x_0 : t^{a_1}x_1 : \cdots : t^{a_n}x_n]$$

then at $\lim_{t\to 0}\lambda_{[0:\cdots:1:\cdots:0]}(t) = [0 : \cdots : 1 : \cdots : 0]$. For $\mathbb{G}_m$ acts on $f : \mathrm{Span}_k\{(0 : \cdots : 1 : \cdots : 0)\} \longrightarrow k$ with $t \cdot f(v) = f(t^{-1} \cdot v)$, its weight is $-a_i$.

Now, we are going to rephrase this in terms of the weights for the action on affine cone. Pick a non-zero lift $\widetilde{x} \in \widetilde{X}$ of $x \in X$. Then the map,

$$\lambda_{\widetilde{x}} := \lambda(-) \cdot \widetilde{x} : \mathbb{G}_m \longrightarrow \widetilde{X}$$

may no longer extend to $\mathbb{P}^1$, as $\widetilde{X}$ is not proper. If it extends to zero (or infinity), we will denote the limit by $\lim_{t\to 0}\lambda(t) \cdot \widetilde{x}$ (resp. $\lim_{t\to\infty}\lambda(t) \cdot \widetilde{x}$). Any point in the boundary $\overline{\lambda_{\widetilde{x}}(\mathbb{G}_m)} \setminus \lambda_{\widetilde{x}}(\mathbb{G}_m)$ must be equal to either of these limit points.

By the representation of $\mathbb{G}_m$, we can pick $e_0, e_1, \cdots, e_n$ of k^{n+1} such that

$$\lambda(t) \cdot e_i = t^{r_i}e_i$$

for some $r_i \in \mathbb{Z}$. We call the integers r_i, the λ-weights of the action on $\mathbb{A}^{n+1}$. For $x \in X$, $\widetilde{x} \in \widetilde{X}$, write $\widetilde{x} = \sum_{i=1}^n x_i e_i$ w.r.t. this basis, then

$$\lambda(t) \cdot \widetilde{x} = \sum_{i=0}^{n} t^{\lambda_i}x_i e_i$$

Definition 1.47. The Hilbert-Mumford weight of x at λ is defined to be

$$\mu(x, \lambda) := -\min\{r_i \big| x_i \neq 0\}$$

Lemma 1.48. *The two definitions for Hilbert-Mumford weight agree:*

$$\mu^{\mathcal{O}(1)}(x, \lambda) = \mu(x, \lambda)$$

Proof. Pick a non-zero lift of $\widetilde{x}$ in the affine cone lying over x. Assume that we have taken coordinates over $\mathbb{A}^{n+1}$ such that

$$\lambda(t) \cdot \widetilde{x} = (t^{r_0}x_0, \cdots, t^{r_n}x)$$

Since $\mu^{\mathcal{O}(1)}(x, \lambda) + r_i \geq 0$, for all i such that $x_i \neq 0$, with equality for at least one i with $x_i \neq 0$,

$$\widetilde{y} = \lim_{t\to 0} t^{\mu^{\mathcal{O}(1)}}(\lambda(t) \cdot \widetilde{x}) = (y_0, \cdots, y_n)$$

exists and it is non-zero.

Furthermore, $\widetilde{y}$ lies over $y := \lim_{t\to 0}\lambda(t) \cdot x$ and the weight of the λ-action on $\widetilde{y}$ is $-\mu^{\mathcal{O}(1)}$. Since $\mathcal{O}(1)_y = \mathrm{Hom}(k\widetilde{y}, k)$, the action of $\lambda(t)$ over $\mathcal{O}(1)_y = \mathrm{Hom}(k\widetilde{y}, k)$ is just $\mu^{\mathcal{O}(1)}$, which is, by definition $\mu(x, \lambda)$. □

Remark. $\mu^{\mathcal{O}(1)}(x, \mu)$ depends on the choice of linearization of X, whereas $\mu(x, \lambda)$ is defined canonically. By 1.48, $\mu^{\mathcal{O}(1)}$ is also canonically. In most situations, however, $\mu^{\mathcal{O}(1)}(x, \lambda)$ is much easier to compute.

Lemma 1.49. *Let λ be a 1-PS of G and let $x \in X$. We diagonalize the $\lambda(\mathbb{G}_m)$-action on the affine cone as above and let $\widetilde{x} = \sum_{i=0}^{n} x_i e_i$ be a non-zero lift of x. Then*

1). *$\mu(x,\lambda) < 0$ if and only if $\widetilde{x} = \sum_{r_i>0} x_i e_i$ if and only if $\lim_{t\to 0}\lambda(t)\cdot\widetilde{x} = 0$;*

2). *$\mu(x,\lambda) = 0$ if and only if $\widetilde{x} = \sum_{r_i\geq 0} x_i e_i$ and there exists $r_i = 0$ such that $x_i \neq 0$ if and only if $\lim_{t\to 0}\lambda(t)\cdot\widetilde{x}$ exists and is non-zero;*

3). *$\mu(x,\lambda) > 0$ if and only if $\widetilde{x} = \sum_{r_i} x_i e_i$ and there exists $r_i < 0$ such that $x_i \neq 0$ if and only if $\lim_{t\to 0}\lambda(t)\cdot\widetilde{x}$ does not exist.*

Proof. We only prove 1). $\mu(x,\lambda) < 0 \iff \lambda_{\widetilde{x}}(t) = \sum_{r_i>0} t^{r_i} x_i e_i \iff \lim_{t\to 0} \lambda(t)\cdot\widetilde{x} = 0.$ □

Lemma 1.50. *Let G be a reductive group acting linearly on a projective scheme $X \subset \mathbb{P}^n$. For $x \in X$,*

1). *x is semi-stable for the action $\lambda(\mathbb{G}_m)$ if and only if $\mu(x,\lambda) \geq 0$, $\mu(x,\lambda^{-1}) \geq 0$;*

2). *x is stable for the action $\lambda(\mathbb{G}_m)$ if and only if $\mu(x,\lambda) > 0$ and $\mu(x,\lambda^{-1}) > 0$.*

Proof. Before proving this lemma, we need:

Lemma 1.51 (Topological criterion)**.** *Let $x \in X$ and choose a non-zero lift $\widetilde{x} \in \widetilde{X}$ of x. Then*

1). *x is semi-stable if and only if $0 \notin \overline{G\cdot\widetilde{x}}$;*

2). *x is stable if and only if $\dim G_{\widetilde{x}} = 0$ and $G\cdot\widetilde{x}$ is closed in $\widetilde{X}$.*

Proof. [Hos15, Proposition 6.1] □

1). By topological criterion 1.51, x is semi-stable for $\lambda(\mathbb{G}_m)$ if and only if $0 \notin \overline{\lambda(\mathbb{G}_m)\cdot\widetilde{x}}$ where $\widetilde{x} \in \widetilde{X}$ is a point lying over x. Since any point in the boundary $\overline{\lambda(\mathbb{G}_m)\cdot\widetilde{x}} \setminus \lambda(\mathbb{G}_m)\cdot\widetilde{x}$ is either

$$\lim_{t\to 0} \lambda(t)\cdot\widetilde{x} \text{ or } \lim_{t\to\infty} \lambda(t)\cdot\widetilde{x} = \lim_{t\to 0} \lambda^{-1}(t)\cdot\widetilde{x}$$

By 1.49, the condition that these two limits are not zero is equivariant to $\mu(x,\lambda) \geq 0$ and $\mu(x,\lambda^{-1}) \geq 0$.

2). By topological criteria, x is stable if and only if $\dim \lambda(\mathbb{G}_m)\cdot\widetilde{x} = 0$ and $\lambda(\mathbb{G}_m)\cdot\widetilde{x}$ is closed, where the second condition holds if and only if $\overline{\lambda(\mathbb{G}_m)\cdot\widetilde{x}} \setminus \lambda(\mathbb{G}_m)\cdot\widetilde{x}$ is empty, that is, $\mu(x,\lambda) > 0$ and $\mu(x,\lambda^{-1}) > 0$.

Moreover, if $\mu(x,\lambda) > 0$ and $\lambda(t)\cdot\widetilde{x} = \widetilde{x} = \sum t^{p_i} x_i e_i = \sum x_i e_i$, these t forms a dimension zero subgroup of $\mathbb{G}_m$. Thus, $\dim \lambda(\mathbb{G}_m)_{\widetilde{x}} < 1$, which implies $\dim \lambda(\mathbb{G}_m)_{\widetilde{x}} = 0$. □

Theorem 1.52 (Hilbert-Mumford)**.** *Let G be a reductive group acting linearly on a projective scheme $X \subset \mathbb{P}^n$. For $x \in X$,*

1). *$x \in X^{ss}$ if and only if $\mu(x,\lambda) \geq 0$ for all 1-PS λ of G;*

2). *$x \in X^{s}$ if and only if $\mu(x,\lambda) > 0$ for all 1-PS λ of G.*

Proof. ($\Longrightarrow$) If x is semi-stable for G, then it is semi-stable for all subgroups of G as every G-invariant function is also H invariant.

Moreover, since $\dim H_x \leq \dim G_x$ for any subgroup H of G, x is stable for G then it is stable for all subgroups of G.

To prove this, we need:

Theorem 1.53 (Fundamental Theorem in GIT)**.** *Let G be a reductive group acting on $\mathbb{A}^{n+1}$. If $x \in \mathbb{A}^{n+1}$ is a closed point and $y \in \overline{G \cdot x}$, there exists an 1-PS λ of G such that $\lim_{t\to 0} \lambda(t) \cdot x = y$.*

Proof. [Hos15, Theorem 6.13] □

If $x \notin X^{ss}$, then $0 \in \overline{G \cdot \widetilde{x}}$ because of topological criteria 1.51. So, there exists 1-PS λ of G such that $\lim_{t\to 0} \lambda(t) \cdot \widetilde{x} = 0$, that is, $\mu(x, \lambda) < 0$ 1.49.

If $G \cdot \widetilde{x}$ is not closed, 1.53 implies that there exists $y \in \overline{G \cdot \widetilde{x}}$ such that $\lim_{t\to 0} \lambda_{\widetilde{x}}(t) = y$. Thus, $\mu(x, \lambda) = 0$.

Also if $\dim G \cdot \widetilde{x} = 0$, then $\lambda_{\widetilde{x}} : t \longmapsto \lambda(t) \cdot \widetilde{x}$ is constant. Thus, $\mu(x, \lambda) = 0$ by 1.49. □

Example 1.54. For $\mathbb{G}_m$ acts on $\mathbb{P}^2$ by

$$t \cdot [x_0 : x_1 : x_2] = [tx_0 : t^{-1}x_1 : x_2]$$

Every 1-PS of $\mathbb{G}_m$ can by written as $\lambda_n : t \longmapsto t^n$. Taking $n \in \mathbb{N}$, consider the following cases:

1). $x_0 = 0, x_1 \neq 0, x_2 \neq 0$, then $\mu(x, \lambda_n) = n$ and $\lambda(x, \lambda_n^{-n}) = 0$;

2). $x_0 = 0, x_1 \neq 0, x_2 = 0$, $\mu(x, \lambda_n) = -n$, $\mu(x, \lambda_n^{-1}) = n$;

3). $x_0 = 0, x_1 = 0, x_2 \neq 0$, $\mu(x, \lambda_n) = 0$;

4). $x_0 \neq 0, x_1 \neq 0, x_2 \neq 0$, then $\mu(x, \lambda_n) = n$ and $\mu(x, \lambda_n) = n$;

5). $x_0 \neq 0, x_1 \neq 0, x_2 = 0$, $\mu(x, \lambda_n) = n$ and $\mu(x, \lambda_n) = n$;

6). $x_0 \neq 0, x_1 = 0, x_2 \neq 0$, $\mu(x, \lambda_n) = n$ and $\mu(x, \lambda_n^{-1}) = 0$;

7). $x_0 \neq 0, x_1 = 0, x_2 = 0$, $\mu(x, \lambda_n) = n$ and $\mu(x, \lambda_n^{-1}) = -n$.

Hence, the unstable points are $[1 : 0 : 0]$ and $[0 : 1 : 0]$ and the stable points are $[x_0 : x_1 : x_2]$ with $x_0 x_1 \neq 0$.

For X a projective scheme with the action of a reductive group G, choose a linearization of X with respect to an ample line bundle $\mathcal{L}$.

Definition 1.55. The Hilbert-Mumford weight of 1-PS λ and $x \in X$ with respect to L is

$$\mu^L(x, \lambda) := -r$$

where r is the weight of the $\lambda(\mathbb{G}_m)$-action on the fiber $\mathcal{L}_y$ over the fixed point $y = \lim_{t\to 0} \lambda(t) \cdot x$.

For an ample line bundle $\mathcal{L}$, it determines a closed embedding

$$i : X \hookrightarrow \mathbb{P}^n$$

and $i^*\mathcal{O}(1) = \mathcal{L}$. For any $x \in X$, we have $\mu^{\mathcal{L}}(x, \lambda) = \mu^{\mathcal{O}(1)}(i(x), \lambda)$. So the following result holds:

Theorem 1.56 (Hilbert-Mumford)**.** *Let G be a reductive group acting on a projective scheme X and L be an ample linearization of this action. Then, for $x \in X$, we have*

$$x \in X^{ss}(\mathcal{L}) \iff \mu^{\mathcal{L}}(x, \lambda) \geq 0 \text{ for all 1-Ps } \lambda \text{ of } G$$

$$x \in X^{s}(\mathcal{L}) \iff \mu^{\mathcal{L}}(x, \lambda) > 0 \text{ for all 1-Ps } \lambda \text{ of } G$$

Affine Varieties/Schemes

Let X be an affine variety with the action of a reductive group G. We can linearize X over it trivial line bundle with respect to some character $\chi : G \longrightarrow \mathbb{G}_m$.

Definition 1.57. Take a 1-PS λ. For $x \in X$ and $\lim_{t\to 0} \lambda(t)\cdot x$ exists, the Hilbert-Mumford weight is defined to be

$$\mu^{\chi}(x,\lambda) = -\langle \chi, \lambda\rangle$$

Remark. Set $y := \lim_{t\to 0}\lambda(t)\cdot x$. y is fixed by the 1-PS λ. Now, consider the action of G on $X \times k$,

$$g\cdot(x,v) = (g\cdot x, \chi(g)^{-1}v)$$

If $g = \lambda(t)$, we get

$$\lambda(t)\cdot(y,v) = (\lambda(t)\cdot y, \chi(\lambda(t))^{-1}v) = (y, t^{-\langle\chi,\lambda\rangle}v)$$

Thus, we can see that λ acts on the fiber of y with weight $-\langle\chi,\lambda\rangle$, which coincides the definition of Hilbert-Mumford weight with respect to some very ample line bundle.

Before giving the Hilbert-Mumford criterion over the affine variety X with an action of reductive group, we need the following lemma:

Lemma 1.58. *$x \notin X^{ss}(\chi)$ if and only if $\overline{G\cdot(x,1)} \cap X\times\{0\} \neq \emptyset$ in $Y = X\times k = \mathrm{Spec}(\mathcal{O}(X)[x])$.*

Proof. For $x \in X^{ss}(\chi)$, there exists $f \in \mathcal{O}(X)^{\chi^n}$ such that $f(x) \neq 0$. Define $F(x,z) = f(x)z^n \in \mathcal{O}(Y) = \mathcal{O}(X)[z]$. Then $F(x,z)$ is G-invariant. Note that $F(x,1) = f(x) \neq 0$ and $F(x,0) = 0$, that is, x is semi-stable then there exists a G-equivariant function F separates $(x,1)$ and $X\times\{0\}$.

If $\overline{G\cdot(x,1)} \cap X\times\{0\} \neq \emptyset$, for any $F \in \mathcal{O}(Y)^G$, $F(x,1) = F(x,0)$ since F is constant along the closed of G-orbits. Thus, x can not be semi-stable at this case.

Conversely, if $x \notin X^{ss}(\chi)$, then for any $f \in \mathcal{O}(X)^{\chi^n}$, $f(x) = 0$. Again, take $F = fz^n$ and F is G-equivariant. Since $F(x,1) = 0$ and $F(x,0) = 0$, $(x,0) \in \overline{G\cdot(x,1)}$, which implies $\overline{G\cdot(x,1)}\cap X\times\{0\} \neq \emptyset$. □

Theorem 1.59 (Hilbert-Mumford affine)**.** *Let X be an affine variety with the action of a reductive group G and a linearization of the trivial bundle via χ. Then*

$$x \in X^{ss}(\chi) \iff \mu(x,\lambda) \leq 0 \text{ for all 1-PS } \lambda \text{ such that } \lim_{t\to 0}\lambda(t)\cdot x \text{ exists}$$

$$x \in X^{s}(\chi) \iff \mu(x,\lambda) < 0 \text{ for all 1-PS } \lambda \text{ such that } \lim_{t\to 0}\lambda(t)\cdot x \text{ exists}$$

Proof. We first prove the result for semi-stability:

For x is semi-stable, 1.35 implies x is semi-stable if and only if there exists $f \in \mathcal{O}(X)^{\chi^n}$ such that $f(x) \neq 0$. Fix such f. Let λ be a 1-PS such that $y := \lim_{t\to 0}\lambda(t)\cdot x$ exists. Then

$$f(\lambda(t)\cdot x) = \chi(\lambda(t))^n f(x) = t^{-n\mu(x,\lambda)}f(x)$$

Because $f(x) \neq 0$, $\lim_{t\to 0} f(\lambda(t)\cdot x) = f(y)$ exists if and only if $\mu(x,\lambda) \leq 0$.

Conversely, suppose that x is not semi-stable, by 1.58 and the Fundamental Theorem in GIT 1.53, there exists 1-PS λ such that the closure of $\lambda(\mathbb{G}_m)$ contains points in $X\times\{0\}$. Suppose λ is such 1-PS. Then

$$\lambda(t)\cdot(x,1) = (\lambda(x)\cdot x, \chi(\lambda(t)^{-1})) = (\lambda(x), t^{\mu(x,\lambda)})$$

we see that $\mu(x,\lambda) > 0$.

Now, suppose that x is stable. Then it is also semi-stale. If there exists some 1-PS λ such that $\mu(x,\lambda)=0$ and $y:=\lim_{t\to 0}\lambda(t)\cdot x$ exists, for $f\in\mathcal{O}(X)^{\chi^n}$ such that $f(x)\neq 0$, $f(y)\neq 0$ for $f(y)=\lim_{t\to 0}t^{-n\mu(x,\lambda)}f(x)$.

Since $G\cdot x$ is closed over X_f, there exists $g\in G$ such that $g\cdot x=y$ and so y is also stable. However, $\lambda(t)\cdot y=y$, which implies $\dim G_y\geq 1$ which leads to a contradiction. Thus, $\mu(x,\lambda)\neq 0$.

Conversely, suppose that $\mu(x,\lambda)<0$ for all 1-PS λ such that $\lim_{t\to 0}\lambda(t)\cdot x$ exists. Then x is obviously semi-stable by the argument for semi-stability. Assume that x is not stable.

$G\cdot x$ is not closed in some X_f with $f\in\mathcal{O}(X)^{\chi^n}$. We can take $y\in\overline{G\cdot x}\cap X_f$. By Fundamental Theorem on GIT, there exists a 1-PS such that $\lim_{t\to 0}\lambda(t)\cdot x=y$ and $f(y)=\lim_{t\to 0}t^{-n\mu(x,\lambda)}f(x)\neq 0$. However since $\mu(x,\lambda)<0$, $f(y)=0$ which leads to a contradiction.

Assume that $G\cdot x$ is closed over X_f but $\dim G_x\geq 1$. Since G_x is reductive, G_x contains a non-trivial 1-PS λ Some reference. Note that at this case both $\lim_{t\to 0}\lambda(t)\cdot x$ and $\lim_{t\to 0}\lambda^{-1}(t)\cdot x$ exists. However, $\mu(x,\lambda)=-\mu(x,\lambda^{-1})$, impossible.

□

Example 1.60. Consider the action of $G=GL(V_1)\times\mathrm{GL}(V_2)$ over $\mathrm{Hom}(V_1,V_2)$ with $\dim V_i=n_i$ by

$$(g_1,g_2)\cdot f=g_2\circ f\circ g_1^{-1}$$

with linearization character $\chi:G\to\mathbb{G}_m, g\mapsto\det(g_1)^{-1}\det(g_2)$.

Fix $f\in\mathrm{Hom}(V_1,V_2)$. For any 1-PS $\lambda:t\mapsto(\lambda_1(t),\lambda_2(t))$, we can diagonalize V_1,V_2 into

$$V_1=\bigoplus_{m\in\mathbb{Z}}V_1(m),\ V_2=\bigoplus_{m\in\mathbb{Z}}V_2(m)$$

Write $f=\sum f_{n,m}$ with $f_{n,m}:V_1(m)\longrightarrow V_2(n)$. Then

$$\lambda(t)\cdot f=\sum_{m,n}t^{n-m}f_{n,m}$$

Thus, $\lim_{t\to 0}\lambda(t)\cdot f$ exists if and only if $n\geq m$ for $f_{n,m}\neq 0$ if and only if $f_{n,m}=0$ for all $n<m$. Let

$$V_1^{\geq m}=\bigoplus_{i\geq m}V_1(m), V_2^{\geq n}=\bigoplus_{j\geq n}V_2(n)$$

$f_{n,m}=0$ for all $n<m$ if and only if $f(V_1^{\geq m})\subset V_2^{\geq m}$. At this case

$$\mu(\lambda,f)=-\left\{-\sum m\dim V_1(m)+\dim nV_2(n)\right\}=\sum_m\dim V_1^{\geq m}-\dim V_2^{\geq m}$$

Thus, f is semi-stable if and only if $\sum_m\dim V_1^{\geq m}-\dim V_2^{\geq m}\leq 0$ for all 1-PS such that $f(V_1^{\geq m})\subset V_2^{\geq m}$. Following this result, if for all (W_1,W_2) such that $f(W_1)\subset W_2$, $\dim W_1\leq\dim W_2$, then f is semi-stable.

Conversely, if f is semi-stable, for any (W_1,W_2), take 1-PS

$$\lambda(t):t\longmapsto\begin{pmatrix}t\mathrm{id}_{W_1}&0\\0&\mathrm{id}\end{pmatrix}\times\begin{pmatrix}t\mathrm{id}_{W_2}&0\\0&\mathrm{id}\end{pmatrix}$$

Then $\lambda(t)\cdot f\underset{t\to 0}{\text{exists}}$ for $f(W_1)\subset W_2$. Now, $\mu(f,\lambda)=\dim W_1-\dim W_2$. By semi-stability condition, $\dim W_1\leq\dim W_2$.

To summerize, we have

$$f\text{ is semi-stable}\iff\dim W_1\leq\dim W_2\text{ for all }f(W_1)\subset W_2$$

2 Hesselink–Kempf–Kirwan–Ness stratification

Let G be a connected reductive group acting on a scheme X, and let $\mathcal{L}$ be an G-linearized line bundle on X. Then X admits a finite decomposition

$$X = X^{ss}(\mathcal{L}) \sqcup \bigsqcup_{\beta \in \mathcal{B}} S_\beta,$$

where each S_β is a locally closed G-invariant subscheme of X. The open stratum is the semistable locus $X^{ss}(\mathcal{L})$, while the other strata consist of unstable points grouped according to their optimal destabilizing one-parameter subgroups.

Equivalently, after choosing an ordering of the indices β by increasing instability, one obtains a finite G-invariant filtration by closed subschemes

$$X^{ss}(\mathcal{L}) \subset X_{\leq \beta_1} \subset X_{\leq \beta_2} \subset \cdots \subset X,$$

such that

$$X_{\leq \beta_i} \setminus X_{<\beta_i} = S_{\beta_i}.$$

As β increases, the stratum S_β consists of points which are "more unstable" in the sense that their optimal Hilbert–Mumford numerical invariant is larger.

2.1 HKKN Decomposition

Let X be a projective variety with the action of a redcutive group G. Fix $\lambda : \mathbb{G}_m \longrightarrow G$ and consider the action of G on G via conjugation.

Definition 2.1. We say

1). $P(\lambda) := \{g \in G \big| \lim_{t\to 0} \lambda(t) g \lambda^{-1}(t)\}$ is parabolic;

2). $U(\lambda) := \{g \in G \big| \lim_{t\to 0} \lambda(t) g \lambda^{-1}(t) = 1\}$ is unitary;

3). $L(\lambda) := \{g \in G \big| \lim_{t\to 0} \lambda(t) g \lambda^{-1}(t) = g\}$ is Levi.

Example 2.2. For $G = \mathrm{GL}_n$, take $\lambda(t) = \mathrm{diag}(t^{a_1}, \cdots, t^{a_n})$. Let $g = (g_{ij}) \in \mathrm{GL}_n$ with $a_1 > a_2 > \cdots > a_n$. Then

$$\lambda(t) g \lambda^{-1}(t) = (t^{a_i - a_j} g_{ij})$$

Then $\lim_{t\to 0} \lambda(t) g \lambda^{-1}(t)$ exists if and only if $a_{ij} = 0$ when $i > j$. Thus,

$$P(\lambda) = \{g = (g_{ij}) \mid g_{ij} = 0, i > j\}$$

$$U(\lambda) = \{g = (g_{ij}) \mid g_{ii} = 1, g_{ij} = 0, g_{ij} = 0, i > j\}$$

$$L(\lambda) = \{g = (g_{ij}) \mid g_{ij} = 0, i \neq j\}$$

For example, when $G = \mathrm{GL}_2$, take $\lambda(t) = \mathrm{diag}(t, 1)$. Then

$$P(\lambda) = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \right\}; U(\lambda) = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \right\}; L(\lambda) = \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \right\}$$

For general $GL(V)$, write $V = \bigoplus_r V_r$, the weight decomposition of λ. Then take $F_k = \bigoplus_{r \geq k} V_r$. Then

$$P(\lambda) = \{g \in \mathrm{GL}(V) \mid g(F_k) \subset F_k, \forall k \in \mathbb{Z}\}$$

$$U(\lambda) = \{g \in \mathrm{GL}(V) \mid g(v) \equiv v \mod F_k\}$$

$$L(\lambda) = \prod_r \mathrm{GL}(V_r)$$

Lemma 2.3. *The map*

$$\rho_\lambda : P(\lambda) \longrightarrow L(\lambda), p \longmapsto \lim_{t\to 0} \lambda(t)p\lambda^{-1}(t)$$

is a homomorphism.

Lemma 2.4. *There is a short exact squence*

$$0 \longrightarrow U(\lambda) \longrightarrow P(\lambda) \xrightarrow{\rho} L(\lambda) \longrightarrow 0$$

Then group $U(\lambda)$ is the niplotent radical of $P(\lambda)$ and $L(\lambda)$ is reductive.

Lemma 2.5. *Let $\lambda : \mathbb{G}_m \longrightarrow G$ be a 1-PS, $x \in X^\lambda$ be a fixed point of λ and $\mathcal{L}$ be a G-line bundle. Then*

$$\mu^\lambda(\mathcal{L}, x) = \mu^\lambda(\mathcal{L}, x')$$

if x, x' lie in the same connected component of X^λ.

Proof. TBC □

Lemma 2.6. *Let G act on X and Y. Assume $f : X \longrightarrow Y$ is G-equivariant. Let λ be a 1-PS, $x \in X$, $y \in Y$ be a fixed points of λ. Let $\mathcal{L}, \mathcal{L}'$ be G-equivariant line bundles on Y. Then*

1). $\mu^{g^{-1}\lambda g}(\mathcal{L}, y) = \mu^\lambda(\mathcal{L}, g \cdot y)$ *for any $g \in G$;*

2). $\mu^\lambda(f^*\mathcal{L}, x) = \mu^\lambda(\mathcal{L}, f(x))$;

3). $\mu^\lambda(\mathcal{L} \otimes \mathcal{L}', y) = \mu^\lambda(\mathcal{L}, y) + \mu^\lambda(\mathcal{L}', y)$

Proof. This follows form the definition of Hilbert-Mumford weight. □

Definition 2.7. Let $\lambda : \mathbb{G}_m \longrightarrow G$ be a 1-PS. Let Z_λ be the connected component of X^λ. Let

$$Y_\lambda := \{x \in X \mid \lim_{t\to 0} \lambda(t) \cdot x \in Z_\lambda\}$$

We call Y_λ the contradicting variety associated to Z_λ.

Definition 2.8. We say the that a $\mathbb{G}_m$-equivariant morphism $f : X \longrightarrow Y$ is an $\mathbb{A}$-filtration if for any points if for any point $y \in Y$, and there exists a Zariski neighborhood U and a linear action on $\mathbb{A}^n$ for some n such that

$$f|_{f^{-1}(U)} : f^{-1}(U) \longrightarrow U \cong p : U \times \mathbb{A}^n \longrightarrow U$$

as $\mathbb{G}_m$-equivariant morphisms.

Definition 2.9. Let X be a smooth quasi-projective variety equipped with a G action. Let λ be a 1-PS. Then X^λ is a smooth closed subvariety of X and Z_λ is a smooth locally-closed subvariety of X. Moreover, the function

$$p_\lambda : Y_\lambda \longrightarrow Z_\lambda, x \longmapsto \lim_{t\to 0} \lambda(t) \cdot x$$

is an $\mathbb{A}$-filtration.

Proof. This is Bialynicki-Birula theory. See [HH22, Section 2.1]. □

Proposition 2.10. *Let X be a quasi-projective variety equipped with a G-action. Let $\lambda : \mathbb{G}_m \longrightarrow G$ be a 1-PS. There is a natural action of $P(\lambda)$ on Y_λ and a natural action of $L(\lambda)$ on Z_λ. The morphism*

$$p_\lambda : Y_\lambda \longrightarrow Z_\lambda, x \longmapsto \lim_{t\to 0} \lambda(t)\cdot x$$

is equivariant with respect to these actions and the homomorphism ρ_λ, that is, the following diagram commutes:

$$\begin{array}{ccc} Y_\lambda \times P_\lambda & \xrightarrow{\sigma} & Y_\lambda \\ \downarrow{\scriptstyle p_\lambda \times \rho_\lambda} & & \downarrow{\scriptstyle p_\lambda} \\ Z_\lambda \times L(\lambda) & \xrightarrow{\sigma} & Z_\lambda \end{array}$$

Proof. Suppose $y \in Y_\lambda$ and $p_\lambda(y) =: z$ while $p \in P(\lambda)$ and $l = \rho_\lambda(p)$. Then we have

$$\lambda(t)p\cdot y = \lambda(t)p\lambda^{-1}(t)\cdot(\lambda(t)\cdot y) \longrightarrow l\cdot z$$

Since G is connected, so is L_λ. For Z_λ is connected and $e \in L_\lambda$, Z_λ is stable under the action of L_λ. Thus, $l\cdot z \in Z_\lambda$ and $p\cdot y \in Y_\lambda$. □

Remark. When G is not connected, we can give a more general construct. Take $Y_\lambda^0 = \{x \in X \mid \lim_{t\to 0}\lambda(t)\cdot x \text{ exist}\}$. Then there is a natural projection:

$$p_\lambda : Y_\lambda^0 \longrightarrow X^\lambda$$

Take Z_λ to be an open subset of X^λ and $Y_\lambda = p_\lambda^{-1}(Z_\lambda)$. Then we first define L_λ to be the subgroup centralizes λ and preserves Z_λ. Let P_λ be the subgroup such that $\lim_{t\to 0}\lambda(t)p\lambda(t)^{-1} \in L_\lambda$. Then we have the same result as Proposition 2.10.

Let $S_\lambda = G\cdot Y_\lambda$. Then we have a natural map:

$$G\times Y_\lambda \longrightarrow S_\lambda, (g,z)\longmapsto g\cdot z$$

Since $g\cdot(p\cdot y) = gp\cdot y$, (g,y) and (gp^{-1}, py) represents the same points of S_λ. Therefore, we consider the quotient $G\times_{P(\lambda)} Y_\lambda$. In fact Need some reference, we have

$$S_i \cong G\times_{P(\lambda)} Y_\lambda$$

Definition 2.11. Assume that X is a smooth variety with a G-action. A Hesselink-Kermpf-Kirean-Ness stratification of X is the following datum

1) A sequence of open subvarities

$$X = X_0 \supset X_1 \supset \cdots \supset X_n$$

2) 1-PS's $\lambda_i : \mathbb{G}_m \longrightarrow G$, $1 \le i \le n$;

3) Choices of conncetd component Z_{λ_j} of fixed locus on X_{j-i}, $1 \le j \le n$.

such that

i) $X_i = X_{j-1} \setminus S_{\lambda_j}$;

ii) For each j, $G\times^{P(\lambda_j)} Z_{\lambda_j} \xrightarrow{\sim} S_j$ is an isomorphism;

iii) For each j, S_{λ_j} is a closed subvary in X_{j-1}.

Additionally, by the Bialynicki-Birula decomposition, we can taking S_i such that the conormal sheaf $\mathcal{N}^\vee_{S_i/V}$ restricted to Z_i has positive weights with respect to λ_i. We call this condition iv).

Example 2.12. Consider $\mathbb{G}_m$ acts on $\mathbb{P}^n$ by

$$t \cdot [x_0 : x_1 : \cdots : x_n] = [t^{a_0}x_0 : t^{a_1}x_1 : \cdots : t^{a_n}x_n]$$

with $a_0 > a_1 > \cdots > a_n > 0$. Then fixed points are $x_i = [0 : \cdot : 0 : 1 : 0 : \cdots : 0]$. Then

$$S_i = Y_i = \{[x_0 : x_1 : \cdot : x_n] \mid x_i \neq 0, x_{i+1} = x_{i+2} = \cdots = x_n = 0\}$$

So the HKKN stratum are

$$X = \{x_n = 0\} \supset \{x_n = x_{n-1} = 0\} \supset \cdots \supset \{x_n = x_{n-1} = \cdots = x_1 = 0\}$$

and we take all $\lambda_i = \mathrm{id}_{\mathbb{G}_m}$. Note that $\{x_n = x_{n-1} = \cdots = x_1 = 0\} = (P^n)^{ss}(\mathcal{O}_X(1))$.

In fact, we'll show that

Theorem 2.13. *Let G be a reductive group and let X be a smooth G-variety which is projective over an affine G-variety. Let $\mathcal{L}$ be a G-linearized relatively ample line bundle on X. Then, for this linearization, X admits a HKKN stratification*

$$X = X_0 \supset X_1 \supset \cdots \supset X_n = X^{ss}(\mathcal{L}).$$

2.2 Construction of HKKN Decomposition

Let $X(G)$ be the set of all 1-PSs.

Definition 2.14. A map $||\cdot|| : X(G) \longrightarrow \mathbb{Z}_{\geq 0}$ is said to be a norm if

1). $||\lambda|| > 0$ if $\lambda \neq 0$;

2). $||n\lambda|| = |n|||\lambda||$;

3). $||g^{-1}\lambda g|| = ||\lambda||$ for all $g \in G$, that is, $||\cdot||$ is G-invariant.

Usually, we can first define a norm on a maximal torus $T \subset G$ which is invariant under the Weyl group. Then it extends to all of $X(G)$, because every 1-PS is conjugate into T. Strictly,

Proposition 2.15. *Every W-invariant norm on $X(T)$ extends uniquely to a G-invariant norm on $X(G)$.*

Example 2.16. To be continued

Now, fix 1-PS $\lambda : \mathbb{G}_m \longrightarrow G$ and a connected component $Z \subset X^\lambda$ of fixed locus. Define $Y_{\lambda,Z} := \{x \in X \big| \lim_{t\to 0}\lambda(t) \cdot x \in Z\}$, called the attracting set. There is a canonical morphism $p_{\lambda,Z} : Y_{\lambda,Z} \longrightarrow Z$ given by $x \longmapsto \lim_{t\to 0}\lambda(t) \cdot x$.

3 Derived Category of G-equivariant Sheaves

Let G be a group scheme acting on a scheme X. Denote the action by

$$\sigma : G \times X \longrightarrow X, \ (g, x) \longmapsto g \cdot x$$

Let $p_2 : G \times X \longrightarrow X, \ (g, x) \longmapsto x$ be the projection onto the second factor.

Definition 3.1. A G-equivariant quasi-coherent sheaf on X is a pair $(\mathcal{F}, \theta)$, where $\mathcal{F}$ is a quasi-coherent sheaf on X and $\theta : p_2^*\mathcal{F} \xrightarrow{\sim} \sigma^*\mathcal{F}$ is an isomorphism of quasi-coherent sheaves on $G \times X$ such that θ satisfies:

1). The cocycle condition:

$$(\mathrm{id} \times \sigma)^*\theta \circ p_{23}^*\theta = (m \times \mathrm{id}_X)^*\theta$$

where $(\mathrm{id} \times \sigma) : G \times G \times X \longrightarrow G \times X,\ (g, h, x) \longmapsto (g, h \cdot x)$ and $p_{23} : G \times G \times X \longrightarrow G \times X,\ (g, h, x) \longmapsto (h, x)$.

2). The unit condition

$$s^*\theta = \mathrm{id}_{\mathcal{F}}$$

where $s : X \longrightarrow G \times X,\ x \longmapsto (e, x)$.

Remark. Let's explain the geometric meaning of θ, the cocycle condition and the unit condition:

For a point $(g, x) \in G \times X$, the fiber of $p_2^*\mathcal{F}$ is $\mathcal{F}_x$ while the fiber of $\sigma^*\mathcal{F}$ at (g, x) is $\mathcal{F}_{g\cdot x}$. Then θ gives isomorphisms $\theta_{g,x} : \mathcal{F}_x \xrightarrow{\sim} \mathcal{F}_{g\cdot x}$.

Then the cocycle condition says the following diagram commutes:

$$\mathcal{F}_x \xrightarrow[\theta_{h,x}]{} \mathcal{F}_{h\cdot x} \xrightarrow[\theta_{g,h\cdot x}]{} \mathcal{F}_{gh\cdot x}, \qquad \mathcal{F}_x \xrightarrow{\theta_{gh,x}} \mathcal{F}_{gh\cdot x}$$

and the unit condition says $\theta_{e,x} = \mathrm{id}_{\mathcal{F}}$.

Example 3.2. 1). Let G acts on X. Take $\mathcal{F} = \mathcal{O}_X$. Then

$$p_2^*\mathcal{O}_X = \mathcal{O}_{G\times X} = \sigma^*\mathcal{O}_X$$

So we may take $\theta = \mathrm{id}_{\mathcal{O}_{G\times X}}$.

2). Let $G = \mathrm{GL}_n$ and $X = \mathbb{A}^n$. The action is the standard linear action give by $g \cdot v = gv$. Consider the tangent bundle $\mathcal{T}_X$. Since $\sigma_g : X \longrightarrow X$ is linear, we get $d\sigma_g : \mathcal{T}_x X \longrightarrow \mathcal{T}_{gx} X$. These glue into an isomorphism $\theta : p_2^*\mathcal{T}_X \longrightarrow \sigma^*\mathcal{T}_X$. So $\mathcal{T}_X$ is naturally GL_n-equivariant.

3). Consider $G = \mathrm{GL}_{n+1}$ acts on $X = \mathbb{P}^n$. The action is induced from the linear action on k^{n+1}. For $\mathcal{O}_X(1)$, a point of the fiber over $[v] \in X$ is a linear form on the linear kv. If $g \in \mathrm{GL}_{n+1}$, then g sends the line kv to kgv. Hence, this is an isomorphism $\theta_{g,[v]} : \mathcal{O}_{[v]} \longrightarrow \mathcal{O}_{[gv]}$, which glues to the equivariant structure.

4). Suppose G acts trivially on X. Then $\sigma(g, x) = x = p_2(g, x)$. So $a^*\mathcal{F} = p_2^*\mathcal{F}$. Thus, each G-equivariant coherent sheaf $(\mathcal{F}, \theta)$ is given by $\mathcal{F} \in \mathrm{Coh}(X)$ and

$$\theta : p_2^*\mathcal{F} \longrightarrow \overset{*}{\sqrt{^2}}\mathcal{F}$$

For each fiber over g, we get $\theta_g : \mathcal{F} \longrightarrow \mathcal{F}$ with $\theta_g \in \mathrm{Aut}(\mathcal{F})$ and $\theta_{gh} = \theta_g \circ \theta_h$.

For every open set U, set $M_U := \Gamma(U, \mathcal{F})$. Then it becomes a representation of G under θ_g. Then

$$M_U = \bigoplus M_{U,\lambda} \otimes V_\lambda$$

where V_λ runs through all irreducible representations of G and $M_{U,\lambda} = \mathrm{Hom}(M_U, V_\lambda)$. Since G-action commutes with the A-action, with $A = \Gamma(U, X)$. $M_{U,\lambda}$ glues to a subsheaf, denoted by $\mathcal{F}_\lambda$. Therefore,

$$\mathcal{F} \cong \bigoplus_\lambda \mathcal{F}_\lambda \otimes V_\lambda$$

Thus, $\mathrm{Coh}_G(X) \simeq \mathrm{Coh}(X) \otimes \mathrm{Rep}(G)$. In particular, when X is a point, $\mathrm{Coh}_G(\mathrm{pt}) = \mathrm{Rep}(G)$.

Definition 3.3. Let $(\mathcal{F}m, \theta_{\mathcal{F}})$, $(\mathcal{G}, \theta_{\mathcal{G}})$ be a morphism of quasi-coherent sheaves $\varphi : \mathcal{F} \longrightarrow \mathcal{G}$ such that the following diagram on $G \times X$ commutes:

$$\begin{array}{ccc} p_2^*\mathcal{F} & \xrightarrow{\theta_{\mathcal{F}}} & \sigma^*\mathcal{F} \\ \downarrow{\scriptstyle p_2^*\varphi} & & \downarrow{\scriptstyle \sigma^*\mathcal{F}} \\ p_2^*\mathcal{G} & \xrightarrow{\theta_{\mathcal{G}}} & \sigma^*\mathcal{G} \end{array}$$

Remark. Without any confusion, over $(g, x) \in G \times X$, we have

$$\begin{array}{ccc} \mathcal{F}_x & \xrightarrow{\theta_{g,x}} & \mathcal{F}_{g\cdot x} \\ \downarrow{\scriptstyle \varphi_x} & & \downarrow{\scriptstyle \phi_{g\cdot x}} \\ \mathcal{G}_x & \xrightarrow{\theta'_{g,x}} & \mathcal{G}_{g\cdot x} \end{array}$$

That is, $\varphi(g \cdot x) = g \cdot \varphi(s)$ for $s \in \mathcal{F}_x$ in some sense.

Suppose we have $(\mathcal{F}, \theta_{\mathcal{F}}), (\mathcal{G}, \theta_{\mathcal{G}}) \in \mathrm{QCoh}_G(X)$. There is an action of G on $\mathrm{Hom}_G(\mathcal{F}, \mathcal{G}) := \mathrm{Hom}_{\mathrm{QCoh}}(\mathcal{F}, \mathcal{G})$, given by

$$g \cdot f = \theta_{\mathcal{G},g} \circ f \circ \theta_{\mathcal{F},g}^{-1}$$

where $\theta_{\mathcal{F},g} : g_X^*\mathcal{F} \xrightarrow{\sim} \mathcal{F}$, with $g_X : X \longrightarrow X$ given by the action of $g \in G$. In fact, consider this fiberwise, we have

$$g \cdot f_x := \theta_{\mathcal{G},g,x} \circ f_{g\cdot x} \circ \theta_{\mathcal{F},g,x}^{-1}$$

When X is a point, this is just the action of G over the morphism spaces of two G-representations.

Proposition 3.4. *For any G-equivariant quasi-coherent sheaves $\mathcal{F}, \mathcal{G}$, we have*

$$\mathrm{Hom}_G(\mathcal{F}, \mathcal{G}) = \mathrm{Hom}_X(\mathcal{F}, \mathcal{G})^G$$

Proof. TBC □

Corollary 3.5. *Assume that G is reductive and let $\mathcal{F}, \mathcal{G}$ be two G-equivariant quasi-coherent sheaves on X. Then there are quasi-coherent isomorphisms:*

$$\mathrm{Ext}_G^i(\mathcal{F}, \mathcal{G}) \cong \mathrm{Ext}_X^i(\mathcal{E}, \mathcal{F})^G$$

Moreover, if $\mathcal{F}^\bullet$ and $\mathcal{G}^\bullet$ are compplexes of G-equivariant quasi-coherent sheaves on X. Then

$$\mathrm{Hom}_G(\mathcal{F}^\bullet, \mathcal{G}^\bullet[i]) \cong \mathrm{Ext}_X^i(\mathcal{F}^\bullet, \mathcal{G}^\bullet)$$

where the morphism spaces are taken in the derived category.

Proof. TBC □

Now, we denote the category of G-equivariant quasi-coherent sheaves by $\mathrm{QCoh}_G(X)$ and denote $D(\mathrm{QCoh}_G(X))$ the derived category of $\mathrm{QCoh}_G(X)$.

Similarly, we can define the G-equivariant coherent sheaves over X and denote the corresponding category by $\mathrm{Coh}_G(X)$ and the corresponding derived categroy consisting bounded sequences by $D_G^b(X)$.

Now, we're going to use the language of stack to describe these objects.

Definition 3.6. A principal G-bundle over a scheme T is $\pi : P \longrightarrow T$ with a right G-action on P

$$\sigma_P : P \times G \longrightarrow P, (p, g) \longmapsto p \cdot g$$

such that

1). G-action preserves fibers $\pi(p \cdot g) = \pi(p)$;

2). There exists an étale or fppf cover $\{U_i \longrightarrow T\}$ such that $P|_{U_i} \cong U_i \times G$l

3). The map $P \times G \longrightarrow P \times_T P,\ (p, g) \longmapsto (p, p \cdot g)$ is an isomorphism.

Remark. The last point is equivalent to the condition that if p_1, p_2 are in the same fiber then there exist unique $g \in G$ such that $p_2 = p_1 \cdot g$.

Definition 3.7. The stack $[X/G]$ is a functor from **Sch** to **Cat** such that

For each $[X/G](T)$, an object is a pair (P, f) where $P \longrightarrow T$ is a principal G-bundle and $f : P \longrightarrow X$ is G-equivariant. Diagrammatically,

$$\begin{array}{ccc} P & \xrightarrow{f} & X \\ \downarrow{\scriptstyle \pi} & & \\ T & & \end{array}$$

while a morphis $(P, f) \longrightarrow (P', f')$ is an isomorphism of principal bundles $\varphi : P \longrightarrow P'$ such that $f = f' \circ \varphi$.

Remark. Since we consider the right action of G on P. Thus, $f : P \longrightarrow G$ is G-equivariant if and only if

$$f(p \cdot g) = g^{-1} \cdot f(p)$$

A quasi-coherent sheaf over $[X/G]$ is defined to be a G-equivariant quasi-coherent sheaf over X. Similarly for coherent sheaves over $[X/G]$. Moreover, we denote the corresponding derived category by $D^b(X/G)$.

Remark. In fact, there is a natural definition of quasi-coherent sheaves, coherent sheaves over any stack. Let $\mathrm{QCoh}(X/G)$ (resp.$\mathrm{Coh}(X/G)$) denote the category of quais-coherent (resp. coherent) sheaves over $[X/G]$. Then we can show that there are equivalence of categories:

$$\mathrm{QCoh}_G(X) \cong \mathrm{QCoh}(X/G),\ \mathrm{Coh}_G(X) \cong \mathrm{Coh}(X/G)$$

So from now, on we just denote $\mathrm{QCoh}_G(X), \mathrm{Coh}_G(X), \mathrm{Hom}_G(\mathcal{F}, \mathcal{G})$ by $\mathrm{QCoh}(X/G), \mathrm{Coh}(X/G)$ and $\mathrm{Hom}_{[X/G]}(\mathcal{F}, \mathcal{G})$.

Now, let's consider the functoriality. Suppose $f : X \longrightarrow Y$ is G-equivariant, that is,

$$f \circ \sigma_X = \sigma_Y(\mathrm{id}_G \times f)$$

We can define the pullback and pushforward related to f.

Pullback: Let $(\mathcal{F}, \theta) \in \mathrm{QCoh}_G(Y)$. Define

$$f^*(\mathcal{F}, \theta) := (f^*\mathcal{F}, f^*\theta)$$

Then we have

$$f^*\theta : \sigma_X^* f^*\mathcal{F} = (\mathrm{id}_G \times f)^* \sigma_Y^*(\mathcal{E}) \xrightarrow[\sim]{(\mathrm{id}\times f)^*\theta} (\mathrm{id}_G \times f)^* p_{2,Y}^* \mathcal{F} = p_{2,X}^* f^*\mathcal{F}$$

It also satisfies cocycle condition and unit condition. Hence, we get $f^* : \mathrm{QCoh}(Y/G) \longrightarrow \mathrm{QCoh}(X/G)$ and then $Lf^* : D^b(\mathrm{QCoh}(Y/G)) \longrightarrow D^b(\mathrm{QCoh}(X/G))$.

Pushforward: Let $(\mathcal{F},\theta)\in \mathrm{QCoh}(X/G)$. Because f is G-equivariant, we have

$$\begin{array}{ccc} G\times X & \xrightarrow{\mathrm{id}\times f} & G\times Y \\ \downarrow{\scriptstyle \sigma_X} & & \downarrow{\scriptstyle \sigma_Y} \\ X & \xrightarrow{\quad f\quad} & Y \end{array}$$

Using base change (σ_Y is smooth), $\sigma_Y^* f_*\mathcal{F}\xrightarrow{\sim}(\mathrm{id}\times f)_*\sigma_X^*\mathcal{F}$. Applying θ, we get

$$(\mathrm{id}\times f)_*\sigma_X^*\mathcal{F}\xrightarrow[\sim]{(\mathrm{id}\times f)_*\theta}(\mathrm{id}\times f)_*p_{2,X}^*\mathcal{F}$$

Since projection is flat and $p_{2,Y}\circ(\mathrm{id}\times f)=f\circ p_{2,X}$, after base change, $(\mathrm{id}\times f)_*p_{2,X}^*\mathcal{F}\xrightarrow{\sim}p_{2,Y}^*f_*\mathcal{F}$. Thus, we define

$$\theta'_f:\sigma_Y^*f_*\mathcal{F}\xrightarrow[\sim]{b.c.}(\mathrm{id}\times f)_*\sigma_X^*\mathcal{F}\xrightarrow[\sim]{(\mathrm{id}\times f)_*\theta}(\mathrm{id}\times f)_*p_{2,X}^*\mathcal{F}\xrightarrow[\sim]{b.c.}p_{2,Y}^*f_*\mathcal{F}$$

Also, it satisfies cocycle conditions and unit conditions. Hence, we get

$$\begin{aligned} f_*:\mathrm{QCoh}(X/G)&\longrightarrow \mathrm{QCoh}(Y/G)\\ (\mathcal{F},\theta)&\longrightarrow (f_*\mathcal{F},\theta'_f)\end{aligned}$$

and the derived functors $Rf_*L:\mathrm{D}^b(\mathrm{QCoh}(X/G))\longrightarrow D^b(\mathrm{QCoh}(Y/G))$.

Always, we have $f^*:\mathrm{Coh}(Y/G)\longrightarrow\mathrm{Coh}(X/G)$. However, to obtain $Lf^*:D^b(X/G)\longrightarrow D^b(Y/G)$, we need the usual coherent preservation conditions, i.e., f is smooth, flat, ect. Also to obtain $f_*:D^b(X/G)\longrightarrow D^b(Y/G)$, we need condition like f is proper, etc. [Huy06, Section 3.3].

Now, let X be a representation of a reductive group G and write $X=\mathrm{Spec}(A)$, with $A=\mathrm{Sym}(X^\vee)$. Then

$$\mathrm{Coh}(X/G)\simeq A\text{-mod}_{f.g.}^G$$

where $A\text{-mod}_{f.g.}^G$ represents the finitely generated A-mods with a compatible G-action.

Proposition 3.8. *Let X be a representation of a reductive group G. Then the family $\{\mathcal{O}_X\otimes V\}_{V\in\mathrm{Rep}(G)}$ is a set of projective generators of* $\mathrm{Coh}(X/G)$.

Proof. Proposition 3.4 implies $\mathrm{Hom}_{[X/G]}(V\otimes\mathcal{O}_X,\mathcal{F})=\mathrm{Hom}_X(V\otimes\mathcal{O}_X,\mathcal{G})^G$. Since V is a linear space, $\mathrm{Hom}_X(V\otimes\mathcal{O}_X,-)$ is exact, while G is reductive implies $(-)^G$ is exact. Thus,

$$\mathcal{F}\longmapsto\mathrm{Hom}_X(V\otimes\mathcal{O}_X,\mathcal{F})^G=\mathrm{Hom}_{[X/G]}(V\otimes\mathcal{O}_X,\mathcal{F})$$

is exact. Thus, $V\otimes\mathcal{O}_X$ is projective.

Now, take $M\in A\text{-mod}_{\mathrm{f.g.}}^G$. Choose finitely many A-module generators $m_1,m_2,...,m_r$. By [Bor91, Section 1.1.9], $m_1,m_2,...,m_r$ lie in some finitely dimensional representation of G. Take $W\subset M$ to be such representation. Then we have

$$A\otimes W\longrightarrow M$$

is a surjection. □

Definition 3.9. For $\mathcal{T}$ a subcategory, a full subcategory $\mathcal{S}\subset\mathcal{T}$ is said to be thick, if

1) For $A\in\mathcal{S}$, $A[n]\in\mathcal{S}$ for any $n\in\mathcal{S}$;
2) For distinguished sequence $A\to B\to C\to A[1]$, if any two of them lie in $\mathcal{S}$, so does the third;
3) $\mathcal{S}$ is closed under direct summand, that is, if $A=B\oplus C\in\mathcal{S}$, then $B,C\in\mathcal{S}$.

Let E be a set of objects in $\mathcal{T}$. Let $\mathrm{thick}(E)$ be the smallest thick category containing E. We say $\mathcal{T}$ is generated by E if $\mathcal{T} = \mathrm{thick}(E)$.

Now, with Proposition 3.8, we can give the generators of $D^b(X/G)$ when X is a representation of G.

Corollary 3.10. *$D^b(X/G)$ is generated by the objects $\mathcal{O}_X \otimes V$.*

Proof. Set $d = \dim X$. Let $\mathcal{F} \in \mathrm{Coh}(X/G)$, then we have a projective resolution:

$$\cdots \longrightarrow \mathcal{O}_X \otimes V_1 \longrightarrow \mathcal{O}_X \otimes V_0 \longrightarrow \mathcal{F} \longrightarrow 0$$

And let $K_d = \ker(\mathcal{O}_X \otimes V_{d-1} \longrightarrow \mathcal{O}_X \otimes V_d)$. By general homological algebra,

$$\mathrm{Ext}^1(K_d, \mathcal{G}) = \mathrm{Ext}^{d+1}(\mathcal{F}, \mathcal{G}) = 0$$

for any G-equivariant quasi-coherent sheaf $\mathcal{G}$. Thus, K_d is projective.

Since K_d is projective and there exists $V \otimes \mathcal{O}_X \twoheadrightarrow \mathcal{F}$, $\mathcal{F}$ is a direct summand of $\mathcal{O}_X \otimes V$. Every $\mathcal{F} \in \mathrm{Coh}(X/G)$ is reprsented in the derived category by a bounded complex built form the projective generators $\mathcal{O}_X \otimes V$. Hence, $\mathcal{F} \in \mathrm{Thick}(\mathcal{O}_X \otimes V)$. Moreover, for any $E \in D^b(X/G)$, we have

$$E_{\leq n-1} \longrightarrow E_{\leq n} \longrightarrow H^n(E)[-n] \longrightarrow E_{\leq n-1}[1]$$

which implies every object of $D^b(X/G)$ can be built form $\mathrm{Coh}(X/G)$ by finitely many cones and shifts. Thus, $D^b(X/G) = \mathrm{Thick}(\mathcal{O}_X \otimes V)$, that is, $D^b(X/G)$ is generated by $\{\mathcal{O}_X \otimes V\}_{V \in \mathrm{Rep}(G)}$. □

Example 3.11. Consider $G = \mathbb{G}_m$ acts on $X = \mathbb{A}^{n+1}$ with weight 1. Let $S = k[x_0, x_1, ..., x_n]$. Then

$$\mathrm{Coh}_{\mathbb{G}_m}(\mathbb{A}^{n+1}) \simeq S\text{-mod}^{\mathbb{G}_m}_{\text{f.g}} \simeq S\text{-grmod}_{\text{f.g.}}$$

The basic generators are $S(m)$ where $S(m)_d = S_{d+m}$. Geometrically,

$$\widetilde{S(m)} = \mathcal{O}_{\mathbb{A}^{n+1}} \otimes k_m$$

with $k_m : t \longrightarrow t^m$, a representation of $\mathbb{G}_m$. Moreover, we set

$$\mathcal{O}_{\mathbb{A}^{n+1}}(m) := \mathcal{O}_{\mathbb{A}^{n+1}} \otimes k_m$$

Then $\mathrm{D}^b(\mathbb{A}^{n+1}/\mathbb{G}_m)$ is generated by $\mathcal{O}_{\mathbb{A}^{n+1}}(m)$, $m \in \mathbb{Z}$.

4 Derived Category of GIT

For X a smooth projective-over-affine variety with the action of a connected reductive group G, fixing a G very ample line bundle, there exists a HKKN stratification over X by Theorem 2.13:

$$X : X_0 \supset X_1 \supset X_2 \supset \cdots \supset X_n = X^{ss}$$

Set $X^{us} = X \setminus X^{ss}$. We can give a description of $D^b(X/G)$ with respect to X^{us}, called derived Kirwan surjectivity.

4.1 Derived Kirwan Surjectivity

For each HKKN stratum, let $\sigma_i : Z_i \hookrightarrow S_i$ and $j_i : S_i \hookrightarrow X$ denotes the respective inclusion, there exists a pullback functor $j^! : D^b(X/G) \longrightarrow D^b(S_i/G)$. Let $D^b_{X^{us}}(X/G)$ be the full subcategory of $D^b(X/G)$ consisting the elements supported at X^{us}. Then we define

Definition 4.1. For each HKKN stratum, choose $w_i \in \mathbb{Z}$ and denote the corresponding function $w : \{1, \cdots, n\} \longrightarrow \mathbb{Z}$. Define the full subcategories of $D^b(X/G)$:

$$D^b_{X^{us}}(X/G)_{\geq w} := \left\{F^\bullet \in D^b_{X^{us}}(X/G) \mid \forall i, \lambda_i \text{ weights of } \mathcal{H}^*(\sigma_i^* j_i^* \mathcal{F}^\bullet) \text{ are } \geq w_i\right\}$$

$$D^b_{X^{us}}(X/G)_{< w} := \left\{F^\bullet \in D^b_{X^{us}}(X/G) \mid \forall i, \lambda_i \text{ weights of } \mathcal{H}^*(\sigma_i^* j_i^! \mathcal{F}^\bullet) \text{ are } < w_i\right\}$$

and $G_w := D^b_{X^{us}}(X/G)_{\geq w} \cap D^b_{X^{us}}(X/G)_{< w}$.

Since X is smooth in a neighborhood of each Z_i, $\mathcal{N}^\vee_{S_i} X$ is a locally free sheaf. In the case, $\det(\mathcal{N}^\vee_{S_i} X)$ is an equivariant invertible sheaf and its restriction to Z_i is concentrated in a single non-negative weight with respect to λ_i. Define

$$\eta_i := \mathrm{wt}_{\lambda_i} \det(\mathcal{N}^\vee_{S_i} X)|_{Z_i}$$

Then we can rewrite G_w as following:

Lemma 4.2. *If X is smooth in a neighborhood of each Z_i. Then*

$$D^b_{X^{us}}(X/G)_{< w} := \{\mathcal{F}^\bullet \in D^b_{X^{us}}(X/G) \mid \forall i, \lambda_i \textit{weight of } \mathcal{H}^*(\sigma_i^* j_i^* \mathcal{F}_i) \textit{ are } < w_i + \eta_i\}$$

$$G_w := \{\mathcal{F}^\bullet \in D^b_{X^{us}}(X/G) \mid \forall i, \lambda_i \textit{ weights of } \mathcal{H}^*(\sigma_i^* j_i^* \mathcal{F}^\bullet) \textit{ lie in } [w_i, w_i + \eta_i)\}$$

Proof. The inclusion $i : S_i \hookrightarrow X$ is a regular embedding, so $j_i^! = \det(\mathcal{N}_{S_i/X}) \otimes j_i^*$ and note that $\mathrm{wt}_{\lambda_i} \det(\mathcal{N}_{S_i/X})|_{Z_i} = \eta_i$. □

Now, for X a smooth projective-over-affine variety with the actions of a connected reductive group G. Let $\mathcal{L}$ be an ample G line bundle over X. Take the HKKN stratum such that the canonical sheaf $\mathcal{N}^\vee_{S_i/X}$ restricted to Z_i has positive weights with respect to λ_i. Then we have:

Theorem 4.3. *[Derived Kirwan Surjectivity] There are semi-orthogonal decompositions*

$$D^b_{X^{us}}(X/G) = \langle D^b_{X^{us}}(X/G)_{< w}, D^b_{X^{us}}(X/G)_{\geq w} \rangle$$

$$D^b(X/G) = \langle D^b_{X^{us}}(X/G)_{< w}, \mathbf{G}_w, D^b_{X^{us}}(X/G)_{\geq w} \rangle$$

and the restricting functor $i^* : \mathbf{G}_w \to D^b(X^{ss}(\mathcal{L})/G)$ is an equivalence of categories.

Sketch of The Proof. We apply induction on the asceneding sequence of open substack, with $X_0/G = X^{ss}(\mathcal{L})/G$ and $X_i/G = X_{i-1}/G \sqcup S_i/G$ for $i = 1, 2, \cdots, n$. Also define $X^{us}_n/G = S_i/G \cup \cdots \cup S_n/G \subset X_n/G$.

Consider the closed HKKN stratum $S_n/G \subset X_n/G$, by [HL15, Section 3], there exists a SOD

$$D^b(X_n/G) = \langle D^b_{X^{us}_n}(X_n/G), \widetilde{G}_w, D^b_{X^{us}_n}(X_n/G)_{\geq w} \rangle$$

where $\widetilde{G}$ is mapped isomorphically onto $D^b(X_{n-1}/G)$ via restriction. Then we use induction hypothesis:

$$D^b(X_n/G) = \langle D^b_{S_n}(X_n/G), D^b_{X^{us}_{n-1}}(X_{n-1}/G), G_w, D^b_{X^{us}_{n-1}}(X_{n-1}/G), D^b_{S_n}(S_n/G) \rangle$$

where

$$D^b_{X^{us}_n}(X_n/G)_{< w} \simeq \langle D^b_{S_n}(X_n/G)_{< w}, D^b_{X^{us}_{n-1}}(X_{n-1}/G) \rangle$$

and similarly for $D^b_{X^{us}_n}(X_n/G)_{\geq w}$. □

Example 4.4. For $\mathbb{G}_m$ acts on $\mathbb{A}^{n+1}$ with weight 1. Take $\lambda(t) = t^{-1}$. Then $Z = Y = S = \{0\} \subset \mathbb{A}^{n+1}$. Then we have

$$\mathcal{N}^{\vee}_{0/\mathbb{A}^{n+1}} \cong \mathbb{C}^{n+1}$$

where $\mathbb{G}_m$ acts on it with weight 1. Then $X = \mathbb{G}_m$ acts on $\det(\mathcal{N}_{0/\mathbb{A}^{n+1}})$ with weight $n+1$. Thus, $\eta_\lambda = n+1$. By definition,

$$\mathbf{G}_w = \{\mathcal{F}^\bullet \mid \mathrm{wt}_\lambda \mathcal{H}^*(\mathcal{F}^\bullet) \text{ lies in } [w, w+\eta_\lambda)\} = \langle \mathcal{O}_{\mathbb{A}^{n+1}}(w), \cdots, \mathcal{O}_{\mathbb{A}^{n+1}}(w+n)\rangle$$

Recall that, $X^{ss}(\lambda)/G \cong \mathbb{P}^n$ and $D^b(\mathbb{P}^n) = \langle \mathcal{O}_{\mathbb{P}^1}(-n), \mathcal{O}_{\mathbb{P}^n}(-n+1), \cdots, \mathcal{O}_{\mathbb{P}^n}\rangle$.

In fact, we can achieve X_+ and X_- by taking different stability conditions. Take $\delta_k : t \mapsto t^k$. Then

$$X_+ = X^{ss}(\delta_k \mid k > 0), X_- = X^{ss}(\delta_k \mid k < 0)$$

4.2 Application: Variation of GIT

We start by considering the rational maps between two smooth varieties.

Definition 4.5. Let $\phi : X_1 \dashrightarrow X_2$ be a birational map between smooth complete varieties. A birational cobordism for ϕ is a normal variety X equipped with a $\mathbb{G}_m$-action such that

$$X_-/\mathbb{G}_m \cong X_1, X_+/\mathbb{G}_m \cong X_2$$

where $X_+\{x \in X \mid \lim_{t\to\infty} t \cdot x \text{ doesn' t exists}\}$ and $X_-\{x \in X \mid \lim_{t\to 0} t \cdot x \text{ does not exist}\}$.

By Theorem 1.52, the open subset $X_+ \cap X_+$ has a geometric quotient by $\mathbb{G}_m$, and the induced birational map between the two quotient is exactly ϕ.

Example 4.6. Take $V = \mathbb{A}^{n+1}_x \oplus \mathbb{A}^{n+1}_y$ and $\mathbb{G}_m$-action on V by weight 1 on x and weight -1 on y. Then

$$V^+ = \{y = 0\}, V^-\{x = 0\}$$

where $X_+ = V \setminus V^+$ and $Y^+ = V \setminus V^-$. Then

$$X_+/\mathbb{G}_m \cong \mathrm{Tot}(\mathcal{O}_{\mathbb{P}^n}(-1)^{\oplus n+1}), X_- \cong \{\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^n}(-1))^{\oplus n+1}\}$$

are birational. $X_+ \dashrightarrow X_-$ is the Atiyah flop.

Moreover, we have:

Theorem 4.7. *Let X_1 and X_2 be two birational projective varieties. Then there is a birational cobordism $X/\mathbb{G}_m$ between X_1 and X_2 are smooth, then by equivariant resolution of singularities X can be chosen to be smooth.*

So in order to understand the derived category of X_1, X_2, should consider what happen when we change the stability conditions.

Suppose X is smooth and $\mathbb{G}_m$ acts on X. By Bialynicki-Birula decomposition [HH22, Section 2], Then

$$X^{\mathbb{G}_m} = \bigsqcup_i Z_i$$

with Z_i smooth connected components. For each $z \in Z$, $T_z X = T_z Z \oplus \mathcal{N}^+_{i,x} \oplus \mathcal{N}^-_{i,x}$, where $\mathcal{N}^+_i$ are the positive weights part, similarly for $\mathcal{N}^-_{i,x}$. Define

$$S^+_i = \{x \in X \mid \lim_{t\to 0} t \cdot x \in Z_i\}, S^-_i = \{x \in X \mid \lim_{t\to\infty} t \cdot x \in Z_i\}$$

the Białynicki-Birula cells. Also, we have $S^+_i \cong \mathcal{N}^+_i$ and $S^-_i \cong \mathcal{N}^-_i$.

Example 4.8. For $\mathbb{G}_m$ acts on $\mathbb{P}^n$ via

$$t\dot{[}x_0 : x_1 : \cdots, x_n] = [t^{a_0}x_0 : t^{a_1}x_1 : \cdots : t^{a_n}x_n]$$

with $a_0 > a_1 > \cdots > a_n$. The fixed points are $p_i = [0 : \cdots : 1 : \cdots : 0]$. Then

$$X_i^+ = \{[x] \mid x_k = 0, 0 < k < i, x_i \neq 0\}, X_- = \{[x] \mid x_k = 0 \mid x_k = 0, n \geq k > i, x_i \neq 0\}$$

Obviously, $\mathbb{P}^n = \bigsqcup X_i^+ = \bigsqcup X_i^-$.

Let $\mathcal{L}$ be a very ample G line bundle. Define $\mu_i \in \mathbb{Z}$ to be the weight of $\mathcal{L}|_{Z_i}$. Then set

$$S_i = S_i^+, \text{ if } \mu_i < 0; S_i = S_i^-, \text{ if } \mu_i > 0$$

so that the normal bundle $\mathcal{N}_{S_i/Z_i}^\vee$ also has a positive weights. More precisely,

$$\Omega_{S_i}|_{Z_i} = \Omega_{Z_i} \oplus \mathcal{N}_i^{+\vee}, \mu_i < 0; \qquad \Omega_{S_i}|_{Z_i} = \Omega_{Z_i} \oplus \mathcal{N}_i^{-\vee}, \mu_i > 0;$$

Thus, we have

$$\eta_i = \mathrm{wt}_{\lambda_i} \det(\mathcal{N}_i^+), \mu_i < 0; \qquad \eta_i = -\mathrm{wt}_{\lambda_i} \det(\mathcal{N}_i^-), \mu_i > 0$$

In symplectic setting, let (X, ω) be a kähler manifold with a Hamiltonian $\mathbb{G}_m$-action and a moment map $\mu : X \longrightarrow \mathfrak{g}_m$. The fundamental vector field generated by $\mathfrak{g}_m$ is

$$\xi_X(x) = \frac{d}{dt}|_{t=0} e^{it} \cdot x, \qquad d\mu_X(-) = \omega(\xi_X(x), -)$$

Then $d\mu_x = 0$ if and only if $\xi_X(x) = 0$. By definition, $\xi_X(x) = 0$ means x is fixed by $\mathbb{G}_m$-action. Thus, $\mathrm{Crit}(\mu) = X^{\mathbb{G}_m}$. Suppose $Z \subset X^{\mathbb{G}_m}$ is a connected component. Then every point of Z is critical, that $\mu|_Z$ is constant.

Example 4.9. Consider the action of $\mathbb{G}_m$ on $\mathbb{P}^2$ by $t \cdot [x : y : z] = [x : ty : t^2 z]$. The moment map is

$$\mu : [x : y : z] \longmapsto \frac{|y|^2 + 2|z|^2}{|x|^2 + |y|^2 + |z|^2}$$

Critical values are $\mu_0 = 0, \mu_1 = 1, \mu_2 = 2$ while fixed points are $p_0 = [1 : 0 : 0]$, $p_1 = [0 : 1 : 0]$ and $p_2 = [0 : 0 : 1]$.

Now, verify $\mathcal{L}$ by twisting by the char $k_{-r} : t \mapsto t^{-r}$, denoted by $\mathcal{L}(r)$ and shift the moment map by r, denoted by μ_r. Then the zero fiber of μ_r corresponds to the fiber of μ^{-1} at $-r$. The most important this is that the semi-stable points change when we twist k_{-r} over $\mathcal{L}$. In the following contexts, we will discuss the change of derived category of semi-stable points.

Remark. Suppose $\mathcal{L}$ is a $\mathbb{G}_m$-linearized line bundle. Twisting a $\mathbb{G}_m$-linearized line bundle by a character leaves the $\mathbb{G}_m$-action on X unchanged, but shifts the $\mathbb{G}_m$-weights on the fibers. Therefore the underlying line bundle and base action are the same, while the GIT semistable locus may change.

For example, we consider $\mathbb{G}_m$ acts on $\mathbb{P}^1$ via $t \cdot [x_0 : x_1] = [x_0 : tx_1]$. Twisting k_{-1} on $\mathcal{O}_X(1)$, the action becomes $t \cdot [x_0 : x_1] = [t^{-1}x_0 : x_1]$. These two actions are the same but the action on fibers are different. Also, note that at beginning, $X^{ss}(\mathcal{L}) = \{[0 : 1]\}$ while $X^{ss}(\mathcal{L}(1)) = \{[1 : 0]\}$.

As r increases, when the zero fiber of μ_r crosses Z_i with $\mathrm{wt}_{\lambda_i}\mathcal{L}(r)|_{\mathbb{Z}_i} = 0$, the sign of $\mu_i = \mathrm{wt}_{\lambda_i}\mathcal{L}(t_i)|_Z$ changes. Suppose $\mu_i(r-1) > 0 > \mu_i(r+1)$. Then the maximal destabilizing 1-PS λ_i flips from λ_- to λ_+. The unstable stratum $S_i = S_i^-$ files to $S_i = S_i^-$ by the settings before. Moreover,

$$\eta_i(r-1) - \eta_i(r+1) = \mathrm{wt}_{\lambda_i}\omega_X|_{Z_i}$$

by the result in Białynicki-Birula decomposition.

Recall that η_i are the length of the widows. If weight of $\omega_X|_{Z_i}$ is zero and Z_i is the only critical set for which $\mu_r = 0$, then $\mathbf{G}(r+1)_w = \mathbf{G}(r-1)_w$. By Theorem 4.3,

$$D^b(X^{ss}(\mathcal{L}(r-1))) \simeq \mathbf{G}(r-1)_w = \mathbf{G}(r+1)_w \simeq D^b(X^{ss}(\mathcal{L}(r+1)))$$

To summarize we have

Proposition 4.10. *Let $\mathcal{L}$ be a critical linearization of $X/\mathbb{G}_m$ and assume that Z_i is the only critical set for which $\mu_i = 0$. Let a be the weight of $\omega_{X/Z}$ and let $\epsilon > 0$ be a small rational number. Then*

1). *If $a < 0$, there exists a fully faithful embedding*

$$D^b(X^{ss}(\mathcal{L}(\epsilon))) \subseteq D^b(X^{ss}(\mathcal{L}(-\epsilon)))$$

2). *If $a = 0$, there exists an equivalence*

$$D^b(X^{ss}(\mathcal{L}(\epsilon))) \simeq D^b(X^{ss}(\mathcal{L}(-\epsilon)));$$

3). *If $a > 0$, there exists a fully faithful embedding*

$$D^b(X^{ss}(\mathcal{L}(\epsilon))) \supseteq D^b(X^{ss}(\mathcal{L}(-\epsilon)))$$

Now, we apply this proposition for Atiyah flop:

Example 4.11. Take $X = \mathbb{A}^4$ with $\mathbb{G}_m$ action of weights $+1, +1, -1, -1$. Then $Z = \{0\} \subset \mathbb{A}^4$. So

$$\mathrm{wt}_\lambda \det(T_0 X) = \mathrm{wt}_\lambda \det(\mathbb{A}^4) = 0$$

Now, we apply the proposition above:

$$D^b(\mathrm{Tot}(\mathrm{O}_{\mathbb{P}^1}(-1)^{\oplus 2})) = D^b(X^{ss}(+)) \simeq D^b(X^{ss}(-)) = D^b(\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}))$$

Recall that $\mathbb{A}^4//\mathbb{G}_m = \mathrm{Spec}(\mathbb{C}[x,y,z,w]/(xw-yz)) = V(xw-yz) =: Z$. Then by GIT we have

$$X^{ss}(X(+)) \xrightarrow{\pi_1} Z, \qquad X^{ss}(X(-)) \xrightarrow{\pi_2} Z$$

Although $X^{ss}(+)$ and $X^{ss}(-)$ are isomorphic as varieties, π_1 and π_2 are not isomorphic, that is, $X(+)$ and $X(-)$ are not isomorphic as Z-schemes. Moreover, consider the fiber product

$$\begin{array}{ccc} X^{ss}(+) \times_Z X^{ss}(-) & \xrightarrow{p_1} & X^{ss}(+) \\ \downarrow{\scriptstyle p_2} & & \downarrow{\scriptstyle \pi_1} \\ X^{ss}(-) & \xrightarrow{\pi_2} & Z \end{array}$$

$X^{ss}(+) \times_Z X^{ss}(-) \cong \mathrm{Bl}_{\mathbb{P}^1}\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2})$ and the equivalence of the derived categories is given $p_2^* p_{1*} : D^b(X^{ss}(-)) \longrightarrow D^b(X^{ss}(+))$ which agree the Atiyah flop given in [Huy06].

5 Matrix Factorization

Let X be a smooth variety and let $w : X \longrightarrow k$ be a rational function. Set $Y = \{w = 0\} \subset X$. We want to understand the coherent sheaves and singularities of Y. We know that when a variety is smooth, every coherent sheaf admits a finite locally free resolution [Huy06, Section 3.2]. However, for $A = k[x]/(x^2)$, consider $k(x) = A/(x)$. It admits an infinite free resolution

$$\cdots \xrightarrow{x} A \xrightarrow{x} A \xrightarrow{x} A \xrightarrow{x} k(x) \longrightarrow 0$$

However, its resolution is period.

Eisenbud [Eis80] discovered that over $A = R/(W)$, minimal free resolutions essentially become 2-period, that is,

$$\cdots \xrightarrow{\phi} P_1 \xrightarrow{\varphi} P_0 \xrightarrow{\phi} P_1 \xrightarrow{\varphi} P_0$$

with $\phi \circ \psi = W \cdot \mathrm{id}$ and $\psi \circ \phi = W \cdot \mathrm{id}$, which leads to the definition of matrix factorization.

5.1 Derived Category of Singularities

We start with the definition of Verdier quotient.

Definition 5.1. A family of morphism Σ in a category is called a multiplicative system if it satisfies

1). All identical morphisms $\mathrm{id}_X \in \Sigma$;

2). The composition of two elements of Σ belongs to Σ;

3). Any diagram $Y \xrightarrow{f} X \xleftarrow{u} X'$ with $s \in \Sigma$ can be completed to a commutative square

$$\begin{array}{ccc} Y' & \xrightarrow{u'} & Y \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{u} & X \end{array}$$

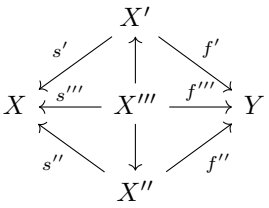
with $u' \in \Sigma$;

4). For any two morphisms f, g, the existence of $s \in \Sigma$ with $f \circ s = g \circ s$ is equivalent to the existence of $t \in \Sigma$ with $t \circ f = t \circ g$.

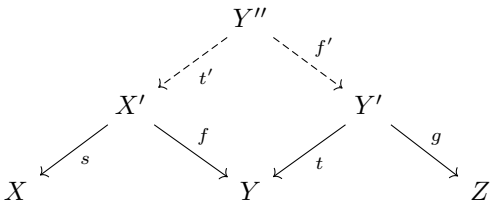
Now, let's define $\mathcal{C}(\sum^{-1})$. Define the objects in $\mathcal{C}(\sum^{-1})$ to be the objects in $\mathcal{C}$. For any $X, Y \in \mathcal{C}(\Sigma^{-1})$, $\mathrm{Mor}(X, Y)$ contains pairs (s, f), denote by $f \circ s^{-1}$, in $\mathcal{C}$ of the form

$$X \xleftarrow{s} X' \xrightarrow{f} Y$$

where $f \circ s^{-1} \sim f' \circ s'^{-1}$ if and only if there is a commutative diagram:



with some $s''' \in \Sigma$. The composition of morphisms $f \circ t^{-1}$ and $g \circ s^{-1}$ is a morphism defined by



that is, $(g \circ f') \circ (s \circ t')^{-1}$. One can check that the composition of $f \circ s^{-1}$ and $g \circ t^{-1}$ are well-defined. There always exists a quotient functor

$$\begin{aligned} Q : \mathcal{C} &\longrightarrow \mathcal{C}(\Sigma^{-1}) \\ X &\longmapsto X \\ f &\longmapsto (f, 1) \end{aligned}$$

Remark. This is a general notation for the definition of morphims in the derived category. See [Huy06, P.32].

Let $\mathcal{C}$ be a triangulated category and $\mathcal{N} \subset \mathcal{C}$ be a full triangulated subcategory of $\mathcal{C}$. Denote by $\Sigma(\mathcal{N})$ a class of morphisms s in $\mathcal{C}$ embedding into an exact triangle

$$X \xrightarrow{s} Y \to N \to X[1]$$

with $N \in \mathcal{N}$. Then we define $\mathcal{C}/\mathcal{N} := \mathcal{C}[\Sigma(\mathcal{N})^{-1}]$.

Example 5.2. Consider the homotopy category $K(X)$ of $\mathrm{Coh}(X)$. Let $K_{ac}(X)$ be the full triangulated subcategory of $K(X)$ containing all acyclic chains. Then

$$D(X) = K(X)/K_{ac}(X)$$

See [Huy06, P.32].

Lemma 5.3. *The followings hold:*

1). *The category $\mathcal{C}/\mathcal{N}$ becomes a triangulated category by taking for distinguished sequence such that are isomorphic to the image of distinguished sequence in $\mathcal{C}$;*

2). *The quotient functor $Q : \mathcal{C} \longrightarrow \mathcal{C}/\mathcal{N}$ annihilates $\mathcal{N}$;*

3). *Any exact functor $F : \mathcal{C} \longrightarrow \mathcal{D}$ of triangulated categories such that $F(X) \cong 0$ for any $X \in \mathcal{N}$ factors uniquely through Q.*

Directly form 2), we have

Lemma 5.4. *Take $f : X \longrightarrow Y \in \mathcal{C}$, if* $\mathrm{Cone}(f) \in \mathcal{C}$, *then*

$$Q(f) : Q(X) \longrightarrow Q(Y)$$

is an isomorphism in $\mathcal{C}/\mathcal{N}$.

Lemma 5.5. *Let $\mathcal{N}$ and $\mathcal{M}$ be full triangulated subcategories of $\mathcal{C}$ and $\mathcal{D}$, respectively. Let $F : \mathcal{C} \longrightarrow \mathcal{D}$ and $G : \mathcal{D} \longrightarrow \mathcal{C}$ be adjoint pair of exact functors such that $F(\mathcal{M}) \subset \mathcal{N}$. Then they induce functors:*

$$\overline{F} : \mathcal{C}/\mathcal{N} \to \mathcal{D}/\mathcal{M}, \qquad \overline{G} : \mathcal{D}/\mathcal{M} \to \mathcal{C}/\mathcal{N}$$

which are adjoint.

Assume that X is separated, Noetherian and of finite Krull dimension and the category of coherent sheaves $\mathrm{Coh}(X)$ has enough locally free sheaves, which implies

Lemma 5.6. *For any bounded above complex of quasi-coherent sheaves $C^\bullet$ on X, there is a bounded above complex of locally free sheaves $P^\bullet$ and a quasi-isomorphism of complex $P^\bullet \xrightarrow{\sim} C^\bullet$.*

Moreover, if $C^\bullet \in D^b(X)$, there is a bounded above complex of locally free sheaves of finite rank $P^\bullet$ with a quasi-isomorphism $P^\bullet \xrightarrow{\sim} C^\bullet$.

Given $C^\bullet \in D(\mathrm{QCoh}(X))$, there are two different truncation functors: The good truncation:

$$\tau^{\geq k} C^\bullet = [\cdots \to 0 \to \mathrm{im}\ dC^{k-1} \to C^k \to \cdots]$$

There is a quotient $C^\bullet \twoheadrightarrow \tau^{\geq k} C^\bullet$ which induces an isomorphism on the $\mathcal{H}^n(C^\bullet)$ and $\mathcal{H}^n(\tau^{\geq k} C^\bullet)$ for all $n \geq k$. The other truncation functor is brutal truncation:

$$\sigma^{\geq k} C^\bullet := [\cdots \to 0 \to C^k \to C^{k+1} \to \cdots]$$

Definition 5.7. A bounded complex of coherent sheaves will be called a perfert if it is quasi-isomorphic to a complex of locally free sheaves in $D^b(X)$.

Example 5.8. For $x \in X$, we have

$$[0 \to k(x) \hookrightarrow k(x) \oplus \mathcal{O}_X \twoheadrightarrow 0] \xrightarrow{\sim} [0 \to 0 \to \mathcal{O}_X \to 0]$$

this $[0 \to k(x) \hookrightarrow k(x) \oplus \mathcal{O}_X \twoheadrightarrow 0]$ is perfect.

Recall that for a smooth scheme X, any coherent sheaf $\mathcal{F}$ admits a bounded resolution of locally free sheaves [Huy06, Prop 3.18]. Thus, $\mathrm{Perf}(X) = D^b(X)$, which motivates us to define the triangulated category of singularities of X.

Definition 5.9. We define the triangulated category of $D_{sg}(X)$ as the quotient of the triangulated category $D^b(X)$ by the full triangulated category $\mathrm{Perf}(X)$ and call it the triangulated category of singularities of X.

Lemma 5.10. *Let $\mathcal{F}$ be a coherent sheaf on X such that for any point $x \in X_{sg}$, it is locally free in some neighbourhood of x. Then it belongs to* $\mathrm{Perf}(X)$.

Proof. By Lemma 5.6, there is a bounded above complex $Q^\bullet$ of vector bundles and quasi-isomorphism $Q^\bullet \longrightarrow \mathcal{F}$. Consider a good truncation, $\tau^{\geq k}Q^\bullet$ for sufficiently small k. Then there is a morphism $r : \tau^{\geq k}Q^\bullet \longrightarrow \mathcal{F}$ that is also a quasi-isomorphism. It suffices to show $\tau^{\geq k}$ is a complex of locally free sheaf, more precisely, to show that the leftmost term $d(\mathrm{im}C^{k-1})$ is locally free.

For any $x \in X_{sg}$, the leftmost term is locally free in some neighbourhood of x because $\mathcal{F}$ is locally free there. If $x \in X_{sg}$, there is a neighborhood U of x which is regular. Hence, the leftmost term is locally free on U, under the assumption $-k > \dim X$, by Syzygy theorem. □

Remark. About Syzygy theorem

Proposition 5.11. *For a scheme X as above, and let $j : U \hookrightarrow X$ be an open embedding such that $X_{sg} \subset U$. Then the functor*

$$j^* : D_{sg}(X) \longrightarrow D_{sg}(U)$$

is an equivalence.

Proof. For any object $A^\bullet \in D^b(X)$, we have $\mu_A : A^\bullet \to j_*j^*A^\bullet$. Since $j : U \longrightarrow X$ is an open immersion, j_* and j^* are exact. Thus, $\mathrm{Cone}(\mu_A)$ is an object whose supported belongs to $X \setminus U$, doesn't intersect X_{sg}, since $j^*\mathrm{Cone}(\mu_A) \cong 0$ in $D^b(U)$. By Lemma 5.10, $\mathrm{Cone}(\mu_A) \in \mathrm{Perf}(X)$. Thus, $Q(A) = Q(j_*j^*A^\bullet)$, that is, $A \cong j_*j^*A$ in $D_{sg}(X)$ by Lemma 5.4. By Lemma 5.5 and $j^*j_* = \mathrm{id}$, $A \cong j_*j^*A$ in $D_{sg}(X)$ implies

$$\mathrm{Hom}_{D_{sg}(X)}(A, B) = \mathrm{Hom}_{D_{sg}(X)}(j_*j^*A, j_*j^*B) = \mathrm{Hom}_{D_{sg}(U)}(j^*A, j^*B)$$

So j^* is fully faithful.

Now, since for any $\mathcal{F}_U \in \mathrm{Coh}(U)$, there exists $\mathcal{F} \in \mathrm{Coh}(X)$ such that $j^*\mathcal{F} = \mathcal{F}|_U$ by [Har77, Exe II.5.18]. j^* is essentially surjective. □

5.2 Matrix Factorization

Let X be a smooth quasi-isomorphism variety over $\mathbb{C}$ and $w : X \longrightarrow \mathbb{A}^1$, called superpotential.

Definition 5.12. A matrix factorization is a pair $\mathbf{E} = (E_0, E_1)$ of vector bundles E_0, E_1 and

$$E_0 \xrightarrow{d_0} E_1, E_1 \xrightarrow{d_1} E_0$$

such that $d_0 \circ d_1 = W\mathrm{id}_{E_1}$ and $d_1 \circ d_0 = W\mathrm{id}_{E_0}$.

Remark. This is equivalent to giving a 2-period complex

$$\cdots \to E_0 \xrightarrow{d_0} E_1 \xrightarrow{d_1} E_0 \xrightarrow{d_0} E_1 \xrightarrow{d_1} E_0 \to \cdots$$

such that $d_0 \circ d_1 = W\mathrm{id}$ and $d_1 \circ d_0 = W\mathrm{id}$.

Definition 5.13. A morphism between $\mathbf{E}, \mathbf{F}$ are morphism $\mathbf{f} = (f_0, f_1)$ such that

$$\begin{array}{ccccc} E_0 & \xrightarrow{d_0^E} & E_1 & \xrightarrow{d_1^E} & E_0 \\ \downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0} \\ F_0 & \xrightarrow{d_0^F} & F_1 & \xrightarrow{d_1^F} & F_0 \end{array}$$

Definition 5.14. We say $\mathbf{f}$ and $\mathbf{g} \in \mathrm{Mor}(\mathbf{E}, \mathbf{F})$ are homotopic is there exists a $\mathbf{h}$ with $h_0 : E_0 \longrightarrow E_1, h_1 : E_1 \to E_0$ such that

$$\mathbf{f} - \mathbf{g} = (d_0^F \circ h_0 + h_1 \circ d_0^E, d_1^F \circ h_1 + h_0 \circ d_1^E)$$

Definition 5.15. The derived category of matrix factorizations of w, denoted by $\mathrm{MF}(X, w)$, is a triangulated category whose objects are matrix factorizations and morphisms are morphisms between MFs module null-homotopic morphisms such that for any $\mathbf{E}$

$$\mathbf{E}[1]_0 = E_1, \mathbf{E}[1]_1 = E_0, d_{\mathbf{E}[1]} = -d_{\mathbf{E}}$$

Let $X_0 = W^{-1}(0)$ and let $\mathbf{E}$ be a MF of W, that is

$$d_0 \circ d_1 = d_1 \circ d_0 = W\mathrm{id}$$

On X_0, $W = 0$, so we get

$$d_0|_{X_0} \circ d_1|_{X_0} = 0, \quad d_1|_{X_0} \circ d_0|_{X_0} = 0$$

Hence, restricting to X_0 gives a 2-periodic complex

$$\cdots \to E_1|_{X_0} \xrightarrow{d_1|_{X_0}} E_0|_{X_0} \xrightarrow{d_0|_{X_0}} E_1|_{X_0} \to \cdots$$

Away from X_0, W is invertible, hence d_0, d_1 are isomorphisms. With this, we can show that the restriction of $\mathbf{E}$ gives an exact sequence over X. So we can define a functor

$$\begin{aligned} \mathbf{\Phi} : MF(X, w) &\longrightarrow \mathrm{D}_{sg}(X) \\ \mathbf{E} &\longmapsto \mathrm{coker}(d_1 : E_1 \longrightarrow E_0) \end{aligned}$$

And we can prove that

Theorem 5.16. *Let X be a smooth variety and $X_0 = w^{-1}(0)$ is a Cartier divisor. Then there exists an equivalence*

$$MF(X, w) \simeq D_{sg}(X_0)$$

5.3 Equivariant and Graded Matrix Factorization

Let (X) be a smooth quasi-projective variety equipped with an action of a linear algebraic group (G) over $\mathbb{C}$. Let $w \in \Gamma(X, \mathcal{O}_X)$ be a regular function satisfying $w(g \cdot x) = \chi(g), w(x)$ for some character $\chi : G \to \mathbb{G}_m$.

For example, one may endow the trivial line bundle $\mathcal{O}_X = X \times \mathbb{A}^1$ with the (G)-linearization determined by χ, $g \cdot (x, z) = (g \cdot x, \chi(g)z)$. With respect to this linearization, w may be regarded as a G-invariant section of $\mathcal{O}_X$. At this case, we call w is G-semi-invariant.

Definition 5.17. A gauged LG-model, is the quadruple $(X, G, \mathcal{L}, w)$ as above, commonly denoted by the pair $([X/G], w)$.

Given a G-invariant bundle E, $f : E \longrightarrow F$ define

$$F(n) := E \otimes \mathcal{L}^n, \qquad f(n) := f \otimes \mathrm{id}_{\mathcal{L}^n}$$

Definition 5.18. A (graded) factorization of $([X/G], w)$ consists of a pair of G-equivariant vector bundles E_0, E_1 and a pair of G-equivariant morphisms

$$E_0 \xrightarrow{d_0} E_1, E_1 \xrightarrow{d_1} E_0(1)$$

such that $d_1 \circ d_0 = w\mathrm{id}_{E_0}$ and $d_0 \circ d_1 = w\mathrm{id}_{E_0}$.

This is equivariant to giving a 2-periodic complex:

$$\cdots \xrightarrow{d_1(-1)} E_0 \xrightarrow{d_0} E_1 \xrightarrow{d_1} E_0(1) \xrightarrow{d_0(1)} E_1(1) \to \cdots$$

Definition 5.19. A morphism between $\mathbf{E}, \mathbf{F}$ are morphism $\mathbf{f} = (f_0, f_1)$ such that

$$\begin{array}{ccccc} E_0 & \xrightarrow{d_0^E} & E_1 & \xrightarrow{d_1^E} & E_0(1) \\ \downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0(1)} \\ F_0 & \xrightarrow{d_0^F} & F_1 & \xrightarrow{d_1^F} & F_0(1) \end{array}$$

Definition 5.20. We say $\mathbf{f}$ and $\mathbf{g} \in \mathrm{Mor}(\mathbf{E}, \mathbf{F})$ are homotopic is there exists a $\mathbf{h}$ with $h_0 : E_0 \longrightarrow F_1, h_1 : E_1 \to F_0(-1)$ such that

$$\mathbf{f} - \mathbf{g} = (d_0^F \circ h_0 + h_1(1) \circ d_0^E, d_1^F \circ h_1 + h_0 \circ d_1^E)$$

Definition 5.21. The derived category of matrix factorizations of w, denoted by $\mathrm{MF}(X/G, w)$, is a triangulated category whose objects are matrix factorizations and morphisms are morphisms between MFs module null-homotopic morphisms such that for any $\mathbf{E}$

$$\mathbf{E}[1]_0 = E_1, \mathbf{E}[1]_1 = E_0, d_{\mathbf{E}[1]} = -d_{\mathbf{E}}$$

Also, we have the following result:

Theorem 5.22. *There is an equivalence*

$$\mathrm{MF}(X/G, w) \simeq D_{sg}(w^{-1}(0)/G)$$

5.4 Koszul Equivalence

Now, let's X be a smooth variety equipped with an action of G. Let E be a locally free G-equivariant sheaf on X and let $s \in \Gamma(X, E)$ be a G-invariant regular section. Let $Y = V(s)$, the vanishing locus of s.

On the geometric bundle $\mathrm{Tot}(E^\vee)$, s induces a G-invariant regular function

$$w : \mathrm{Tot}(E^\vee) \longrightarrow k, (x, v) \longmapsto \langle \tau, p^* s \rangle_{(x,v)}$$

where τ be the tautological section over $\mathrm{Tot}(E^\vee)$ and p is the natural projection form $\mathrm{Tot}(E^\vee) \longrightarrow X$.

Example 5.23. Let $X = \mathbb{A}^n$, $E = \mathcal{O}_X^{\oplus t}$ and $s = (f_1, \cdots, f_t)$. Then

$$w : \mathbb{A}^{n+k} \longrightarrow \mathbb{A}^1$$
$$(x_1, \cdots, x_n, t_1, \cdots, t_k) \longmapsto \sum f_i(x_1, x_2, \cdots, x_n)t_i$$

Let $\mathbb{G}_m$ act on $\mathrm{Tot}(E^\vee)$ by scaling the fibers. Consider the following diagram

$$\begin{array}{ccccc} \mathrm{Tot}(E^\vee)|_Y & \xhookrightarrow{i} & \mathrm{Tot}(E^\vee) & \xrightarrow{w} & \mathbb{A}^1 \\ \downarrow{\scriptstyle \pi} & & \downarrow & & \\ Y & \hookrightarrow & X & & \end{array}$$

Theorem 5.24. *Take $s \in \mathrm{H}^0(X, E)$ as above. There exists an equivalence*

$$D^b(s^{-1}(0)/G) \simeq \mathrm{MF}(\mathrm{Tot}_X E^\vee/G \times \mathbb{G}_m, w)$$

Ideal of the proof. Note that $w|_{\mathrm{Tot}(\mathrm{E}^\vee)}|_Y = 0$. Consider $i_*\pi^* : \mathrm{Coh}(Y) \longrightarrow \mathrm{Coh}(w^{-1}(0))$ which can be extended to

$$i_*\pi^* : D^b(Y/G) \to D^b(w^{-1}(0)/G) \xrightarrow{Q} D^b(w^{-1}(0)/G)/\mathrm{Perf}(w^{-1}(0)/G)$$

Thus, we get

$$D^b(Y/G) \longrightarrow D^b_{sg}(w^{-1}(0)/G \times \mathbb{G}_m) \simeq \mathrm{MF}(\mathrm{Tot}E^\vee/G \times \mathbb{G}_m, w)$$

In short we have

$$\begin{aligned} D^b(s^{-1}/G) &\longrightarrow \mathrm{MF}(\mathrm{Tot}(\mathrm{E}^\vee)/\mathrm{G} \times \mathbb{G}_\mathrm{m}, w) \\ \mathcal{F} &\longmapsto (0 \longrightarrow i_*\pi^*\mathcal{F}), (i_*\pi^*\mathcal{F} \longrightarrow 0) \end{aligned}$$

where $i_*\pi^*\mathcal{F}$ has support contained in $w^{-1}(0)$. □

6 Nakajima Varieties

Stability conditions on quiver representations give rise to a rich class of moduli spaces, including Hilbert schemes of points, Nakajima quiver varieties, and many important hyperkähler varieties.

As the stability parameter varies, these moduli spaces undergo wall-crossing phenomena, like subsection 4.2. Different choices of stability may produce distinct symplectic resolutions of the same singular affine variety, giving rise to classical birational transformations such as the Atiyah flop and the Mukai flop. These examples provide some of the fundamental motivations for the study of the D-equivalence conjecture, which predicts that K-equivalent smooth varieties should have equivalent derived categories.

In this section, we briefly review the basic constructions and several fundamental examples of quiver varieties, following Sections 8–11 of [Soi21]. Rather than providing complete proofs, our goal is to collect the main ideas and results that will be used later.

6.1 Semi-stability Of Representation of Quivers

Now, for a quiver $Q = (Q_0, Q_1)$.

Definition 6.1. Let $\theta = (\theta_i) \in \mathbb{Z}^{Q_0}$. A representation of Q with dimension vector $v = (v_i)$ is called θ-semi-stable if

$\theta \cdot \alpha = 0$ and for any subrepresentation $W \subset V$ with dimension vector β, we have $\theta \cdot \beta \leq 0$

We say that V is θ-stable if

$\theta \cdot \alpha = 0$ and for any subrepresentation $W \subset V$ with dimension vector β, we have $\theta \cdot \beta < 0$

Define

$$\begin{aligned}\chi_\theta : \mathrm{GL}(\alpha) &\longrightarrow \mathbb{C}^* \\ (g_i) &\longmapsto (\det(g_i)^{\theta_i})\end{aligned}$$

For $\mathrm{GL}(\alpha)$ acts on $\mathrm{Rep}(Q,\alpha)$, consider the linearisation along χ_θ. By Hilbert-Mumford criterion:

Theorem 6.2. *Let Q be a quiver and let $\theta = (\theta_i) \in \mathbb{Z}^{Q_0}$. Let $\alpha \in \mathbb{Z}_{\geq 0}^{Q_0}$ be a dimension vector such that $\theta \cdot \alpha = 0$. Any $V \in \mathrm{Rep}(Q,\alpha)$ is χ_θ-semi-stable if and only V is θ-semi-stable.*

Proof. [Soi21, Theorem 8.3.3]. □

6.2 Framing and Double Quivers

Let $Q = (Q_0, Q_1)$ be a quiver.

Definition 6.3 (Framing)**.** Let $Q = (Q_0, Q_1)$ be a quiver. A framing of Q is a quiver $Q^{fr} = (Q_0^{fr}, Q_1^{fr})$ such that

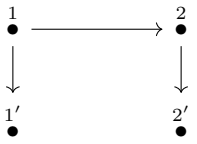
1). $Q_0^{fr} = Q_0 \bigsqcup Q_0'$ together with a bijection between Q_0 and Q_0' via the mapping $i \to i'$;

2). $Q_1^{fr} = Q_1 \bigsqcup \{b\}_{i \in Q_0}$;

3). $t^{fr}(a) = t(a)$ and $s^{fr}(a) = s(a)$ for all $a \in Q_1$;

4). $s^{fr}(b_i) = i$ and $t^{fr}(b_i) = i'$.

Example 6.4. The framing of A_2 is given by



For fixed dimension vector v over Q_0, consider $\mathrm{Rep}(Q^{fr}, v, w)$ with w the dimension vector on Q_0'. We have the following equivalence:

Proposition 6.5. *Let $Q = (Q_0, Q_1)$ be a quiver. Fix a dimension vector v for Q, and let $\hat{Q}$ be the quiver with vertex set $Q_0 \bigsqcup \{\infty\}$ and an arrow set consisting of Q_0 together with w_i additional arrows from each vertex $i \in Q_0$ to ∞. Let the dimension vector $\hat{v}$ be given by $\hat{v}_i = v_i$ for $i \in Q_0$ and $\hat{v}_\infty = 1$. There is a $\mathrm{G}(v)$-equivariant isomorphism $\mathrm{Rep}(Q^{fr}, v, w) \longrightarrow \mathrm{Rep}(\hat{Q}, \hat{v})$*

Proof. The group G(v) acts on $\mathrm{Rep}(\hat{Q}, \hat{v})$ via the embedding $G(v) \hookrightarrow G(\hat{v})$ given by $g = (g_i) \longmapsto (g_i, \mathrm{id})$. Consider the morphism of affine variety

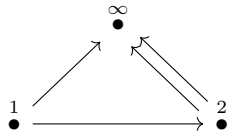
$$\begin{aligned}\mathrm{Rep}(Q^{fr}, v, w) &\longrightarrow \mathrm{Rep}(\hat{Q}, \hat{v}) \\ (\{V_i\}, \{W_i\}, \{f_a\}, \{f_b\}) &\longmapsto (\{V_i\}, \{V_\infty\}, \{f_a\}, \{r_{ij}\})\end{aligned}$$

where $V_\infty = \mathbb{C}$ and r_{ik} is the k-th row of f_{b_i} for $1 \leq k \leq w_i$. The inverse of this map directly given by

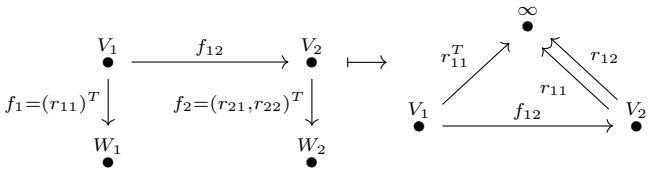
$$(\{V_i\}, \{V_\infty\}, \{f_a\}, \{r_{ik}\}) \longmapsto (\{V_i\}, \{W_i\}, \{f_a\}, \{(r_{i1}, \cdots, r_{iw_i})\})$$

Since $g_{t(a)} f_a g_{s(a)}^{-1} \longmapsto g_{t(a)} f_a g_{s(a)}^{-1}$ and $f_{b_i} g_i^{-1} \longmapsto (r_{ik} g_i^{-1})$, this map is $G(v)$-equivariant. □

Example 6.6. Also, consider A_2, let $v = (2, 3)$ and $w = (1, 2)$. Then the quiver $\hat{A}_2$ is given by



and we have $\mathrm{Rep}(A_2^{fr}, v, w) = \mathrm{Rep}(\hat{A}_2, \hat{v})$. For the $G(v)$-equivariant isomorphism, we have



where r_{ij} are cloume vectors.

Definition 6.7 (Double)**.** Let Q^{op} be the quiver with the same vertex set but with the source and target of each of the arrows revered. The quiver $\overline{Q}$ with the vertex set Q_0 and arrow set $\overline{Q}_1 = Q_1 \bigsqcup Q_1^{op}$ is called the double of Q.

We can see that

$$\mathrm{Rep}(\overline{Q}, v) \cong \mathrm{Rep}(Q, \alpha) \times \mathrm{Rep}(Q^{op}, v)$$

For semi-stability of representation of framing of quivers, under the action of $\mathrm{GL}(v)$, we need to use Proposition 6.5.

Now, let's define the Nakajima varieties. Let Q be q quiver Q and Q^{fr} be its framing. We can identify $T^*\mathrm{Rep}(Q^{fr}, \alpha, \alpha')$ with the representation of the double of the framed quiver representation $\mathrm{Rep}(\overline{Q^{fr}}, v, w)$. Under the identification, the action of the group $G(v)$ on $\mathrm{Rep}(Q^{fr}, v, w)$ naturally lifts to an action on $\mathrm{Rep}(\overline{Q^{fr}}, v, w)$ can be written explicitly by

$$g \cdot (V, W, \{f_a\}, \{f_{a^*}\}, \{f_b\}) = (V, W, \{g_{t(a)} f_a g_{s(a)}^{-1}\}, \{g_{t(a)} f_{a^*} g_{s(a)}^{-1}\}, \{f_b g_{s(b)}^{-1}\}, \{g_{t(b)} f_b\})$$

The corresponding moment map is given by

$$\begin{aligned} \mu : \mathrm{Rep}(\overline{Q^{fr}}, v, w) &\longrightarrow \mathfrak{g}(v) \\ \overline{V^{fr}} &\longmapsto \left(-f_{b_i^*} f_{b_i} + \sum_{\substack{a \in Q_1 \\ t(a)=i}} f_a f_{a^*} - \sum_{\substack{a \in Q_1 \\ s(a)=i}} f_{a^*} f_a \right)_{i \in Q_0} \end{aligned}$$

Definition 6.8. Let $\theta \in \mathbb{Z}^{Q_0}$ and $\lambda \in \mathbb{C}^{Q_0}$. Then algebraic variety

$$\mathfrak{M}_{\theta,\lambda}(v, w) := \mu^{-1}(\lambda) /\!/_{\chi_\theta} G(v)$$

is called a quiver Variety. We denote by $\mathfrak{M}^s_{\theta,\lambda}(v, w)$ the geometric quotient corresponding to the χ_θ-stable points. $\mathfrak{M}^{-1}_{\theta,0}(v, w)$ is called the Nakajima variety.

For the semi-stable condition over quiver variety, we have

Theorem 6.9. *Let $\theta \in \mathbb{Z}^{Q_0}$ and let $\lambda \in \mathbb{C}^{Q_0}$ be such that $\mu^{-1}(\lambda)$ is non-empty. Then representation $\overline{V^{fr}} \in \mu^{-1}(\lambda)$ is χ_θ semi-stable with respect to the action of $G(v)$ if and only if for any subrepresentation $V' \subset \overline{V}$ with dimension vector a' the following conditions hold:*

1). *If* $V_i' \subset \ker f_{b_i}$*, for all* $i \in Q_0$*, then* $\theta \cdot \beta \leq 0$*;*

2). *If* $V_i' \supset \mathrm{Im}(f_{b^*})$ *for all* $i \in I_Q$*, then* $\theta \cdot \beta \leq \theta \cdot \alpha$*.*

The representation $\overline{V^{fr}}$ *is* χ_θ*-stable if the above inequalities are strict with the additional conditions that* W *is non-trivial in the first case and* W *is proper in the second case.*

Proof. [Soi21, Theorem 11.1.2] □

Example 6.10 (Mukai Flop)**.** Consider the framed A_1 quiver

$$A_1^{fr} : \overset{v}{\bullet} \longrightarrow \overset{w}{\bullet}$$

and its doubled quiver

$$\overline{A_1^{fr}} : \overset{v}{\bullet} \rightleftarrows \overset{w}{\bullet}$$

where $\dim W = w$ and $\dim V = v$. Then

$$X := T^*\mathrm{Rep}(A_1^{fr}, v, w) = \mathrm{Hom}(V, W) \oplus \mathrm{Hom}(W, V)$$

and the moment map is

$$\begin{aligned} \mu : T^*\mathrm{Rep}(A_1^{fr}, v, w) &\longrightarrow \mathfrak{gl}(V) \\ (f, f_*) &\longmapsto f_* \circ f \end{aligned}$$

Since the framed vertex is fixed, a subrepresentation is simply a subspace of V. There are two possible stability parameters, namely > 0 and < 0. Suppose $(f, f_*) \in \mu^{-1}(0)$, so that

$$f_* \circ f = 0$$

The subspaces

$$\ker f \subset V, \qquad \mathrm{im}\ f \subset V$$

are subrepresentations. Hence Theorem 6.9 implies

$$\dim(\ker f) \leq 0, \qquad \dim(\mathrm{im}\ f) \leq v.$$

If $\theta > 0$, the first inequality forces $\ker f = 0$ and $\mathrm{im}\ f \leq v$. So f is injective. By Theorem 6.9, we can moreover show that

$$X_0^{ss}(\theta_{>0}) = \{(f, f_*) \mid f \text{ is injective}, f_* \circ f = 0\}.$$

Thus,

$$X^{ss}(\theta_{>0}) \cong \{W' \subset W \mid W' \cong V, f_* : W/W' \longrightarrow W'\}$$

Let $W' = \mathrm{im}\, f$. Then $W' \cong V$, and f_* factors through the quotient $W/W' \longrightarrow V$. Conversely, such data determine (f, f_*). Hence

$$X^{ss}(\theta_{>0}) \cong \{(W', \varphi) \mid W' \subset W,\ \dim W' = v,\ \varphi : W/W' \longrightarrow W'\},$$

which is naturally the total space of the cotangent bundle of the Grassmannian. Therefore

$$\mathfrak{M}_{\theta>0,0}(v, w) \cong T^*\,\mathrm{Gr}(v, W).$$

If $\theta < 0$, the second inequality implies f_* is surjective. Again, $f_* \circ f = 0$ implies $\mathrm{im}\, f \subset \ker f_*$. Thus

$$X^{ss}(\theta_{<0}) = \{(f, f_*) \mid f_* \text{ is surjective},\ \mathrm{im}\, f \subset \ker f_*\}.$$

Writing $W' = \ker f_*$,we obtain $W/W' \cong V$, and f factors through $V \cong W/W' \longrightarrow W'$. Hence

$$X^{ss}(\theta_{<0}) \cong \{(W', \psi) \mid W' \subset W,\ \dim W' = w - v,\ \psi : W/W' \longrightarrow W'\},$$

which identifies with

$$T^* \operatorname{Gr}(w - v, W).$$

Therefore

$$\mathfrak{M}_{\theta<0,0}(v, w) \cong T^* \operatorname{Gr}(w - v, W).$$

When $\theta = 0$, the Nakajima quiver variety is the affine GIT quotient

$$\mathfrak{M}_{0,0}(v, w) = \mu^{-1}(0)//GL(V).$$

Define

$$\begin{aligned} \Phi : \mu^{-1}(0) &\longrightarrow \left\{M \in \operatorname{End}(W) \;\middle|\; M^2 = 0,\ \operatorname{rank} M \le v\right\} =: Z, \\ (f, f_*) &\longmapsto f \circ f_*. \end{aligned}$$

Since $(f \circ f_*)^2 = f(f_* \circ f)f_* = 0$, and $\operatorname{rank}(f \circ f_*) \le \min\{\operatorname{rank} f, \operatorname{rank} f_*\} \le v$, the map is well defined.

Conversely, let $M \in \operatorname{End}(W)$ satisfy $M^2 = 0$ and $\operatorname{rank} M \le v$. Choose an embedding and a projection

$$i : \operatorname{im} M \hookrightarrow V, \qquad p : V \twoheadrightarrow \operatorname{im} M$$

such that$p \circ i = \mathrm{id}_{\operatorname{im} M}$. Define

$$f_* = i \circ M, \qquad f = \iota \circ p$$

where $\iota : \operatorname{im} M \hookrightarrow W$ denotes the inclusion. Then $f \circ f_* = M$, and $f_* \circ f = iMp = 0$ since $\operatorname{im}(Mp) \subset \operatorname{im} M$ and $M^2 = 0$. Thus Φ is surjective.

Moreover, Φ is $GL(V)$-invariant, and one checks that two points of $\mu^{-1}(0)$ have the same image if and only if they lie in the same $GL(V)$-orbit. Consequently,

$$\mathfrak{M}_{0,0}(v, w) \cong Z = \left\{M \in \operatorname{End}(W) \;\middle|\; M^2 = 0,\ \operatorname{rank} M \le v\right\}.$$

By variation of GIT, there are two projective morphisms

$$\pi_+ : \mathfrak{M}_{\theta>0,0}(v, w) \longrightarrow Z, \qquad \pi_- : \mathfrak{M}_{\theta<0,0}(v, w) \longrightarrow Z.$$

Furthermore,

$$\mathfrak{M}_{\theta>0,0}(v, w) \cong T^* \operatorname{Gr}(v, W) \cong T^* \operatorname{Gr}(w - v, W) \cong \mathfrak{M}_{\theta<0,0}(v, w)$$

as abstract varieties, where the middle isomorphism is induced by the canonical duality $\operatorname{Gr}(v, W) \cong \operatorname{Gr}(w - v, W)$. However, the morphisms π_+ and π_- are different. Therefore, $\mathfrak{M}_{\theta>0,0}(v, w)$ and $\mathfrak{M}_{\theta<0,0}(v, w)$ are NOT isomorphic as Z-schemes.

Consider the special case $v = 1$ and $w = n + 1$. Then

$$\mathfrak{M}_{\theta>0,0}(1, n + 1) \cong T^*\mathbb{P}(W), \qquad \mathfrak{M}_{\theta<0,0}(1, n + 1) \cong T^*\mathbb{P}(W^\vee).$$

The resolution π_+ is given by

$$\pi_+ : T^*\mathbb{P}(W) \longrightarrow Z, \qquad (L, \varphi) \longmapsto X : W \twoheadrightarrow W/L \xrightarrow{\varphi} L \hookrightarrow W,$$

where $L \subset W$ is a line and $\varphi \in \operatorname{Hom}(W/L, L)$.

Similarly,

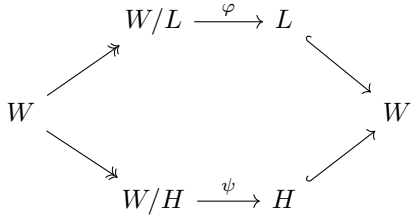
$$\pi_- : T^*\mathbb{P}(W^\vee) \longrightarrow Z, \qquad (H,\psi) \longmapsto X : W \twoheadrightarrow W/H \xrightarrow{\psi} H \hookrightarrow W,$$

where $H \subset W$ is a hyperplane and $\psi \in \mathrm{Hom}(W/H, H)$.

Denote

$$X_0^+ = T^*\mathbb{P}(W), \qquad X_0^- = T^*\mathbb{P}(W^\vee).$$

The fiber product $X_0^+ \times_Z X_0^-$ consists of pairs $\big((L,\varphi),(H,\psi)\big)$ which determine the same nilpotent endomorphism $X \in Z$, that is, the following diagram commutes



If $X \neq 0$, then $L = \mathrm{im}(X), H = \ker(X)$, and the relation $X^2 = 0$ implies $L \subset H$. Over the origin $0 \in Z$, we have $\psi = 0$ and $\varphi = 0$ so that $(\pi_+ \times \pi_-)^{-1}(0) \cong \mathbb{P}(W) \times \mathbb{P}(W^\vee)$.

Consider the projection $p_+ : X_0^+ \times_Z X_0^- \longrightarrow X_0^+$. For $(L,\varphi) \in X_0^+$, the fiber of p_+ is

$$p_+^{-1}(L,\varphi) = \begin{cases} \{\ker(X)\}, & \varphi \neq 0, \\ \{H \subset W \mid L \subset H\} \cong \mathbb{P}((W/L)^\vee), & \varphi = 0. \end{cases}$$

Hence p_+ is an isomorphism away from the zero section and replaces each point of the zero section $\mathbb{P}(W) \subset T^*\mathbb{P}(W)$ by the projective space $\mathbb{P}((W/L)^\vee)$. Therefore

$$X_0^+ \times_Z X_0^- \cong \mathrm{Bl}_{\mathbb{P}(W)}(T^*\mathbb{P}(W))\,.$$

Similarly,

$$X_0^+ \times_Z X_0^- \cong \mathrm{Bl}_{\mathbb{P}(W^\vee)}(T^*\mathbb{P}(W^\vee))\,.$$

Consequently, the Mukai flop is obtained by blowing up the zero section of $T^*\mathbb{P}(W)$ and then blowing down the exceptional divisor in the opposite ruling.

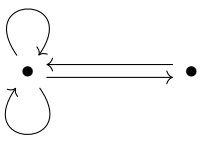
Example 6.11 (Hilbert Scheme)**.** Consider the framed Jordan quiver

J^{fr} :



and its doubled quiver

$\overline{J^{fr}}$:



Let V and W be vector spaces with $\dim V = v$ and $\dim W = w$. Then

$$X := T^*\mathrm{Rep}(J^{\mathrm{fr}}, v, w) = \mathrm{End}(V) \oplus \mathrm{End}(V) \oplus \mathrm{Hom}(W,V) \oplus \mathrm{Hom}(V,W),$$

and a point is written as (B, B_*, f_*, f).

Throughout this section we only consider the case $v = n$ and $w = 1$. The moment map is given by

$$\mu(B, B_*, f_*, f) = [B, B_*] + f_* \circ f \in \mathfrak{gl}(V).$$

Hence $(B, B_*, f_*, f) \in \mu^{-1}(0)$ if and only if $[B, B_*] = -f_* \circ f$. Since W is one-dimensional, the operator $f_* \circ f$ has rank at most one. Therefore $[B, B_*]$ has rank at most one, and we may apply the following lemma.

Lemma 6.12. *Let $A, B \in \mathrm{End}(\mathbb{C}^n)$. If $[A, B]$ has rank at most 1, then there exists a basis of $\mathbb{C}^n$ in which both A and B are upper triangular.*

Proof. See [Soi21, Lemma 11.2.1]. □

Assume that $\theta < 0$. By Theorem 6.9, a point (B, B_*, f_*, f) is θ-semistable if and only if there is no proper subrepresentation of (V, B, B_*) containing $\mathrm{im}\, f_*$. Let $v = f_*(1) \in V$. The subspace generated by all vectors of the form

$$p(B, B_*)v, \qquad p(x, y) \in \mathbb{C}[x, y],$$

where variables in $\mathbb{C}[x, y]$ are ordered so that power of x appear before y, is a non-zero subrepresentation of (V, B, B_*) containing $\mathrm{im}\, f_*$. By semistability, it must coincide with V. Thus v is a cyclic vector for the pair (B, B_*).

Choose a basis in which both B and B_* are upper triangular. Then, for every polynomial $p(x, y) \in \mathbb{C}[x, y]$, the operator $p(B, B_*)$ is also upper triangular. Consequently, $p(B, B_*)[B, B_*]$ is strictly upper triangular, and hence has trace zero. Using the moment map equation, $[B, B_*] = -f_* \circ f$, we obtain $\mathrm{tr}(p(B, B_*)\, f_* \circ f) = 0$. By symmetry of the trace, $\mathrm{tr}(f \circ p(B, B_*) \circ f_*) = 0$. Since $W \cong \mathbb{C}$ is one-dimensional, $f \circ p(B, B_*) \circ f_* \in \mathrm{End}(W) \cong \mathbb{C}$, so vanishing of its trace implies $f \circ p(B, B_*) \circ f_* = 0$ for every polynomial p. As $f_*(1)$ is a cyclic vector, the vectors $p(B, B_*)f_*(1), p \in \mathbb{C}[x, y]$, span V. Hence f vanishes on a spanning set of V, and therefore $f = 0$.

Thus every θ-semistable point is represented by a triple (B, B_*, f_*). Such a triple determines an ideal

$$I_{(B,B_*,f_*)} = \{p \in \mathbb{C}[x, y] \mid p(B, B_*)f_*(1) = 0\}.$$

Moreover, the map

$$\begin{aligned} \mathbb{C}[x, y]/I_{(B,B_*,f_*)} &\longrightarrow V, \\ p(x, y) &\longmapsto p(B, B_*)f_*(1), \end{aligned}$$

is an isomorphism, since $f_*(1)$ is cyclic. In particular, $\dim\big(\mathbb{C}[x, y]/I_{(B,B_*,f_*)}\big) = n$.

Finally, the ideal $I_{(B,B_*,f_*)}$ depends only on the $\mathrm{GL}(V)$-orbit of (B, B_*, f_*). Indeed, if $(B', B'_*, f'_*) = (gBg^{-1}, gB_*g^{-1}, gf_*)$, for some $g \in \mathrm{GL}(V)$, then $I_{(B',B'_*,f'_*)} = I_{(B,B_*,f_*)}$. Hence we obtain a well-defined morphism

$$\mathfrak{M}_{\theta<0}(n, 1) \longrightarrow \left\{I \subset \mathbb{C}[x, y] \;\middle|\; \dim\big(\mathbb{C}[x, y]/I\big) = n\right\}.$$

The target carries a natural algebraic variety structure, namely the Hilbert scheme $\mathrm{Hilb}^n(\mathbb{A}^2)$, which parametrizes zero-dimensional subschemes of $\mathbb{A}^2$ of length n.

Conversely, let $I \in \mathrm{Hilb}^n(\mathbb{A}^2)$. Then the quotient $R := \mathbb{C}[x, y]/I$ is an n-dimensional $\mathbb{C}$-algebra. Choosing an isomorphism $R \cong \mathbb{C}^n$, the multiplication action of $\mathbb{C}[x, y]$ on R gives rise to two commuting endomorphisms $B, B_* \in \mathrm{End}(\mathbb{C}^n)$. Moreover, the image of $1 \in \mathbb{C}[x, y]$ defines a cyclic vector $f_*(1) \in \mathbb{C}^n$, which generates R as a $\mathbb{C}[x, y]$-module. Thus we obtain a triple (B, B_*, f_*) satisfying the $\theta_{<0}$-stability condition. Since B and B_* commute, we have

$$[B, B_*] = 0,$$

and hence $(B, B_*, f_*, 0) \in \mu^{-1}(0)$. Therefore this determines a point of

$$\mathfrak{M}_{\theta_{<0},0}(n,1).$$

The choice of an isomorphism $R \cong \mathbb{C}^n$ amounts to choosing a basis of R, so the resulting quadruple is determined only up to the natural $\mathrm{GL}(n)$-action. Hence we obtain a morphism

$$\mathrm{Hilb}^n(\mathbb{A}^2) \longrightarrow \mathfrak{M}_{\theta_{<0},0}(n,1),$$

which is inverse to the morphism constructed above. Consequently,

$$\mathfrak{M}_{\theta_{<0},0}(n,1) \cong \mathrm{Hilb}^n(\mathbb{A}^2)$$

as algebraic varieties.

When $\theta > 0$, the semistability condition changes. In this case, (B, B_*, f_*, f) is θ-semistable if and only if there is no nonzero B, B_*-invariant subspace of V contained in $\ker f$. Equivalently,

$$\bigcap_{p \in \mathbb{C}[x,y]} \ker\bigl(f \circ p(B, B_*)\bigr) = 0.$$

Indeed, the above intersection is a B, B_*-invariant subspace of V contained in $\ker f$, and hence must be trivial by semistability. This shows that f is a cyclic vector for the dual representation on V^*, that is, $V^* = \mathrm{span}\,\{f \circ p(B, B_*) \mid p \in \mathbb{C}[x,y]\}$. As in the case $\theta < 0$, the trace argument implies that$f_* = 0$. Consequently, the moment map equation reduces to $[B, B_*] = 0$. Now consider the homomorphism

$$\Phi : \mathbb{C}[x,y] \longrightarrow V^*, \qquad p \longmapsto f \circ p(B, B_*),$$

and define $I_{(B,B_*,f)} := \ker \Phi$. Since f is cyclic for the dual representation, Φ is surjective and there is an induced isomorphism $\mathbb{C}[x,y]/I_{(B,B_*,f)} \cong V^*$. Thus $I_{(B,B_*,f)}$ defines a point of $\mathrm{Hilb}^n(\mathbb{A}^2)$, yielding a morphism

$$\mathfrak{M}_{\theta>0,0}(n,1) \longrightarrow \mathrm{Hilb}^n(\mathbb{A}^2).$$

The inverse morphism is constructed exactly as in the case $\theta < 0$ by duality, and therefore

$$\mathfrak{M}_{\theta>0,0}(n,1) \cong \mathrm{Hilb}^n(\mathbb{A}^2).$$

When $\theta = 0$, the Nakajima variety $\mathfrak{M}_{0,0}(n,1)$ is the affine GIT quotient $X_0 /\!/ GL(V)$, which parametrizes the closed $GL(V)$-orbits in $X_0 = \mu^{-1}(0)$. Hence it suffices to study points whose orbits are closed.

Let $(B, B_*, f, f_*) \in X_0$ lie in a closed orbit. Define $W = \mathrm{span}_{\mathbb{C}}\{\, p(B, B_*) f_*(1) \mid p \in \mathbb{C}[x,y] \,\} \subset V$. By construction, W is invariant under both B and B_*, and clearly $\mathrm{im}(f_*) \subset W$. Using the same trace argument as before, one obtains $f(W) = 0$, so that $W \subset \ker(f)$.

Choose a basis of V adapted to the decomposition $V = W \oplus W'$, where W' is any complementary subspace. Relative to this basis,

$$B = \begin{pmatrix} B_{11} & B_{12} \\ 0 & B_{22} \end{pmatrix}, \qquad B_* = \begin{pmatrix} B_{*11} & B_{*12} \\ 0 & B_{*22} \end{pmatrix}, \qquad f_* = \begin{pmatrix} C \\ 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & D \end{pmatrix}.$$

Now consider the one-parameter subgroup

$$\lambda(t) = \begin{pmatrix} I_W & 0 \\ 0 & tI_{W'} \end{pmatrix}.$$

Its action is given by

$$\lambda(t)B\lambda(t)^{-1} = \begin{pmatrix} B_{11} & t^{-1}B_{12} \\ 0 & B_{22} \end{pmatrix}, \qquad \lambda(t)B_*\lambda(t)^{-1} = \begin{pmatrix} B_{*11} & t^{-1}B_{*12} \\ 0 & B_{*22} \end{pmatrix},$$

$$\lambda(t)f_* = \begin{pmatrix} C \\ 0 \end{pmatrix}, \qquad f\lambda(t)^{-1} = \begin{pmatrix} 0 & t^{-1}D \end{pmatrix}.$$

Taking the limit as $t \to \infty$, we obtain

$$\left(\begin{pmatrix} B_{11} & 0 \\ 0 & B_{22} \end{pmatrix}, \begin{pmatrix} B_{*11} & 0 \\ 0 & B_{*22} \end{pmatrix}, \begin{pmatrix} C \\ 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \end{pmatrix} \right).$$

Since the orbit of (B, B_*, f, f_*) is closed, the Hilbert–Mumford criterion implies that this limit belongs to the same $GL(V)$-orbit. Next consider the scalar one-parameter subgroup $\lambda(t) = tI_V$. Since $\lambda(t) \cdot (B, B_*, f_*, f) = (B, B_*, tf_*, t^{-1}f)$, taking the limit as $t \to 0$ yields $(B, B_*, 0, 0)$. Again, the closedness of the orbit implies that this point lies in the same orbit. Consequently, every closed orbit admits a representative of the form $(B, B_*, 0, 0)$, where $[B, B_*] = 0$.

By the simultaneous triangularization lemma, there exists a basis in which both B and B_* are upper triangular. Consider the one-parameter subgroup $\lambda(t) = \mathrm{diag}(t^n, t^{n-1}, \ldots, t)$. Then

$$\lim_{t\to 0} \lambda(t)B\lambda(t)^{-1} \quad \text{and} \quad \lim_{t\to 0} \lambda(t)B_*\lambda(t)^{-1}$$

are the diagonal parts of B and B_*, respectively. Since the orbit is closed, these diagonal matrices lie in the same orbit. Finally, conjugation by permutation matrices permutes the diagonal entries, so the orbit is determined precisely by the unordered pairs of eigenvalues of B and B_*. Therefore,

$$X_0 /\!/ GL(V) \cong (\mathbb{A}^n \times \mathbb{A}^n)/S_n,$$

where S_n acts diagonally by simultaneously permuting the coordinates. Hence

$$\mathfrak{M}_{0,0}(n, 1) \cong (\mathbb{A}^n \times \mathbb{A}^n)/S_n.$$

7 Derived Category of GIT: Combinatorial Constructions

In this section, we focus on derived categories arising from a representation X of G. A fundamental example, discussed earlier, is provided by quiver representations, which lead to the construction of Nakajima varieties.

7.1 Magic Window

Let (X) be a representation of (G). Then (X) admits a weight-space decomposition $X = \bigoplus_{\alpha \in X^*(G)} X_\alpha$, where $X^*(G) = \mathrm{Hom}(G, \mathbb{G}_m)$ is the character lattice of (G), and X_α denotes the weight space of weight α. Also let $X_*(G) = \mathrm{Hom}(\mathbb{G}_m, G)$ be the lattice of one-parameter subgroups cocharacters of G. There is a natural pairing

$$\langle\ ,\ \rangle : X^*(G) \times X_*(G) \longrightarrow \mathbb{Z}$$
$$(\alpha, \lambda) \longmapsto \langle \alpha, \lambda \rangle,$$

where $\langle \alpha, \lambda \rangle \in \mathbb{Z}$ is the unique integer such that

$$(\alpha \circ \lambda)(t) = t^{\langle \alpha, \lambda \rangle} \qquad \text{for all } t \in \mathbb{G}_m.$$

Consider a linearisation determined by a character l of G. Let

$$X^{us} = \bigsqcup_{\lambda} S_\lambda$$

be the HKKN stratification of the unstable locus. For a representation (X) of (G), the stratification is constructed as follows.

Choose a 1-PS $\lambda : \mathbb{G}_m \to G$. Let

$$X = \bigoplus_{\alpha \in X^*(G)} X_\alpha$$

be the weight-space decomposition of the representation X with respect to G. Given a one-parameter subgroup $\lambda : \mathbb{G}_m \longrightarrow G$, the representation X restricts to a $\mathbb{G}_m$-representation via λ. This yields the decomposition

$$X = \bigoplus_{m \in \mathbb{Z}} X(m), \qquad X(m) = \bigoplus_{\langle \alpha, \lambda \rangle = m} X_\alpha,$$

where $X(m)$ denotes the λ-weight-m subspace. Consequently,

$$Z_\lambda = \bigoplus_{\langle \alpha, \lambda \rangle = 0} X_\alpha, \qquad Y_\lambda = \bigoplus_{\langle \alpha, \lambda \rangle \geq 0} X_\alpha.$$

where Z_λ be the corresponding fixed locus and Y_λ is the associated blade. The corresponding HKKN stratum is then $S_\lambda = G \cdot Y_\lambda$. The unstable locus is obtained as the disjoint union of these strata.

We define

$$\eta_\lambda = \mathrm{wt}_\lambda \Big(\det \left(\mathcal{N}^\vee_{S_\lambda / X} \right) \big|_{Z_\lambda} \Big) .$$

There is a natural conormal exact sequence

$$0 \longrightarrow \mathcal{N}^\vee_{S_\lambda / X} \longrightarrow \mathcal{N}^\vee_{Y_\lambda / X} \longrightarrow \mathcal{N}^\vee_{Y_\lambda / S_\lambda} \longrightarrow 0.$$

Since $S_\lambda \cong G \times^{P_\lambda} Y_\lambda$, the stratum S_λ is a fibre bundle over G/P_λ with fibre Y_λ. Consequently,

$$\mathcal{N}_{Y_\lambda / S_\lambda} \cong \mathcal{T}_{S_\lambda}\big|_{Y_\lambda} / \mathcal{T}_{Y_\lambda} \cong \mathfrak{g}/\mathfrak{p}_\lambda \cong \mathfrak{g}_\lambda^{<0},$$

where $\mathfrak{g}_\lambda^{<0} = \bigoplus_{\langle \alpha, \lambda \rangle < 0} \mathfrak{g}_\alpha$. Hence

$$\mathcal{N}^\vee_{Y_\lambda / S_\lambda} \cong \left(\mathfrak{g}_\lambda^{<0} \right)^\vee \cong \mathfrak{g}_\lambda^{>0}.$$

Moreover, $\mathcal{N}_{Y_\lambda / X} \cong X_\lambda^{<0}$, and therefore

$$\eta_\lambda = \mathrm{wt}_\lambda \big(\det \big((X_\lambda^{<0})^\vee \big) \big) - \mathrm{wt}_\lambda \big(\det \big((\mathfrak{g}_\lambda^{<0})^\vee \big) \big) .$$

Equivalently,

$$\eta_\lambda = \mathrm{wt}_\lambda \big(\det \big((X_\lambda^{>0})^\vee \big) \big) - \mathrm{wt}_\lambda \big(\det \big((\mathfrak{g}_\lambda^{>0})^\vee \big) \big) .$$

The expression for η_λ motivates the definition

$$\mathbb{L}^{\lambda > 0} = \sum_{\langle \lambda, \alpha \rangle > 0} \dim X_\alpha \cdot \alpha - \sum_{\langle \lambda, \alpha \rangle > 0} \dim \mathfrak{g}_\alpha \cdot \alpha$$

Then $\eta_\lambda = \langle \lambda, \mathbb{L}^{\lambda > 0} \rangle$, which agrees with the definition given in [HLS20, Section 2].

Recall that for an interval of integers $w = (w_0, \ldots, w_n)$, we defined the corresponding window category in theorem 4.2. For a representation X of G, Corollary 3.10 shows it is generated by $\mathcal{O}_X \otimes V$ with V representations of G. So fix $\{\lambda_1, \cdots, \lambda_n\}$ to be the 1-PSs related to the HKKN stratification, $\mathcal{O}_X \otimes V$ lies in $\mathcal{G}^w$ if and only if

$$V = \bigoplus_{\alpha} V_\alpha, \ \langle \alpha, \lambda_i \rangle \in [w_i, w_i + \eta_i) \text{ if } V_\alpha \neq 0$$

We now introduce the following family of polytopes.

Definition 7.1. Let $\epsilon \in X^*(G)_{\mathbb{R}}$ be a character. Define the polytope

$$\overline{\nabla} := \left\{ \chi \in X^*(G)_{\mathbb{R}} \;\middle|\; -\frac{\eta_\lambda}{2} \leq \langle \lambda, \chi \rangle \leq \frac{\eta_\lambda}{2} \text{ for every 1-PS } \lambda \right\}.$$

Remark. At first glance, $\overline{\nabla}$ appears to be defined by infinitely many inequalities. However, the authors of [HLS20] show that these inequalities are redundant. In fact, only finitely many inequalities are needed to define $\overline{\nabla}$, making it possible to relate $\overline{\nabla}$ to the window category.

Definition 7.2. Let $\Omega \subset X^*(T)_{\mathbb{R}}$ be a region. Define $\mathcal{M}(\Omega) \subset D^b(X/G)$ to be the full triangulated subcategory generated by objects of the form

$$\mathcal{O}_X \otimes V,$$

where V is a finite-dimensional G-representation whose T-weights all lie in Ω.

However, we first introduce a class of representations that will play an important role throughout this section.

Definition 7.3. A representation X of G is called *quasi-symmetric* if for every line $L \subset X^*(T)_{\mathbb{R}}$, the sum of the weights of X lying on L, counted with multiplicity, is zero; that is,

$$\sum_{\substack{\alpha \in L \\ X_\alpha \neq 0}} \dim X_\alpha \cdot \alpha = 0.$$

Example 7.4. For any representation X of G, the cotangent representation $T^*X \cong X \oplus X^\vee$ is quasi-symmetric, since every weight α of X is paired with the weight $-\alpha$ of $X^\vee$ with the same multiplicity.

In particular, for a quiver Q, we have

$$T^*\mathrm{Rep}(Q, v) \cong \mathrm{Rep}(Q, v) \oplus \mathrm{Rep}(Q, v)^\vee \cong \mathrm{Rep}(\overline{Q}, v),$$

where $\overline{Q}$ denotes the doubled quiver. Hence $\mathrm{Rep}(\overline{Q}, v)$ is always a quasi-symmetric representation.

From now on, all representations are assumed to be quasi-symmetric.

Theorem 7.5. *Let δ and l be characters of G, and suppose that $\partial(\delta + \overline{\nabla}) \cap X^*(T) = \varnothing$. Then the restriction functor*

$$\mathrm{Res}_{X^{ss}(l)} : \mathcal{M}(\delta + \overline{\nabla}) \longrightarrow D^b(X^{ss}(l)/G)$$

is fully faithful. Moreover, if $X^{ss}(l) = X^s(l)$,then it is an equivalence.

Proof. See [HLS20, Section 2, Section 3]. □

Remark. $\partial(\delta + \overline{\nabla}) \cap X^*(T) = \varnothing$. is the genericity condition ensuring no lattice point lies on the boundary of the translated polytope. Consequently, every lattice point of $\delta + \overline{\nabla}$ is either strictly inside or strictly outside, so the window is unambiguously determined by the lattice points contained in the polytope.

7.2 Wall Crossing

Let us discuss the condition $X^{ss}(l) = X^s(l)$ when X is a quasi-symmetric representation.

Let $\beta_1, \ldots, \beta_n$ denote the T-weights of $X^\vee$. Define the zonotope

$$\overline{\Sigma} := \left\{ \sum_{i=1}^n a_i \beta_i \;\middle|\; a_i \in [-1, 0] \right\} \subset X^*(T)_{\mathbb{R}}.$$

We say that a character $l \in X^*(T)_{\mathbb{R}}$ is *generic for* $\overline{\Sigma}$ if it is not parallel to any proper face of $\overline{\Sigma}$.

Proposition 7.6. *Let X be a quasi-symmetric representation of a reductive group G, and assume that the action of G on X has generically finite stabilizers. Let $l \in X^*(G)_{\mathbb{R}} \subset X^*(T)_{\mathbb{R}}$ be a character whose image in $X^*(T)_{\mathbb{R}}$ is generic for $\overline{\Sigma}$. Then $X^{ss}(l) = X^s(l)$.*

There is a useful relation between $\overline{\nabla}$ and $\overline{\Sigma}$.

Lemma 7.7. *Assume that $\epsilon \in X^*(G)_{\mathbb{R}}$ is generic with respect to $\overline{\Sigma}$. Then*

$$X^*(T)^+_{\mathbb{R}} \cap \left(-\rho + \frac{1}{2}\overline{\Sigma}\right) = X^*(T)^+_{\mathbb{R}} \cap \overline{\nabla}.$$

Proof. [HLS20, Lemma 2.9, Corollary 2.10] □

In the following examples, we consider actions of $\mathbb{G}_m$. In this case, the generic characters are precisely the points of $\mathbb{R} \setminus \{0\}$.

Example 7.8. Consider the action of $\mathbb{G}_m$ on

$$X = \mathbb{A}^n \oplus \mathbb{A}^n,$$

where $\mathbb{G}_m$ acts with weights $+1$ and -1 on the two summands, respectively. Then

$$X^{ss}_+ = (\mathbb{A}^n \setminus \{0\}) \oplus \mathbb{A}^n, \qquad X^{ss}_= \mathbb{A}^n \oplus (\mathbb{A}^n \setminus \{0\}).$$

Consequently,

$$X^{ss}_+ /\!/ \mathbb{G}_m \simeq \mathrm{Tot}_{\mathbb{P}^{n-1}}\big(\mathcal{O}_{\mathbb{P}^{n-1}}(-1)^{\oplus n}\big),, \qquad X^{ss}_- /\!/ \mathbb{G}_m \simeq \mathrm{Tot}_{\mathbb{P}^{n-1}}\big(\mathcal{O}_{\mathbb{P}^{n-1}}(-1)^{\oplus n}\big).$$

These fit into the diagram

$$X^{ss}_+ /\!/ \mathbb{G}_m \xrightarrow{\pi_+} X^{ss}(0) /\!/ \mathbb{G}_m \xleftarrow{\pi_-} X^{ss}_- /\!/ \mathbb{G}_m,$$

which is the Atiyah flop.

Take $\nabla = [-1, 1], \delta = \frac{1}{2}$. Then the window category $\mathcal{M}(\delta + \nabla)$ identifies with the derived categories on both sides:

$$\begin{array}{ccccc} D^b(X^{ss}(+)/\mathbb{G}_m) & \longleftarrow & \mathcal{M}(\delta + \nabla) & \longrightarrow & D^b(X^{ss}(-)/\mathbb{G}_m) \\ \pi^*\mathcal{O}_{\mathbb{P}^1} & \longleftarrow\!\shortmid & \mathcal{O}_X \otimes k_0 & \shortmid\!\longrightarrow & \pi^*\mathcal{O}_{\mathbb{P}^1} \\ \pi^*\mathcal{O}_{\mathbb{P}^1}(-1) & \longleftarrow\!\shortmid & \mathcal{O}_X \otimes k_1 & \shortmid\!\longrightarrow & \pi^*\mathcal{O}_{\mathbb{P}^1}(1). \end{array}$$

This recovers the derived equivalence described in Example **??**.

Example 7.9. Consider the action of $\mathbb{G}_m$ on

$$X = \mathbb{A}^1 \oplus \mathbb{A}^n,$$

where the two summands have weights $+1$ and $-n$, respectively. Since the weights are $\{1, -n, \ldots, -n\}$, the representation is quasi-symmetric.

The semistable loci are

$$X^{ss}_+ = (\mathbb{A}^1 \setminus \{0\}) \oplus \mathbb{A}^n, \qquad X^{ss}_- = \mathbb{A}^1 \oplus (\mathbb{A}^n \setminus \{0\}).$$

Hence

$$X^{ss}_+ /\!/ \mathbb{G}_m \simeq \mathrm{Tot}_{\mathbb{P}^{n-1}}\big(\mathcal{O}_{\mathbb{P}^{n-1}}(-n)\big),, \qquad X^{ss}_- /\!/ \mathbb{G}_m \simeq [\mathbb{A}^n/(\mathbb{Z}/n\mathbb{Z})].$$

Taking $\nabla = [-n/2, n/2], \delta = (n-1)/2$, the window theorem yields an equivalence

$$\mathcal{M}(\delta + \nabla) \xrightarrow{\sim} D^b(X^{ss}(+)/\mathbb{G}_m).$$

Under this equivalence,

$$\mathcal{O}_X \otimes k_i \longmapsto \pi^* \mathcal{O}_{\mathbb{P}^{n-1}}(i), \qquad 0 \leq i \leq n-1,$$

recovering the standard exceptional collection on $\mathbb{P}^{n-1}$.

Proposition 7.6 shows that every connected component of $(X^*(G)_{\mathbb{R}})_{\overline{\Sigma}\text{-gen}}$ is contained in a single GIT chamber. Since the complement of $(X^*(G)_{\mathbb{R}})_{\overline{\Sigma}\text{-gen}}$ is a locally finite union of proper affine hyperplanes, the $\overline{\Sigma}$-generic locus is open and dense in $X^*(G)_{\mathbb{R}}$.

Consequently, we obtain the following corollary.

Corollary 7.10. *Assume that $(X^*(G)_{\mathbb{R}})_{\overline{\Sigma}\text{-}gen} \neq \varnothing$. Then for every $l \in X^*(G)_{\mathbb{R}}$, there exists a character l' arbitrarily close to l such that $X^{ss}(l') = X^s(l')$.*

7.3 Derived Category of Symplectic Resolution

Definition 7.11. A symplectic linear representation X of a reductive group G is

1). A vector space equipped with a non-degenerate skew-symmetric bilinear form $\omega \in \bigwedge^2 X^\vee$;

2). G acts linearly on X and preserving ω.

For example, we can consider the representation of quivers, which naturally carry a symplectic structure.

Let X be a symplectic representation of a reductive group G with symplectic form ω, and let $\mu : X \longrightarrow \mathfrak{g}^\vee$ be the corresponding moment map, normalized so that $\mu(t \cdot x) = t^2 \mu(x)$ for $t \in \mathbb{G}_m$. Using the Killing form, we identify $\mathfrak{g} \simeq \mathfrak{g}^\vee$, and define

$$W : X \times \mathfrak{g} \longrightarrow \mathbb{A}^1, \qquad W(v, \xi) = \langle \mu(v), \xi \rangle.$$

The group G acts on $X \times \mathfrak{g}$ by $g \cdot (v, \xi) = (g \cdot v, \mathrm{Ad}(g)\xi)$. In addition, $\mathbb{G}_m$ acts by scalar multiplication, $t \cdot (v, \xi) = (tv, t\xi)$. Hence $X \times \mathfrak{g}$ becomes a representation of $G \times \mathbb{G}_m$.

For $g \in G$,

$$W(g \cdot (v, \xi)) = \langle \mu(gv), \mathrm{Ad}(g)\xi \rangle.$$

Since μ is equivariant and the Killing form is Ad-invariant,

$$W(g \cdot (v, \xi)) = \langle \mathrm{Ad}(g)\mu(v), \mathrm{Ad}(g)\xi \rangle = W(v, \xi).$$

Similarly,

$$W(t \cdot (v, \xi)) = \langle \mu(tv), t\xi \rangle = \langle t^2 \mu(v), t\xi \rangle = t^3 W(v, \xi).$$

Therefore W is semi-invariant of weight 3. Now, W is a superpotient on $X \times \mathfrak{g}$. We now compare the categories of matrix factorizations associated with different GIT quotients.

Lemma 7.12. *Let $W : X \longrightarrow \mathbb{A}^1$ be a G-invariant and $\mathbb{G}_m$-equivariant morphism. Let $\overline{\Sigma}$ be the zonotope corresponding to the character of the quasi-symmetry representation X. Assume $\overline{\Sigma}$ spans $X^*(T)_{\mathbb{R}}$ and $(X^*(G)_{\mathbb{R}})_{\overline{\sum}-\text{gen}} \neq \emptyset$. Then for any $l, l' \in X^*(G)_{\mathbb{R}}$ such that $X^{ss}(l) = X^s(l)$ and $X^{ss}(l') = X^s(l')$, we have*

$$MF(X^{ss}(l)/G \times \mathbb{G}_m, W) = MF(X^{ss}(l')/G \times \mathbb{G}_m, W)$$

Proof. By Theorem 5.16, we may identify the category of matrix factorizations with the category of singularities:

$$\mathrm{MF}(X/G, W) \simeq D_{\mathrm{sg}}(W^{-1}(0)) \simeq D^b\big(W^{-1}(0)\big)\big/\operatorname{Perf}\big(W^{-1}(0)\big).$$

Therefore, it suffices to show that the following equivalences hold:

$$D^b\big(X_0^{ss}(l)/(G\times\mathbb{G}_m)\big) \simeq D^b\big(X_0^{ss}(l')/(G\times\mathbb{G}_m)\big), \qquad \operatorname{Perf}\big(X_0^{ss}(l)/(G\times\mathbb{G}_m)\big) \simeq \operatorname{Perf}\big(X_0^{ss}(l')/(G\times\mathbb{G}_m)\big).$$

Recall that when we write $X^{ss}(l)/(G\times\mathbb{G}_m)$, we are considering the linearisation given by the character $(l,0)\in X^*(G\times\mathbb{G}_m)$. Suppose that the HKKN stratification of X with respect to l is

$$X^{us} = \bigsqcup_{\lambda_i} S_{\lambda_i},$$

with $\eta_{\lambda_i} = \mathrm{wt}_{\lambda_i}\Big(\det \mathcal{N}^\vee_{S_{\lambda_i}/X}\big|_{Z_{\lambda_i}}\Big)$.

Now consider the action of $G\times\mathbb{G}_m$ with linearisation $(l,0)$. For any one-parameter subgroup $(\alpha,m):\mathbb{G}_m\longrightarrow G\times\mathbb{G}_m$, we have

$$\langle (l,0),(\alpha,m)\rangle = \langle l,\alpha\rangle.$$

Hence the Hilbert–Mumford numerical function is independent of the second factor. Consequently, the semistable locus is unchanged, and the HKKN stratification for the action of $G\times\mathbb{G}_m$ coincides with that for the action of G. In particular, the unstable strata are indexed by the same destabilising one-parameter subgroups, which we identify with $(\lambda_i,0)$.

Therefore, the corresponding window category is

$$\mathcal{G}^w = \left\{\mathcal{F}^\bullet\in D^b(X/(G\times\mathbb{G}_m)) \;\middle|\; \mathrm{wt}_{\lambda_i}\big(\mathcal{H}^*(\mathcal{F}^\bullet)\big)\in[w_i, w_i+\eta_{\lambda_i}) \text{ for all } i\right\},$$

and Theorem 4.3 yields an equivalence

$$\mathcal{G}^w \simeq D^b\big(X^{ss}(l)/(G\times\mathbb{G}_m)\big).$$

This motivates the consideration of the full triangulated subcategory of $D^b(X/(G\times\mathbb{G}_m))$ generated by objects of the form $\mathcal{O}_X\otimes(U\boxtimes k_m)$, where U is a G-representation whose weights lie in $\delta+\overline{\nabla}$, and k_m denotes the one-dimensional representation of $\mathbb{G}_m$ of weight m.

Now let $\delta\in X^*(G)_{\mathbb{R}}$ be such that $\partial(\delta+\overline{\nabla})\cap M=\varnothing$. Then Theorem 7.5 yields an equivalence

$$F_{l,l',\delta}: D^b(X^{ss}(l)/(G\times\mathbb{G}_m)) \xrightarrow{\mathrm{res}^{-1}_{X^{ss}(l)}} \mathcal{M}(\delta+\overline{\nabla}) \xrightarrow{\mathrm{res}_{X^{ss}(l')}} D^b(X^{ss}(l')/(G\times\mathbb{G}_m)).$$

Moreover, this equivalence restricts to an equivalence

$$\operatorname{Perf}(X^{ss}(l)/(G\times\mathbb{G}_m)) \simeq \operatorname{Perf}(X^{ss}(l')/(G\times\mathbb{G}_m)).$$

Since $W: X/G\times\mathbb{G}_m\to\mathbb{A}^1/\mathbb{G}_m$ is equivariant, pullback along W endows $\mathcal{M}(\delta+\overline{\nabla})$ with the structure of a $\operatorname{Perf}(\mathbb{A}^1/\mathbb{G}_m)$-module category. Explicitly, for $E\in\operatorname{Perf}(\mathbb{A}^1/\mathbb{G}_m)$ and $\mathcal{F}\in\mathcal{M}(\delta+\overline{\nabla})$, the action is given by

$$\mathcal{F}\mapsto\mathcal{F}\otimes W^*E.$$

Applying the base-change theorem for derived and perfect categories, we obtain

$$\operatorname{Perf}(\{0\})\otimes_{\operatorname{Perf}(\mathbb{A}^1/\mathbb{G}_m)} D^b(X^{ss}(l)/(G\times\mathbb{G}_m)) \simeq D^b(X_0^{ss}(l)/(G\times\mathbb{G}_m)),$$

and

$$\operatorname{Perf}(\{0\})\otimes_{\operatorname{Perf}(\mathbb{A}^1/\mathbb{G}_m)} \operatorname{Perf}(X^{ss}(l)/(G\times\mathbb{G}_m)) \simeq \operatorname{Perf}(X_0^{ss}(l)/(G\times\mathbb{G}_m)).$$

Combining these equivalences with $F_{l,l',\delta}$, we deduce

$$D^b\big(X_0^{ss}(l)/(G\times\mathbb{G}_m)\big) \simeq D^b\big(X_0^{ss}(l')/(G\times\mathbb{G}_m)\big), \qquad \operatorname{Perf}\big(X_0^{ss}(l)/(G\times\mathbb{G}_m)\big) \simeq \operatorname{Perf}\big(X_0^{ss}(l')/(G\times\mathbb{G}_m)\big).$$

□

Lemma 7.13. *Assume $X^{ss}(l) = X^s(l)$. Then we have*

$$\mathrm{Crit}(W) \cap (X \times \mathfrak{g})^{ss}(l) = \mathrm{Crit}(W) \cap X^{ss}(l) \times \mathfrak{g}$$

Proof. For $(u, \eta) \in T_x X \oplus \mathfrak{g}$,

$$dW_{(x,\xi)}(u, \eta) = \langle d\mu_x(u), \xi \rangle + \langle \mu(x), \eta \rangle.$$

Using the defining property of the moment map, $\langle d\mu_x(u), \xi \rangle = dH_\xi = \omega(\xi_M(x), u)$, hence

$$dW_{(x,\xi)}(u, \eta) = \omega(\xi_M(x), u) + \langle \mu(x), \eta \rangle.$$

Since ω and $\langle -, - \rangle$ are non-degenerate,

$$dW_{(x,\xi)} = 0 \iff \mu(x) = 0 \text{ and } \xi_M(x) = 0.$$

By Theorem 1.52, $\mathrm{Crit}(W) \cap (X \times \mathfrak{g})^{ss}(l) \supset \mathrm{Crit}(W) \cap X^{ss}(l) \times \mathfrak{g}$ is obvious.

Suppose that x is unstable. Then there exists a one-parameter subgroup $\lambda : \mathbb{G}_m \to G$ such that $\lim_{t\to 0} \lambda(t) \cdot x$ exists and $\mu^l(x, \lambda) > 0$. Since $H_\xi(x) = 0$, the infinitesimal action of ξ vanishes at x. Hence $\xi \in \mathrm{Lie}(\mathrm{Stab}(x))$. Since $\lim_{t\to 0} \lambda(t) \cdot x$ exists, the stabiliser of x is contained in the associated parabolic subgroup P_λ. Consequently, $\xi \in \mathfrak{p}_\lambda := \mathrm{Lie}(P_\lambda)$.

Write $\mathfrak{g} = \bigoplus_{m\in\mathbb{Z}} \mathfrak{g}_m$ for the λ-weight decomposition. Since $\xi \in \mathfrak{p}_\lambda = \bigoplus_{m\geq 0} \mathfrak{g}_m$, we may write $\xi = \sum_{m\geq 0} \xi_m, \xi_m \in \mathfrak{g}_m$. Then

$$\mathrm{Ad}(\lambda(t))\xi = \sum_{m\geq 0} t^m \xi_m,$$

which admits a limit as $t \to 0$. Since the linearisation on $X \times \mathfrak{g}$ is the pullback of the linearisation on X, say $(l, 0)$, the Hilbert–Mumford numerical function satisfies

$$\mu^{(l,0)}(x, \xi, \lambda) = \mu^l(x, \lambda).$$

Hence the same one-parameter subgroup λ destabilises (x, ξ). Therefore $\lim_{t\to 0} \lambda(t) \cdot (x, \xi)$ exists and $\mu^{(l,0)}(x, \xi, \lambda) = \mu^{l)}(x, \lambda) > 0$ Thus (x, ξ) is unstable.

Consequently, if (x, ξ) is semistable, then x must be semistable. This proves the reverse inclusion and hence the lemma. □

Theorem 7.14. *Let X be a symplectic representation of a reductive group G, μ be the unique moment map $\mu : X \longrightarrow \mathfrak{g}^\vee$ such that $\mu(t \cdot x) = t^2\mu(x)$ and let $\overline{\Sigma}$ be the zonotope corresponding to the character of the linear representation $X \oplus \mathfrak{g}$. If $(X^*(G)_\mathbb{R})_{\overline{\sum}-\mathrm{gen}} \neq \emptyset$, then for any $l, l' \in X^*(G)_\mathbb{R}$ such that $X_0^{ss}(l) = X_0^s(l)$ and $X_0^{ss}(l') = X_0^s(l')$, one has a derived equivalence of hyperkähler quotients*

$$D^b(X_0^{ss}(l)/G) \simeq D^b(X_0^{ss}(l')/G)$$

Proof. Applying this Lemma 7.12 to $X \times \mathfrak{g}$, we get

$$\mathrm{MF}\left((X \times \mathfrak{g})^{ss}(l')/G \times \mathbb{G}_m, W\right) \cong \mathrm{MF}\left((X \times \mathfrak{g})^{ss}(l)/G \times \mathbb{G}_m, W\right)$$

By Proposition 5.11, we aim to analysis the singular points in $(X \times \mathfrak{g}^{ss}(l)$.

By Lemma 7.13, the critical locus of W inside $(X \times \mathfrak{g})^{ss}(l)$ coincides with its critical locus inside $X^{ss}(l) \times \mathfrak{g}$. Therefore, Proposition 5.11 implies that

$$\mathrm{MF}((X \times \mathfrak{g})^{ss}(l)/(G \times \mathbb{G}_m), W) \simeq \mathrm{MF}((X^{ss}(l) \times \mathfrak{g})/(G \times \mathbb{G}_m), W)\,.$$

Consider the vector bundle $E := (X^{ss}(l) \times \mathfrak{g})/G \longrightarrow X^{ss}(l)/G$. Its dual bundle is $E^\vee = (X^{ss}(l) \times \mathfrak{g}^\vee)/G$. The moment map $\mu : X^{ss}(l) \longrightarrow \mathfrak{g}^\vee$ is G-equivariant and therefore determines a section $s_\mu \in H^0(X^{ss}(l)/G, E^\vee)$. Under the identification $\mathrm{Tot}(E) = (X^{ss}(l) \times \mathfrak{g})/G$, the function

$$W(v, \xi) = \langle \mu(v), \xi \rangle$$

is precisely the canonical pairing between the tautological fiber coordinate ξ and the section s_μ. Therefore Theorem 5.24 yields

$$\mathrm{MF}((X^{ss}(l)\times\mathfrak{g})/(G\times\mathbb{G}_m),W)\simeq D^b\big(Z(s_\mu)\big).$$

Since $Z(s_\mu)=\mu^{-1}(0)^{ss}(l)/G=X_0^{ss}(l)/G$, we obtain

$$\mathrm{MF}((X^{ss}(l)\times\mathfrak{g})/(G\times\mathbb{G}_m),W)\simeq D^b(X_0^{ss}(l)/G).$$

□

Example 7.15. 1). Let us first consider the Mukai flop from Example 6.10. Recall the two symplectic resolutions

$$\pi_+:\mathfrak{M}_{\theta>0,0}(v,w)\longrightarrow Z,\qquad \pi_-:\mathfrak{M}_{\theta<0,0}(v,w)\longrightarrow Z,$$

where

$$\mathfrak{M}_{\theta>0,0}(v,w)\cong T^*\operatorname{Gr}(v,W),\qquad \mathfrak{M}_{\theta<0,0}(v,w)\cong T^*\operatorname{Gr}(w-v,W),$$

and

$$Z=\mathfrak{M}_{0,0}(v,w)=\left\{M\in\operatorname{End}(W)\;\middle|\;M^2=0,\ \operatorname{rank}(M)\le v\right\}.$$

By Theorem 7.14, there is an equivalence of derived categories

$$D^b\big(\mathfrak{M}_{\theta>0,0}(v,w)\big)\simeq D^b\big(\mathfrak{M}_{\theta<0,0}(v,w)\big).$$

Moreover, in the special case $v=1$ and $w=n+1$, Example 6.10 identifies the correspondence

$$T^*\mathbb{P}(W)\ \xleftarrow{p_+}\ X_0^+\times_Z X_0^-\ \xrightarrow{p_-}\ T^*\mathbb{P}(W^\vee),$$

where

$$X_0^+\times_Z X_0^-\cong\operatorname{Bl}_{\mathbb{P}(W)}\big(T^*\mathbb{P}(W)\big)\cong\operatorname{Bl}_{\mathbb{P}(W^\vee)}\big(T^*\mathbb{P}(W^\vee)\big).$$

The induced Fourier–Mukai functor

$$\Phi=Rp_{-*}\circ Lp_+^*:D^b\big(\mathfrak{M}_{\theta>0,0}(1,n+1)\big)\longrightarrow D^b\big(\mathfrak{M}_{\theta<0,0}(1,n+1)\big)$$

is an equivalence, as proved in [Huy06, Section 11].

In fact this can be achieved via some equivalence via window category. See [ST25].

7.4 Action of Kähler Groupiod

In Theorem 7.5, one must choose a character δ such that

$$\partial(\delta+\overline{\nabla})\cap M=\varnothing.$$

In fact, the set of admissible δ is the complement of a periodic hyperplane arrangement.

Lemma 7.16. *There exists a locally finite periodic hyperplane arrangement in $X^*(T)_\mathbb{R}$ such that, for every δ lying outside this arrangement,*

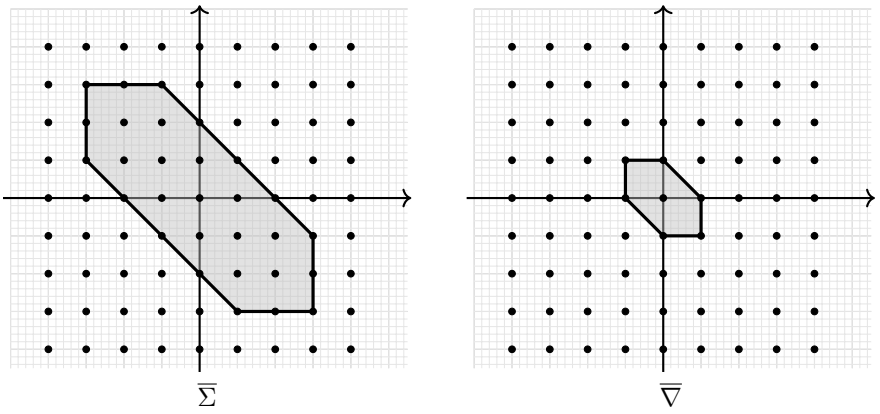
$$\partial(\delta+\overline{\nabla})\cap M=\varnothing.$$

Proof. Taking $H_1,\cdots,H_n$ are the hyperplanes that $H_i\cap\overline{\nabla}$ is a facet, then taking the hyperplane arrangement to be the set of $m-H_i$ with $m\in X^*(T)$. Moreover, since M is a lattice, such arrangement is locally finite. □

Example 7.17. Consider the action of GL_2 acts on $X := T^*\mathrm{Rep}(J^{fr}, 2, 1)$. Then it has weights:

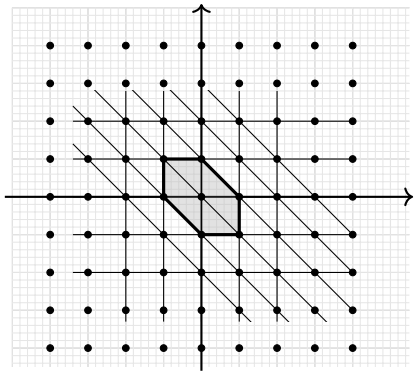
$$e_1, -e_1, e_2, -e_2, e_1 - e_2, e_2 - e_1, e_1 - e_2, e_2 - e_1$$

Then $\overline{\Sigma}$ and $\overline{\nabla}$ related to this action are given by



$\overline{\Sigma}$ $\qquad$ $\overline{\nabla}$

Then by lemma 7.16, we get the hyper surfaces are



In this section, we study how the window category

$$\mathcal{M}(\delta + \overline{\nabla})$$

changes when δ crosses one of the hyperplanes in the arrangement. This wall-crossing behavior will lead to semiorthogonal decompositions and derived equivalences between different windows.

Let V be a real vector space along with a locally finite hyperplane arrangement $\mathcal{A} = \{H_\alpha \subset V\}$. Fix a subset $V_{gen} \subset V$ with the following property:

1). For any collection of hyperplanes $H_{\alpha_0}, \cdots, H_{\alpha_n} \in \mathcal{A}$, let $H'_{\alpha_0}, \cdots, H'_{\alpha_n}$ be the same hypersurfaces translated so that they pass through the origin. Then there is some $l \in V_{gen}$ lying in each connected;

2). V_{gen} is symmetric with respect to the origin.

Consider the directed graph whose vertices are the chambers $[\delta]$, for $\delta \in V$ not lying on any hyperplane of $\mathcal{A}$, and whose edges

$$[\delta] \xrightarrow{l} [\delta']$$

are labelled by elements $l \in V_{\mathrm{gen}}$, where l is not parallel to any hyperplane of $\mathcal{A}$ meeting the line segment

$$[(\delta, \delta')] = \{(1-t)\delta + t\delta' \mid t \in [0,1]\}.$$

Definition 7.18. The groupoid $\Gamma(V, \mathcal{A})$ is the quotient of the free groupoid on this directed graph by the congruence generated by the following relations:

1) If no hyperplane of $\mathcal{A}$ meets the line segment $[(\delta, \delta')]$, then the arrow

$$[\delta] \xrightarrow{l} [\delta']$$

is identified with the identity morphism.

2) Whenever $[\delta] \xrightarrow{l} [\delta'], [\delta'] \xrightarrow{l} [\delta'']$ are valid arrows, one has

$$[\delta] \xrightarrow{l} [\delta''] \ \sim \ \big([\delta'] \xrightarrow{l} [\delta'']\big) \circ \big([\delta] \xrightarrow{l} [\delta']\big).$$

3) For any pair of valid arrows $[\delta] \xrightarrow{l} [\delta'], [\delta] \xrightarrow{l'} [\delta']$, they are identified whenever l and l' induce the same orientation with respect to every hyperplane of $\mathcal{A}$ meeting the segment $[(\delta, \delta')] = \{(1-t)\delta + t\delta' \mid t \in [0,1]\}$.

Remark. By [HLS20, Section 6], there exists a family of equivalences

$$\Phi_{\delta,\delta'} : \mathcal{M}(\delta + \overline{\nabla}) \longrightarrow \mathcal{M}(\delta' + \overline{\nabla}), \qquad \delta, \delta' \in V_{\text{gen}},$$

satisfying the following properties:

1) For every $\delta \in V_{\text{gen}}$, $\Phi_{\delta,\delta} = \text{id}$.

2) For any $\delta, \delta', \delta'' \in V_{\text{gen}}$, $\Phi_{\delta',\delta''} \circ \Phi_{\delta,\delta'} = \Phi_{\delta,\delta''}$.

3) If the wall-crossing paths from δ to δ' determined by l and l' induce the same orientation with respect to every hyperplane crossed by the segment joining δ and δ', then the resulting equivalences coincide.

These properties motivate the relations imposed in the definition of the groupoid $\Gamma(V, \mathcal{A})$.

Consider the fundamental groupiod Π_1 of $V \otimes_{\mathbb{R}} \mathbb{C} \setminus \bigcup_\alpha H_\alpha \otimes_{\mathbb{R}} \mathbb{C}$. For every object $[\delta] \in \Gamma(V, \mathcal{A})$, we assign the point $\delta + 0i \in V \otimes_{\mathbb{R}} \mathbb{C}$, while for every morphism $[\delta] \xrightarrow{l} [\delta']$, define

$$\gamma(t) = \begin{cases} \delta + 3tli, & t \in [0, \frac{1}{3}] \\ (2-3t)\delta + (3t-1)\delta' + li, & t \in [\frac{1}{3}, \frac{2}{3}] \\ \delta' + (3-3t)li, t \in [\frac{2}{3}, 1] \end{cases}$$

We can show that

Lemma 7.19. *Each of the equivalence of arrows in* $(1), (2), (3)$ *correspond to a homotopy of paths.*

So we have

Proposition 7.20. *The functor* $\Phi : \Gamma(V, \mathcal{A}) \longrightarrow \Pi(V_{\mathbb{C}} \setminus \bigcup H_{\alpha,\mathbb{C}})$ *is an equivalence of groupiod.*

Except wall crossing, tensoring character of G also gives an equivalence between $\mathcal{M}(\delta + \overline{\nabla})$ and $\mathcal{M}(\delta + \chi + \overline{\nabla})$.

Now, let $V = L \otimes_{\mathbb{Z}} \mathbb{R}$ be a real vector space generated by a lattice L, and that locally finite hyperplane arrangement $\{H_\alpha \subset L_{\mathbb{R}}\}$ is invariant under the action of L by translation.

Definition 7.21. Let $\widetilde{\Gamma}(L, \mathcal{A})$ be the enlargement of the groupiod of $\Gamma(L_{\mathbb{R}}, \mathcal{A})$ defined above, which is obtained by formally adding an arrow $[\delta] \rightsquigarrow [\delta + m]$ for every δ in the complement of the H_α and every $m \in L$. We impose the additional relations

1). arrows of the form $\rightsquigarrow$ commutes with those of the form $\xrightarrow{l}$, in the sense that

$$([\delta] \xrightarrow{l} [\delta'] \rightsquigarrow [\delta' + m]) \sim ([\delta] \rightsquigarrow [\delta + m] \xrightarrow{l} [\delta' + m])$$

whenever the arrow $\xrightarrow{l}$ makes sense.

2). The arrows $[\delta] \rightsquigarrow [\delta + m]$ and $[\delta + m] \rightsquigarrow [\delta + m + m']$ compose to $[\delta] \rightsquigarrow [\delta + m + m']$.

Remark. The additional arrows encode the action of the lattice L by translation on chambers. Relation (1) says that this translation action is compatible with wall-crossing: crossing a wall and then translating by $m \in L$ gives the same morphism as first translating by m and then crossing the corresponding translated wall. In other words, tensoring by the character corresponding to m commutes with wall-crossing functors.

Then we can show

Proposition 7.22. *The equivalence $\Phi : \Gamma(L_{\mathbb{R}}, \mathcal{A}) \to \Pi_1 \left(L_{\mathbb{C}} \setminus \bigcup_\alpha H_{\alpha,\mathbb{C}}\right)$ extends to an isomorphism*

$$\Phi : \widetilde{\Gamma}(L, \mathcal{A}) \to \Pi_1 \left((L_{\mathbb{C}} \setminus \bigcup_\alpha H_{\alpha,\mathbb{C}})/L \right)$$

compatible with the canonical faithul functor $\Pi_1 \left(L_{\mathbb{C}} \setminus \bigcup_\alpha H_{\alpha,\mathbb{C}}\right) \subset \Pi_1 \left((L_{\mathbb{C}} \setminus \bigcup_\alpha H_{\alpha,\mathbb{C}})/L\right)$.

For a quasi-symmetric representation X, by Lemma 7.16, we can find a $X^*(T)$-periodic locally finite hyperplanes arrangement $\mathcal{A} = \{H_\alpha \subset X^*(G)_{\mathbb{R}}\}$, which is defined so that $\partial(\delta + \overline{\nabla}) \cap X^*(T) = \emptyset$ for any $\delta \in X^*(G)$ in the complement of this hyperplane arrangment. Then we define

$$\mathcal{K}_X := \left(X^*(G)_{\mathbb{C}} \setminus \bigcup_\alpha H_{\alpha,\mathbb{C}} \right) / X^*(H)$$

Then Proposition 7.22, implies we have

$$\widetilde{\Gamma}(X^*(G), \mathcal{A}) \simeq \Pi_1(\mathcal{K}_X)$$

Now, assume that there exists $l \in X(T)^*_{\mathbb{R}}$ such that $X^s(l) = X^{ss}(l)$. Chose the hypersurface arangement in Lemma 7.16. Applying Theorem 7.5, we have

Theorem 7.23. *Assigning $F([\delta]) := \mathcal{M}(\delta + \overline{\nabla})$ and assign $F([\delta] \xrightarrow{l} [\delta'])$ to the functor*

$$\Phi_{\delta,\delta'} : \mathcal{M}(\delta + \overline{\nabla}) \xrightarrow{Res} D^b(X^{ss}(l)/G) \xrightarrow{Res^{-1}} \mathcal{M}(\delta' + \overline{\nabla})$$

and assiging $F([\delta] \rightsquigarrow [\delta + m])$ to the functor

$$\Phi_{\delta,\delta+m} : \mathcal{M}(\delta + \overline{\nabla}) \xrightarrow{(\mathcal{O}_X \otimes m) \otimes (-)} \mathcal{M}(\delta + m + \overline{\nabla})$$

extends uniquely to a representation of the groupoid $\widetilde{\Gamma}(X^*(T)^W, \mathcal{A})$.

Proof. [HLS20, Proposition 6.6]. □

From a category $\mathcal{C}$, which is a modulo over $\mathrm{Perf}(\mathbb{A}^1/\mathbb{G}_m)$. Define a singularity category of $\mathcal{C}$ by formally inverting the action of line bundle. First, taking the category supported on the zero fiber

$$\mathcal{C}_0 = \mathrm{Fun}_{\mathrm{Perf}(\mathbb{A}^1/\mathbb{G}_m)} \left(\mathrm{Perf}(\{0\}/\mathbb{G}_m), \mathcal{C}\right)$$

There is a natural endofunctor

$$T(E) = E \otimes \mathbb{A}_1\langle 1 \rangle[2]$$

Define $D_{sg}(\mathcal{C})$ to be the local obtained by forcing $T \simeq \mathrm{id}$ equation for objects E, F,

$$\mathrm{Hom}_{D_{sg}(\mathcal{C})}(E,F) = \varinjlim_{p} \mathrm{Hom}_{\mathcal{C}_0}(E, F \otimes \mathcal{O}\langle p\rangle[2p])$$

Now, we apply the construction of $D_{sg}(-)$ on the magic window. Then it gives an equivalence

$$D_{sg}(\mathcal{M}(\delta+\overline{\nabla})) \simeq D_{sg}(D^b(X^{ss}(l)/G\times\mathbb{G}_m)) \simeq D^b_{sg}(X^{ss}(l)/G\times\mathbb{G}_m)$$

Corollary 7.24. *Let X be a linear representation of $G\times\mathbb{G}_m$, which is quasi-symmetric as a representation of G, and $W: X\longrightarrow\mathbb{A}^1$ be a G-invariant and $\mathbb{G}_m$-equivalence map. Then assigning $F(\delta) = D_{sg}(\mathcal{M}(\delta+\overline{\nabla}))$, assigning $F([\delta]\xrightarrow{l}[\delta'])$ to functor*

$$D_{sg}(\mathcal{M}(\delta+\overline{\nabla})) \xrightarrow{Res} D^b_{sg}(X^{ss}(l)/G\times\mathbb{G}_m, W) \xrightarrow{Res^{-1}} D_{sg}(\mathcal{M}(\delta+\overline{\nabla}))$$

and assigning $F([\delta]\rightsquigarrow[\delta+m])$ to functor

$$D_{sg}(\mathcal{M}(\delta+\overline{\nabla})) \xrightarrow{(\mathcal{O}_X\otimes m)\otimes(-)} D_{sg}(\mathcal{M}(\delta+m+\overline{\nabla}))$$

defines a representation of $\widetilde{\Gamma}(X^*(T)^W, \mathcal{A})$, where $\mathcal{A}$ is the $X^*(T)^W$-periodic hyperplane arrangement in $X^*(T)^W$ determined by the polytope $\overline{\nabla}\subset X^*(T)_{\mathbb{R}}$.

Moreover, in the proof of Theorem 7.14, there exists

$$D^b_{sg}(X^{ss}(l)/G\times\mathbb{G}_m, W) \simeq D^b(X_0^{ss}(l)/G)$$

for $l\in X^*(T)_{\mathbb{R}}$ such that $X^{ss}(l) = X^s(l)$. Then we have

Corollary 7.25. *Let X be a symplectic linear representation of a split reductive group G with moment map $\mu: X\longrightarrow\mathfrak{g}^\vee$. Let $W: X\times\mathfrak{g}\longrightarrow\mathbb{A}^1$ be the G-invariant $\mathbb{G}_m$-equivaraint map induced by μ.*

Regarding $X\oplus\mathfrak{g}$ as a quasi-symmetric representation of G with corresponding zonotope $\overline{\Sigma}$, we assume that $\overline{\Sigma}$ spans $X^*(T)_{\mathbb{R}}$ and $\left(X^*(T)^W_{\mathbb{R}}\right)_{\overline{\Sigma}-gen}\neq\emptyset$. Then for any l for which $X_0^{ss}(l) = X_0^s(l)$ and δ such that $\partial(\delta+\overline{\nabla})\cap X^*(T) = \emptyset$, we have

$$D_{sg}(\mathcal{M}(\delta+\overline{\nabla})) \simeq D^b(X_0^{ss}(l)/G)$$

through which gives an action of $\Pi_1(\mathcal{K}_{X\oplus\mathfrak{g}})$ on $D^b(X_0^{ss}(l)/G)$.

To determine $\mathcal{K}_{X\oplus\mathfrak{g}}$, we need to compute $\overline{\nabla}_{X\oplus\mathfrak{g}}$, which turns out to be the same as $\frac{1}{2}\overline{\Sigma}_X$. Namely, using the fact that $\rho = \frac{1}{2}\sum_{\alpha\in\Phi_+}\alpha$ and by lemma 7.7

$$X^*(T)^+\cap\overline{\nabla}_{X\oplus\mathfrak{g}} = X^*(T)^+\cap(\rho+\frac{1}{2}\overline{\Sigma}_{X\oplus\mathfrak{g}}) = X^*(T)^+\cap(\frac{1}{2}\overline{\Sigma}_X)$$

Example 7.26 (Hilbert Scheme)**.** Consider the Nakajima quiver variety associated to the Jordan quiver, in Exmaple 6.11. In this case, G acts on $X\oplus\mathfrak{g}$ is

$$X' := \mathfrak{gl}_n\oplus T^*\left(\mathrm{Hom}(\mathbb{C}^n,\mathbb{C}^n)\oplus\mathrm{Hom}(\mathbb{C}^n,\mathbb{C})\right)$$

Take T to be the diagonal matrices, that is, the maximal torus, and let $e_1,\cdots,e_n$ be the standard basis for the character latiice, so that the weights of X are $\{e_i,-e_i\mid 1\le i\le n\}$ and $\{e_i-e_j\mid 1\le i,j\le n\}$ with the latter set appering with mutilplication 2, while the weight our $\mathfrak{gl}_n$ is $\{e_i-e_j\mid 1\le i,j\le n\}$ with multiplication 1.

Lemma 7.27. *The hyperplanes arrangement associated to* $\overline{\nabla}$ *is given by*

$$\left\{\sum_{i\in S} x_i = c \mid c \in \mathbb{Z}, S \text{ is a nonempty subset of } \{1,\cdots,n\}\right\}$$

Since $X^*(G)$ is the line spanned by $e_1+\cdots+e_n$, so the intersection with this arrangement gives the set

$$\left\{\left(\frac{a}{b}+\frac{1}{2}\right)(e_1+\cdots+e_n) \mid a,b\in\mathbb{Z}, 1\leq b\leq n\right\}$$

Thus

$$\mathcal{K}_{X\oplus\mathfrak{g}} = \left(\mathbb{C}\setminus\left(\mathbb{Z}\cup\frac{1}{2}\mathbb{Z}\cup\cdots\cup\frac{1}{n}\mathbb{Z}\right)/\mathbb{Z}\right)$$

which is identified with

$$\{q\in\mathbb{C}^* \mid q^k\neq 1, \forall k=1,\cdots,n\}$$

via exponential map $t\mapsto e^{2\pi i t-\frac{1}{2}}$.

References

[Bor91] Armand Borel. *Linear Algebraic Groups*, volume 126 of *Graduate Texts in Mathematics*. Springer, New York, 2 edition, 1991.

[Eis80] David Eisenbud. Homological algebra on a complete intersection, with an application to group representations. *Transactions of the American Mathematical Society*, 260(1):35–64, 1980.

[Har77] Robin Hartshorne. *Algebraic Geometry*, volume 52 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1977.

[HH22] Tamás Hausel and Nigel Hitchin. Very stable higgs bundles, equivariant multiplicity and mirror symmetry. *Inventiones Mathematicae*, 228:893–989, 2022.

[HL15] Daniel Halpern-Leistner. The derived category of a git quotient. *Journal of the American Mathematical Society*, 28(3):871–912, 2015.

[HLS20] Daniel Halpern-Leistner and Steven V. Sam. Combinatorial constructions of derived equivalences. *Journal of the American Mathematical Society*, 33(3):735–773, 2020.

[Hos15] Victoria Hoskins. Moduli problems and geometric invariant theory, 2015. Lecture notes.

[Huy06] Daniel Huybrechts. *Fourier–Mukai Transforms in Algebraic Geometry*. Oxford Mathematical Monographs. Oxford University Press, 2006.

[Soi21] Alexander Soibelman. Lecture notes on quiver representations and moduli problems in algebraic geometry, 2021. Version of 16 May 2021.

[ST25] Ed Segal and Wei Tseng. A simpler kernel for stratified mukai flops. 2025.